

# MINIMAL MODELS AND MORI FIBRE SPACES FOR GENERALIZED LOG CANONICAL $\mathbb{Q}$ -PAIRS

OPENAI

ABSTRACT. We resolve the existence form of the minimal-model conjecture for projective generalized log canonical  $\mathbb{Q}$ -pairs over algebraically closed fields of characteristic zero. Such a pair admits a minimal model when its adjoint divisor is pseudo-effective, and a Mori fibre space otherwise, while keeping the nef  $\mathbb{Q}$ -Cartier  $b$ -divisor fixed. Taking the nef part to be zero gives the corresponding result for ordinary log canonical  $\mathbb{Q}$ -pairs.

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## 1. INTRODUCTION

The minimal model program seeks birational models on which the canonical divisor has a definite sign: nef on a minimal model, or negative along the fibres of a Mori fibre space. For a generalized pair, the adjoint divisor also contains the trace of nef data on a higher birational model. This generalized-pair formalism of Birkar–Zhang [12, Definition 1.4] accommodates the moduli terms arising in the canonical bundle formula [4]. The existence question therefore asks for a birational model while keeping those nef data fixed.

We prove the following form of the existence conjecture. Our discrepancy convention is the log convention: the discrepancy of a component of a crepant boundary of coefficient  $b$  is  $1 - b$ .

**Theorem 1.1.** *Let  $k$  be an algebraically closed field of characteristic zero, and let  $(X, B + M)$  be a projective generalized log canonical  $\mathbb{Q}$ -pair. Thus  $X$  is normal and integral,  $B$  is an effective rational boundary, and  $M$  is the trace of a nef rational Cartier divisor on a higher projective birational model, with  $K_X + B + M$   $\mathbb{Q}$ -Cartier.*

*There are a normal integral projective variety  $Y$  and a birational map  $\phi: X \dashrightarrow Y$  whose inverse contracts no divisors such that, writing  $B_Y = \phi_* B$  and taking  $M_Y$  to be the trace of the same nef  $b$ -divisor, the generalized pair  $(Y, B_Y + M_Y)$  is log canonical and*

$$A_{X, B+M}(E) \leq A_{Y, B_Y+M_Y}(E)$$

*for every prime divisor  $E$  over their common function field. The inequality is strict for every divisor on  $X$  contracted by  $\phi$ . More precisely, the following alternatives are determined by  $D = K_X + B + M$ :*

- (i) *if  $D$  is pseudo-effective, then  $K_Y + B_Y + M_Y$  is nef;*
- (ii) *if  $D$  is not pseudo-effective, there is a projective contraction  $h: Y \rightarrow T$  to a normal projective variety with connected fibres such that*

$$\dim T < \dim Y, \quad \rho(Y/T) = 1, \quad -(K_Y + B_Y + M_Y) \text{ is ample over } T.$$

No  $\mathbb{Q}$ -factoriality is imposed on  $X$  or  $Y$ . The discrepancy comparison uses the original nef  $b$ -divisor and the stated log normalization. The assertion is the existence form of the minimal model conjecture [20, Conjecture 1.1], with fixed rational generalized nef data. It gives a terminating choice of birational program; it makes no assertion that every arbitrary MMP terminates. The nef conclusion of Theorem 1.1 does not itself include semiample or the existence of a nonzero pluricanonical section.

**Corollary 1.2** (Ordinary log canonical pairs). *Let  $(X, B)$  be a projective log canonical  $\mathbb{Q}$ -pair over an algebraically closed field of characteristic zero. There is a birational map  $\phi: X \dashrightarrow Y$  extracting no divisors, with  $B_Y = \phi_* B$ , such that  $(Y, B_Y)$  is log canonical and*

$$A_{X, B}(E) \leq A_{Y, B_Y}(E)$$

*for every prime divisor over the common function field, strictly for prime divisors on  $X$  contracted by  $\phi$ . If  $K_X + B$  is pseudo-effective,  $K_Y + B_Y$  is nef; otherwise  $Y$  admits a Mori fibre contraction with relative Picard number one and relatively ample  $-(K_Y + B_Y)$ .*

*Proof.* Apply Theorem 1.1 to the fixed zero nef  $b$ -divisor. □

In the pseudo-effective ordinary case, the companion log abundance theorem [35, Theorem 1.1] makes  $K_Y + B_Y$  semiample on the same endpoint:  $(Y, B_Y)$  is a projective lc pair with effective rational boundary and nef  $\mathbb{Q}$ -Cartier adjoint. The proof of Corollary 1.2(i) in the same reference already uses this application of Corollary 1.2. This separate consequence gives no semiample assertion for nonzero generalized nef data.

The ordinary case illustrates the role of the generalized data. When  $M = 0$ , the crepant equation defining discrepancies contains only the canonical divisor and the boundary. For a generalized pair one pulls back the same nef divisor from its determination on every common resolution; it cancels

in comparisons between successive models. The local phase theorem concerns ordinary Gorenstein klt germs in both cases. The generalized structure enters the transfer from the smooth relative problem to the final pair, where the NQC MMP applies because the fixed nef data are rational.

**1.1. Background and significance.** The extremal-ray and contraction approach to the minimal model problem originates in Mori’s work [34]; the subsequent theory of singular pairs is developed systematically in Kollár–Mori [28]. Birkar–Cascini–Hacon–McKernan [11] established existence and finite generation for klt pairs in the big setting. Their results supply the flips, positive-boundary programs and finiteness of ample models used in this paper. The remaining pseudo-effective case requires an argument that does not assume bigness of the adjoint or boundary.

Weak Zariski decompositions organize one route from that problem to existence. Birkar [7] related weak decompositions and log minimal models under lower-dimensional MMP hypotheses. Han–Li [22] developed the generalized-pair version, and Lazić–Tsakanikas [29] reduced ordinary-lc existence to the relative smooth problem. Their later special-MMP results [30] give a  $\mathbb{Q}$ -factorial NQC precursor to the transfer used here. Tsakanikas–Xie [36] remove that ambient factoriality restriction using the generalized MMP foundations, including the flip results of Hacon–Liu and Liu–Xie and the contractions of Xie [21, 33, 37]. Their terminating programs have explicit existence or non-pseudo-effectivity hypotheses. We establish the smooth input and its relative weak Zariski decompositions before applying those theorems; the transfer itself is credited to them. Hu–Liu subsequently established flips without the NQC hypothesis [23, Theorems 1.1–1.2]; the terminating transfer used here remains in the NQC setting.

The local argument uses the normalized-volume theory initiated by Li [31]. Blum proved existence of minimizing valuations [13, Main Theorem and Section 7], Xu proved their quasi-monomiality [38, Theorem 1.2], and Xu–Zhuang established uniqueness and finite generation [39, 40]. These results supply the invariant minimizing valuation and its finitely generated klt cone. Section 4 establishes the additional equivariant rational regrading and Gorenstein Rees properties needed here. Birkar’s complements, boundedness of Fano varieties and effective birationality for polarized varieties [8, 9, 10] provide the projective numerical inputs. The new product estimate links those inputs to controlled valuation selection and exact character congruences.

The relative estimates build on Ambro’s canonical bundle formula [4] and the b-semiampleness theorem of Bakker–Filipazzi–Mauri–Tsimmerman [5]. Their bounded-family effectiveness results give a related precedent under additional hypotheses, including klt singularities [5, Corollaries 7.8–7.9]; the reference form here may be lc. Each use of the canonical bundle formula checks generic effectivity, discrepancy rank one and the adjoint pullback identity. Uniformity then comes from the finitely many prepared families, not from an unstated uniform index in the b-semiampleness theorem. Toroidal preparation uses Abramovich–Denef–Karu and Abramovich–Karu [2, 3]; the fibre comparisons also use Kollár’s linking of lc centres [27] and Birkar’s very-exceptional-divisor method [6]. Valuative retraction follows the framework of Jonsson–Mustață [24], while logarithmic Cartier and Deligne–Illusie theory supply the positive-characteristic selection step [15, 25, 16]. The global use of asymptotic multiplier ideals comes from Demailly–Ein–Lazarsfeld [17]. These established methods are distinguished below from the uniform lifting, mixing, rounding and termination arguments proved in this paper.

**1.2. The local input and the global contradiction.** The central local statement is Theorem 3.1. For a sequence of Gorenstein klt germs of fixed dimension with cyclic symmetry, it selects divisorial valuations of bounded discrepancy whose values recover, up to one fixed positive integer, the character angles of a chosen group element. The congruence holds for every rational semi-invariant, and the restricted valuation on the invariant field is integral-valued.

To apply it, suppose that a canonical MMP with scaling, starting from a smooth projective variety, is infinite. Uniformly generated asymptotic multiplier ideals bound the accumulated discrepancy changes. On integral-valued divisorial tests of bounded original discrepancy, the set of

finite limits of these changes is countable. The phase theorem on cyclic index-one covers then rules out unbounded canonical indices: otherwise arbitrary character angles would lie in a countable set. With one common Cartier multiple fixed, higher-direct-image vanishing on late flips and effective base point freeness force a divisor to be both globally generated and negative on a contracted curve. This contradiction proves the smooth absolute input. The relative reduction and descent to the stated field are included in Section 3.

**1.3. Uniform estimates and the structure of the proof.** The exposition begins with the global deduction, then develops its local input, and finally proves the deeper uniform estimates used there. There are three useful reading routes.

For the global conclusion, Theorem 3.1 implies smooth absolute termination (Proposition 3.2). Proposition 3.3 supplies the relative smooth weak Zariski decomposition, after which the generalized existence theorems and Proposition 3.4 give Theorem 1.1 over the stated field.

For the phase construction, the cone estimates of Proposition 4.1 and the degree-product estimate of Proposition 5.1 lead to constrained valuation optimization, logarithmic Cartier selection and exact rounding in Sections 6–8. The characteristic reduction occurs only after the germ and its required finite data have been fixed. A finite recursion chooses all bounds before the final integer is chosen, and the resulting congruence holds for every semi-invariant. Integral-valued divisorial valuations in this argument need not be primitive.

For the uniform estimates, bounded presentations, actual trait calculations and lc-centre incidence in Sections 11–13 prove Theorem 10.1. Only its reference form has bounded index. Tower reduction then gives Theorem 9.1, which enters the integral jump estimate and hence the product bound. Ordinary fibre-component labels are retained along with cone weights: weights alone do not identify the required actual valuation. Saturated face lattices eliminate hidden orbit covers. The effective dlt construction in Lemma 13.1 supplies the lc-incidence input by eliminating a very exceptional divisor on a fixed finite model, independently of Theorem 1.1.

The following locators make the deferred proofs explicit.

| Result             | Statement  | Proof          |
|--------------------|------------|----------------|
| Phase theorem      | Section 3  | Sections 4–8   |
| Integral jump      | Section 5  | Section 9      |
| Lifting and Mixing | Section 9  | Section 10     |
| Bounded fibre      | Section 10 | Sections 11–13 |

The arrows in Figure 1 give the logical order of these proofs, including their later-section inputs.

## 2. CONVENTIONS AND PRELIMINARY NOTATION

**2.1. Pairs, nef data and discrepancies.** All varieties are integral. A contraction is a projective surjective morphism  $f$  to a normal variety with  $f_*\mathcal{O} = \mathcal{O}$ . Unless a different field is specified, the local and field-theoretic arguments take place over  $\mathbb{C}$ . Section 3 explains the passage to any algebraically closed field of characteristic zero. Auxiliary function fields are finitely generated over  $\mathbb{C}$  and need not be algebraically closed.

For a generalized pair  $(X, B + M)$ , choose a projective birational model  $\pi: X' \rightarrow X$  carrying the nef rational Cartier divisor  $M'$ . On a smooth model  $p: W \rightarrow X$  dominating  $X'$ , write  $M_W$  for its pullback and define  $B_W$  by

$$K_W + B_W + M_W = p^*(K_X + B + M).$$

For a prime divisor  $E$  on  $W$ , set

$$A_{X,B+M}(E) = 1 - \text{coeff}_E(B_W).$$

This is independent of the choice of higher model. The nef data are held fixed throughout all birational operations on a generalized pair. An ordinary subpair may have negative boundary

coefficients; its discrepancy is defined by the same crepant formula without a nef part. The terms lc and klt refer respectively to nonnegative and positive log discrepancies. Effectivity, when needed, is stated separately.

**2.2. Homogeneous valuations and volume forms.** A divisorial valuation means a positive real multiple  $v = c \operatorname{ord}_E$  of a prime-divisor order, unless an integral or rational normalization is specified. Discrepancies and orders of divisors are extended homogeneously. An integral-valued divisorial valuation need not be primitive. We also use quasi-monomial valuations on fixed log-smooth models, where positive weights on boundary parameters define their values; at a face, zero-weight directions belong to the residue coefficient field. Any finite-group action on functions and differentials uses the same pullback convention. The angle of a character takes values in  $\mathbb{R}/\mathbb{Z}$ , with exponential  $a \mapsto \exp(2\pi i a)$ .

For a real valuation  $v$  centred at a closed point  $x$  of a normal  $n$ -fold,  $n \geq 1$ , put

$$\mathfrak{a}_t(v) = \{g \in \mathcal{O}_{X,x} : v(g) \geq t\}, \quad \mathfrak{a}_{>t}(v) = \{g \in \mathcal{O}_{X,x} : v(g) > t\}.$$

These ideals have finite colength for  $t > 0$ : sufficiently high powers of the maximal ideal lie in them, since its finitely many generators have positive value. We use

$$\operatorname{vol}_{X,x}(v) = \limsup_{t \rightarrow \infty} \frac{n! \operatorname{length}(\mathcal{O}_{X,x}/\mathfrak{a}_t(v))}{t^n}, \quad \operatorname{gr}_v \mathcal{O}_{X,x} = \bigoplus_{t \geq 0} \mathfrak{a}_t(v)/\mathfrak{a}_{>t}(v).$$

For a klt pair  $(X, \Delta)$  and finite log discrepancy, its normalized volume is  $\widehat{\operatorname{vol}}_{X,\Delta}(v) = A_{X,\Delta}(v)^n \operatorname{vol}_{X,x}(v)$ ; it is  $+\infty$  when the discrepancy is infinite [39, Definitions 2.4–2.5]. Thus  $\operatorname{vol}(cv) = c^{-n} \operatorname{vol}(v)$  for  $c > 0$ , and normalized volume is invariant under positive rescaling. The notation  $(v > T)$  means  $\mathfrak{a}_{>T}(v)$ . The inclusions  $\mathfrak{a}_{t+1} \subset \mathfrak{a}_{>t} \subset \mathfrak{a}_t$  show that strict cutoffs give the same normalized limsup. For the finitely generated positive rational gradings used below, Hilbert asymptotics give the limit and the leading colength coefficient  $\operatorname{vol}(v)/n!$ . This numerical valuation volume is distinct from a differential volume form and from the volume of a divisor.

Let  $L/\mathbb{C}$  be a function field of transcendence degree  $d$ . An  $r$ -pluri-volume form is a nonzero element of  $(\bigwedge^d \Omega_{L/\mathbb{C}}^1)^{\otimes r}$ , with  $r$  a positive integer. On a smooth model its associated subpair has boundary  $-r^{-1} \operatorname{div}(\omega)$ ; write  $A_\omega$  for the resulting homogeneous discrepancy. For  $r = 1$  we say volume form. For a volume form  $\eta$ , the index-one formulas are

$$(2.1) \quad A_\eta(c \operatorname{ord}_E) = c(1 + \operatorname{ord}_E(\eta)), \quad A_{h\eta}(u) = A_\eta(u) + u(h).$$

If  $v$  is integral-valued, the discrepancy of a volume form is integral. This index-one fact is used in the integral jump argument.

If two forms  $\omega, \sigma$  are compared at a common positive index  $p$  and  $H = \omega^{\otimes p/r_\omega} / \sigma^{\otimes p/r_\sigma}$ , then

$$A_\omega(v) - A_\sigma(v) = \frac{1}{p} v(H).$$

Here the quotient denotes a rational function after identifying the one-dimensional spaces of top forms. Indeed, the boundary difference is  $-p^{-1} \operatorname{div}(H)$ , which gives the displayed sign. Under a finite characteristic-zero field extension, the discrepancy of a pulled-back form at an upstairs divisorial valuation equals that of the restricted valuation downstairs. Indeed, if a primitive valuation  $\operatorname{ord}_{E'}$  upstairs restricts to  $e \operatorname{ord}_E$  downstairs, pullback of a volume form satisfies

$$\operatorname{ord}_{E'}(\eta) = e \operatorname{ord}_E(\eta) + (e - 1).$$

Adding 1 gives  $A_\eta(\operatorname{ord}_{E'}) = e A_\eta(\operatorname{ord}_E)$ ; homogeneity and tensor powers give the assertion for every normalization and pluri-index. This identity already accounts for the scale of the restricted valuation; no separate rescaling is implicit. The restricted nontrivial valuation is divisorial, since the value-group index and residue-field extension are finite.

For a subfield  $K \subset L$  and a form on  $L$ , a barred discrepancy denotes the infimum over divisorial prolongations with the specified restriction to  $K$ , whenever introduced in the relevant setup. The precise admissible valuations and hypotheses are stated in Section 9. The notation  $A_\omega$  always refers to the form in that local setup, not to a fixed boundary on every model.

**2.3. Positivity, bounds and limits.** Divisor pullbacks are taken for rational Cartier multiples. A degree on an embedded projective variety means intersection with the appropriate power of its hyperplane class. When volumes of divisors are used, roundings and divisibility are taken on the fixed model specified before the limit. Constants in a sequential assertion may become uniform after the stated passage to a subsequence. Each such assertion specifies which dimension, bounded family, fixed comparison constant or earlier choice is permitted to enter the bound.

An NQC nef part is a nonnegative real combination of nef rational Cartier b-divisors. The rational nef data in Theorem 1.1 are therefore NQC. We use the standard klt MMP, relative vanishing and big-boundary existence results as established literature. All stronger uniformity, specialization, valuation-selection and termination statements needed for the proof are established in the sections that follow.

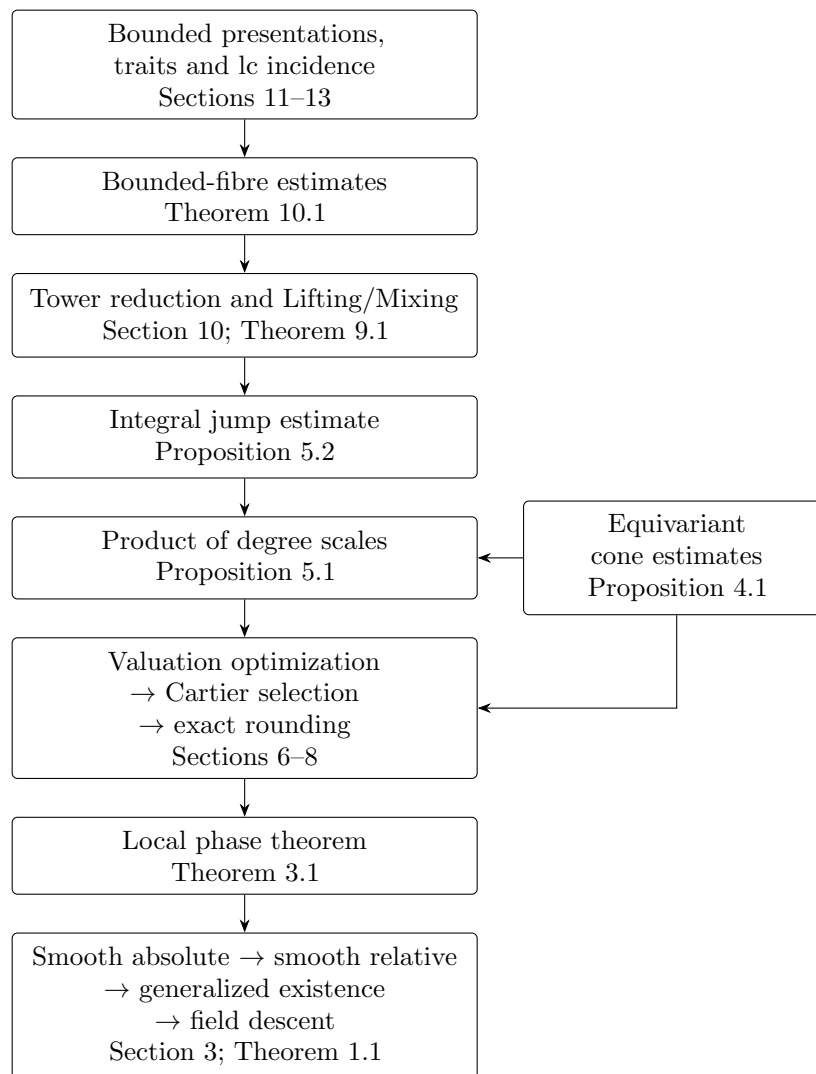


FIGURE 1. Logical dependencies. The preparation group includes the trait minimum identity used for the bounded base form; no bounded moduli index is assumed in proving that identity. The tower box also includes the geometric step used directly in the integral jump argument.

### 3. FROM LOCAL PHASES TO MINIMAL MODELS

We first isolate the local assertion and deduce the global theorem from it. Throughout this section, until Proposition 3.4, the base field is  $\mathbb{C}$ . Log discrepancies of divisorial valuations are homogeneous: if  $u = c \operatorname{ord}_E$  with  $c > 0$ , then  $A(u) = cA(\operatorname{ord}_E)$ . In particular, “integral-valued” does not require a divisorial valuation to be primitive.

At a centre where a volume form  $\eta$  generates the canonical sheaf, its form discrepancy  $A_\eta$  is the ordinary boundary-zero discrepancy. We use (2.1) and its finite-extension compatibility from Section 2, with the actual normalization of the restricted valuation.

For a finite-order character, its angle belongs to  $\mathbb{R}/\mathbb{Z}$ , with the convention that its value is  $\exp(2\pi i \operatorname{angle}(\chi))$ . Characters below describe the actions on functions and forms.

**Theorem 3.1** (Local phase assertion). *Let  $(V_i, o_i)$  be a sequence of pointed algebraic Gorenstein klt germs of a fixed dimension  $n \geq 2$ , with  $o_i$  closed. Suppose that a finite cyclic group  $G_i$  acts on an affine representative of each germ, fixes  $o_i$ , and admits a semi-invariant canonical generator  $\theta_i$ , a volume form with zero divisor near  $o_i$ . Choose an arbitrary element  $\gamma_i \in G_i$  for each  $i$ .*

*After passage to a subsequence, there are divisorial valuations  $w'_i$  centred at  $o_i$ , a constant  $C > 0$ , and a positive integer  $\ell$ , independent of  $i$  on that subsequence, such that*

$$A_{\theta_i}(w'_i) \leq C$$

*and, for every nonzero rational semi-invariant function  $h$  on  $V_i$  of character  $\chi$ ,*

$$(3.1) \quad w'_i(h) \equiv \ell \operatorname{angle}(\chi(\gamma_i)) \pmod{\mathbb{Z}}.$$

*In particular,  $w'_i$  restricts to an integral-valued valuation of  $\mathbb{C}(V_i)^{G_i}$ .*

The proof occupies Sections 4–8, using Proposition 5.2, whose proof is supplied in Section 9 and the subsequent sections. We now give all steps of the global deduction.

#### 3.1. Termination of smooth canonical programs.

**Proposition 3.2.** *Assume Theorem 3.1. Let  $X$  be a smooth projective complex variety with pseudo-effective canonical divisor. A  $K_X$ -MMP with scaling of an effective ample rational divisor  $H$ , chosen so that  $(X, H)$  is klt and  $K_X + H$  is ample, terminates with nef canonical divisor. Consequently, on a smooth projective birational model  $p : U \rightarrow X$  one has*

$$p^*K_X = P + N, \quad P \text{ nef and rational Cartier, } N \geq 0 \text{ rational Cartier.}$$

*Proof.* The assertion is immediate in dimension zero; in dimension one a smooth projective curve with pseudo-effective canonical divisor already has nef canonical divisor. Suppose that  $n = \dim X \geq 2$ . Such an  $H$  is obtained from sufficiently positive general divisors with small coefficients. The ordinary klt MMP, including existence of flips [11], gives a program

$$X = X_0 \dashrightarrow X_1 \dashrightarrow X_2 \dashrightarrow \dots$$

with  $X_i$  projective,  $\mathbb{Q}$ -factorial and klt. Write  $H_i$  for the strict transform of  $H$  and  $\lambda_i$  for the scaling number at its  $i$ th step. If the program is infinite, the  $\lambda_i$  lie in  $(0, 1]$ , are nonincreasing and rational: a contracted curve is  $K_{X_i}$ -negative and  $(K_{X_i} + \lambda_i H_i)$ -trivial. Only finitely many steps can be divisorial, because each such step decreases the Picard number.

For a divisorial valuation  $u$  of the common function field, set

$$(3.2) \quad A_i(u) = A_{X_i}(u), \quad E_i(u) = A_i(u) - A_X(u), \quad \Delta_i(u) = A_{i+1}(u) - A_i(u) \geq 0.$$

We use compatible canonical divisors on every common resolution. At step  $j$  the pullbacks of the adjoints  $K + \lambda_j H$  agree. Their difference is exceptional over the output and numerically trivial there, so this follows from the negativity lemma. The difference of the canonical pullbacks is effective by the same lemma. It follows, by summing the stepwise identities, that for  $0 < t \leq \lambda_i$  the

difference between the pullback of  $K_X + tH$  and that of  $K_{X_i} + tH_i$  is effective and exceptional over  $X_i$ , and its value at  $u$  is

$$(3.3) \quad F_i(t; u) = \sum_{j < i} \left(1 - \frac{t}{\lambda_j}\right) \Delta_j(u).$$

Thus the transformed pair  $(X_i, tH_i)$  is klt. For rational  $t$ , this effective exceptional difference identifies the systems of sufficiently divisible multiples on the two sides after adding their fixed part. In particular,  $K_{X_i} + tH_i$  is big. At  $t = \lambda_i$  it is also nef, and klt base point freeness makes it semiample.

We spell out why

$$(3.4) \quad \lambda_i \longrightarrow 0.$$

Otherwise choose  $j$  after the last divisorial contraction, put  $S = X_j$ , and choose a positive rational  $\beta < \lim_i \lambda_i$ . The tail maps from  $S$  are small. For  $\beta \leq t \leq \lambda_j$ , the boundaries  $tH_j$  are klt and their adjoints are big. Moreover,  $H_j$  is big: sections of sufficiently divisible multiples of  $H$  push to sections of its transform. Choose a decomposition

$$H_j \sim_{\mathbb{Q}} A_1 + \cdots + A_r + G,$$

where  $G \geq 0$  and the  $A_a$  are general, small effective ample rational divisors whose numerical classes span  $N^1(S)_{\mathbb{R}}$ . In each  $tH_j$  replace only a fixed sufficiently small rational portion  $\delta H_j$ , with  $0 < \delta < \beta$ , by this expression. The resulting boundaries remain klt, since the entire compact interval can be checked on one log resolution. Small variations of the coefficients of the  $A_a$  give a rational polytope of klt boundaries with a common ample part and big adjoints. BCHM finiteness of marked ample models [11, Corollary 1.1.5] applies to this polytope.

Small  $\mathbb{Q}$ -factorial modifications identify the numerical divisor spaces. For completeness, if a divisor is numerically trivial on one model, its pullback and the pullback of its transform differ by a divisor exceptional over the other model; that difference is numerically trivial over that model and is zero by negativity. Hence numerical equivalence is transported in both directions. At each scaling value  $\lambda_i$ , the nef class  $K_{X_i} + \lambda_i H_i$  can therefore be made ample by an arbitrarily small rational perturbation in the displayed spanning directions. The perturbation may be chosen within the fixed polytope. Smallness identifies the corresponding section systems, so the marked model  $S \dashrightarrow X_i$  is one of its finitely many ample models. No marked model can recur: discrepancies are nondecreasing at each flip, and some discrepancy increases strictly. Indeed, otherwise the canonical pullbacks would agree, contradicting the negative and positive intersection signs on the two sides of the flip over its contraction base. This proves (3.4).

For rational  $0 < t \leq 1$ , put  $L_t = K_X + tH$ . Its asymptotic base order is

$$\sigma_u(t) = \inf_m \frac{1}{m} u(\mathfrak{b}(|mL_t|)), \quad \sigma_u = \lim_{t \downarrow 0} \sigma_u(t),$$

where  $m$  runs through sufficiently divisible positive integers. The values  $\sigma_u(t)$  increase as  $t$  decreases: a sufficiently divisible multiple of  $(t' - t)H$  is globally generated when  $t' > t$ . At a scaling value there is no asymptotic base order on  $X_i$ , by semiampleness, so (3.3) gives

$$(3.5) \quad \sigma_u(\lambda_i) \leq E_i(u) \leq \sigma_u.$$

The second inequality follows by fixing  $i$ , observing that  $\sigma_u(t) \geq F_i(t; u)$  for  $0 < t \leq \lambda_i$ , and letting  $t$  tend to zero.

*A uniform multiplier-ideal estimate.* For each positive integer  $s$ , consider

$$\mathcal{I}_s(t) = \mathcal{I}(X, s\|L_t\|).$$

These ideals form a decreasing family as  $t \downarrow 0$ , by the same comparison of base ideals. They stabilize to an ideal  $\mathcal{I}_s$ . Here stabilization follows from global generation, not from a descending

chain condition for ideals. Fix a very ample divisor  $H_0$ . For each  $s$ , choose a Cartier divisor  $P_s$  so positive that

$$P_s - jH_0 - K_X - sL_t \text{ is ample for every } 0 \leq t \leq 1 \text{ and } 1 \leq j \leq n.$$

Asymptotic Nadel vanishing and Castelnuovo–Mumford regularity make all  $\mathcal{J}_s(t) \otimes \mathcal{O}_X(P_s)$  globally generated [17, Corollary 1.5 and Theorem 1.8]. Their spaces of sections are nested subspaces of the fixed finite-dimensional vector space  $H^0(X, \mathcal{O}_X(P_s))$ , and therefore eventually constant. Generation then makes the ideals themselves constant for all sufficiently small rational  $t > 0$ .

For every divisorial  $u$  one has

$$(3.6) \quad s\sigma_u(t) - A_X(u) \leq u(\mathcal{J}_s(t)) \leq s\sigma_u(t).$$

The first bound follows from the log-resolution formula for a divisible level computing the asymptotic multiplier ideal. For the second, asymptotic subadditivity on the smooth variety  $X$  [17, Variants 2.4–2.5] gives, for sufficiently divisible  $m$ ,

$$\mathfrak{b}(|msL_t|) \subseteq \mathcal{J}(X, ms\|L_t\|) \subseteq \mathcal{J}_s(t)^m.$$

Taking orders, dividing by  $m$ , and taking the asymptotic infimum proves the bound. Combining stabilization, (3.4), (3.5), and (3.6), we obtain: for every fixed positive integer  $s$ , there is an index  $i(s)$  such that for every  $i \geq i(s)$  and every divisorial valuation  $u$ ,

$$(3.7) \quad \frac{u(\mathcal{J}_s)}{s} \leq E_i(u) \leq \sigma_u \leq \frac{u(\mathcal{J}_s) + A_X(u)}{s}.$$

The index  $i(s)$  is independent of  $u$ . In particular,

$$(3.8) \quad 0 \leq \Delta_i(u) \leq \frac{A_X(u)}{s} \quad (i \geq i(s)).$$

*Countability of finite limiting changes.* We claim that there is a fixed countable subset  $\mathcal{E} \subseteq \mathbb{R}$  containing every finite limit of  $E_{i_a}(u_a)$ , where  $i_a \rightarrow \infty$ , the  $u_a$  are integral-valued divisorial valuations, and  $A_X(u_a) \leq C$  for some constant  $C$  depending on the sequence.

Choose a countable algebraically closed field  $F \subset \mathbb{C}$  over which  $X$  and all the countably many ideals  $\mathcal{J}_s$  are defined. For every  $f \in F(X_F)^*$ , its zero and pole divisors on  $X$  have positive log canonical thresholds when nonzero. Their orders at  $u_a$  are therefore bounded by constants times  $A_X(u_a)$ , so  $u_a(f)$  is bounded. Since these values are integral, diagonal extraction over the countable field makes them eventually constant for every  $f$ . The eventual values define an integral valuation  $u_\infty$  on  $F(X_F)$ , trivial on  $F$ .

We next show that  $u_\infty$  is either trivial or divisorial. If it is nontrivial and its centre is not a divisor, blow up the closure of its centre and continue. At each stage we work near the generic point of the centre, where the ambient variety and that centre can be taken smooth. The next centre lies above this generic smooth neighbourhood, where the blowup is smooth. Taking a projective model and resolving outside it therefore does not change the valuation calculation. A centre of codimension at least two has an ideal of positive integral value. Blowing it up thus increases the value of the relative Jacobian over  $X_F$  by at least one. If  $N$  such blowups were possible, choose a smooth affine neighbourhood  $U = \text{Spec } A$  of the last centre on which this relative Jacobian divisor is principal, with equation  $b$ . Then  $u_\infty(b) \geq N$ .

Choose finitely many  $F$ -algebra generators of  $A$ , including the inverses used to obtain this affine neighbourhood. Their  $u_a$ -values are eventually equal to their nonnegative  $u_\infty$ -values. Because each  $u_a$  is trivial on  $\mathbb{C}$ , every polynomial in these generators with complex coefficients has nonnegative value. Thus  $u_a$  has a centre on  $U_{\mathbb{C}}$  for all sufficiently large  $a$ . Its centre need not be the base change of the centre of  $u_\infty$ . The Jacobian identity holds throughout this chart and nevertheless gives

$$A_X(u_a) = A_{U_{\mathbb{C}}}(u_a) + u_a(b) \geq u_a(b) = u_\infty(b) \geq N.$$

Taking  $N > C$  is impossible. The centre therefore eventually has codimension one. The valuation then dominates its discrete valuation ring, is zero on its units, and is a positive integral multiple of that DVR order.

There are only countably many projective models of  $F(X_F)$  defined over the countable field  $F$ , hence only countably many prime divisors on such models and their integral multiples. Adding the trivial valuation still gives a countable set of possibilities for  $u_\infty$ . For each fixed  $s$ , the preceding finite-chart argument applied to finitely many local generators of  $(\mathcal{J}_s)_F$  gives

$$u_a(\mathcal{J}_s) = u_\infty((\mathcal{J}_s)_F) =: a_s \quad \text{for all sufficiently large } a.$$

If  $E_{i_a}(u_a) \rightarrow e$ , (3.7) therefore implies

$$(3.9) \quad \frac{a_s}{s} \leq e \leq \frac{a_s + C}{s} \quad \text{for every positive integer } s.$$

One limiting valuation determines at most one such  $e$ : a second sequence may have a different bound  $C'$ , but the two resulting limits differ by at most  $\max\{C, C'\}/s$  for every  $s$ . If  $u_\infty$  is trivial, all  $a_s$  are zero and  $e = 0$ . This proves the claimed countability, including its independence from any subsequent choice of phases.

*A common canonical index.* We claim that there is a positive integer  $R$  such that

$$(3.10) \quad RK_{X_i} \text{ is Cartier for every } i.$$

Otherwise, after passage to a subsequence, choose closed points  $x_i \in X_i$  whose local canonical indices  $r_i$  tend to infinity. If the indices were bounded, their least common multiple would give (3.10).

We recall the index-one cover at such a point. Choose a rational top form  $\eta$ , write  $K = \text{div}(\eta)$ , shrink about the point so that  $rK = \text{div}(g)$ , and normalize after adjoining  $z$  with  $z^r = g$ . Minimality of  $r$  and Kummer theory make this an extension of degree  $r$ . It is unramified in codimension one; ramification of discrepancies shows that it is klt. The form

$$\theta = \frac{\pi^* \eta}{z}$$

has zero divisor and has primitive character under the cyclic cover group. The cover is Gorenstein: its canonical sheaf is generated by  $\theta$ , and klt singularities are Cohen–Macaulay.

There is exactly one point over the specified base point. Indeed, a finite group quotient has a transitive action on the points of each geometric fibre. If this orbit had more than one point, a nonconstant character of its cyclic permutation representation would give a regular semi-invariant  $a$ , of character  $z^j$  with  $0 < j < r$ , nonzero at every point of the fibre. One obtains  $a$  by lifting its values on the reduced fibre and projecting to the character space. Its invariant power  $a^r$  is nonzero at the base point, so  $a$  is a unit near the whole fibre. The function  $a/z^j$  is invariant, and its divisor downstairs is  $-jK$ . Thus  $jK$  is principal near the point, contradicting the minimality of  $r$ . The unique point is fixed by the cover group, as required in Theorem 3.1.

Now fix any  $\alpha \in \mathbb{R}/\mathbb{Z}$ . Because  $r_i \rightarrow \infty$  and the character of  $\theta_i$  is primitive, choose group elements  $\gamma_i$  so that the angles of the  $\theta_i$ -characters at  $\gamma_i$  tend to  $\alpha$ . Apply Theorem 3.1, and let  $u_i$  be the restrictions of the resulting  $w'_i$  to the common downstairs field. These restrictions are integral-valued and divisorial, and quasi-étale discrepancy compatibility gives

$$0 < A_X(u_i) \leq A_i(u_i) = A_{\theta_i}(w'_i) \leq C.$$

Choose a rational top form  $\eta_i^0$  downstairs that generates the canonical sheaf at the centre of  $u_i$  on the smooth variety  $X$ . With pullback of forms understood, (2.1) gives

$$(3.11) \quad E_i(u_i) = w'_i(\theta_i/\eta_i^0).$$

The ratio is a rational semi-invariant with the character of  $\theta_i$ . Consequently, after passage to the subsequence in Theorem 3.1, these nonnegative bounded changes have residues tending to  $\ell\alpha$ . Pass

further so that they converge in  $\mathbb{R}$ . Their limit belongs to the fixed countable set  $\mathcal{E}$ , and hence  $\ell\alpha$  belongs to its image  $\bar{\mathcal{E}}$  in  $\mathbb{R}/\mathbb{Z}$ . But

$$\bigcup_{\ell \geq 1} \{\alpha \in \mathbb{R}/\mathbb{Z} : \ell\alpha \in \bar{\mathcal{E}}\}$$

is countable, whereas  $\alpha$  was arbitrary. The possible dependence of  $\ell$  on the subsequence does not change this contradiction. This proves (3.10).

*Vanishing on late flips.* For every fixed positive integer  $m$ , all sufficiently late flips, with negative contraction  $f_i : X_i \rightarrow Z_i$ , satisfy

$$(3.12) \quad R^k f_{i*} \mathcal{O}_{X_i}(mK_{X_i}) = 0 \quad (k > 0).$$

To prove this, take a common smooth resolution  $p : U \rightarrow X_i$ ,  $q : U \rightarrow X_{i+1}$  over  $Z_i$ , sufficiently high that the fractional parts to be rounded have simple normal crossing support, and put

$$\mathcal{L}_m = \mathcal{O}_U(K_U + \lceil (m-1)q^*K_{X_{i+1}} \rceil).$$

The rational divisor  $(m-1)q^*K_{X_{i+1}}$  is nef over both  $X_i$  and  $Z_i$ : a curve contracted by  $p$  is also contracted over  $Z_i$ , and  $K_{X_{i+1}}$  is ample over  $Z_i$ . It is big relative to both bases since the relevant morphisms are birational; this includes  $m=1$ , whose divisor is zero and whose generic fibres are zero-dimensional. Relative Kawamata–Viehweg vanishing [28] therefore gives

$$R^k p_* \mathcal{L}_m = R^k (f_i p)_* \mathcal{L}_m = 0 \quad (k > 0).$$

We claim that  $p_* \mathcal{L}_m = \mathcal{O}_{X_i}(mK_{X_i})$ , where the right side denotes the divisorial sheaf if  $mK_{X_i}$  is not Cartier. One inclusion is checked in codimension one, where the flip and the resolution are isomorphisms. For the reverse inclusion, let a local rational section  $h$  of  $\mathcal{O}_{X_i}(mK_{X_i})$  be given. The divisor

$$\operatorname{div}_U(h) + mp^*K_{X_i}$$

is effective, because  $\operatorname{div}_{X_i}(h) + mK_{X_i}$  is effective and rational Cartier. Before rounding, the difference between the divisor defining  $\mathcal{L}_m$  and  $mp^*K_{X_i}$  has coefficient, at any prime divisor of  $U$ ,

$$A_i - 1 - (m-1)\Delta_i.$$

For  $i \geq i(m)$ , (3.8) implies

$$(3.13) \quad A_i - 1 - (m-1)\Delta_i \geq -1 + \frac{A_X}{m} > -1.$$

Adding this to the effective pullback of the section divisor and taking round-up leaves every integral coefficient nonnegative. Hence  $h$  belongs to  $p_* \mathcal{L}_m$ . The pushforward identity and Leray prove (3.12).

Fix  $R$  as in (3.10). For each late step choose a sufficiently positive very ample divisor  $J$  on  $Z_i$  so that the Cartier divisor

$$D = -RK_{X_i} + f_i^* J$$

and  $D - K_{X_i}$  are ample. This is possible by relative anti-ampleness of  $K_{X_i}$ . Effective base point freeness, applied to the klt pair  $(X_i, 0)$  and the nef Cartier divisor  $D$  with  $D - K_{X_i}$  nef and big, gives an integer  $N = N(n)$  such that  $L = ND$  is globally generated [18, Theorem 1.1 and Remark 1.2]. It is also ample. Only now choose a stage late enough for (3.12) to hold for the finite list

$$m = R + jNR, \quad 1 \leq j \leq n.$$

The integer  $N$  is independent of the step and of  $J$ , so this order of choices is legitimate. For each such step, projection formula, Leray and Serre vanishing on  $Z_i$  give, for all sufficiently large integers  $a$ ,

$$(3.14) \quad H^j(X_i, \mathcal{O}_{X_i}(RK_{X_i} + af_i^* J - jL)) = 0 \quad (j > 0).$$

Indeed the displayed divisor equals  $(R + jNR)K_{X_i} + (a - jN)f_i^* J$ .

Castelnuovo–Mumford regularity with respect to the ample globally generated  $L$  now makes  $\mathcal{O}_{X_i}(RK_{X_i} + af_i^*J)$  globally generated. To use the usual very ample formulation, the morphism defined by  $L$  is finite onto its image, since  $L$  is ample; push the sheaf to projective space, apply regularity there using (3.14), and pull back the generating sections. A globally generated line bundle has nonnegative degree on every curve, contradicting its degree  $RK_{X_i} \cdot C < 0$  on a curve contracted by  $f_i$ . The hypothetical infinite program is impossible.

Pseudo-effectivity of the canonical class persists at every finite step: push forward the big systems of  $K_X + tH$  for arbitrarily small rational  $t > 0$ , and take their limiting numerical classes. It rules out a final negative fibre contraction, since the restriction to a general positive-dimensional fibre would be both pseudo-effective and anti-ample. The terminal model  $Y$  therefore has nef canonical divisor. On a common smooth projective resolution  $p : U \rightarrow X$ ,  $q : U \rightarrow Y$ , the MMP inequalities give

$$p^*K_X = q^*K_Y + N, \quad N \geq 0.$$

Taking  $P = q^*K_Y$  proves the last assertion.  $\square$

**3.2. Relative weak Zariski decompositions.** An NQC weak Zariski decomposition of a rational Cartier divisor  $D$  over  $Z$  means a projective birational model  $p : Y \rightarrow X$  and a numerical decomposition  $p^*D \equiv_Z P + N$ , where  $N \geq 0$  and  $P$  is a nonnegative real linear combination of rational Cartier divisors nef over  $Z$ . The decompositions constructed here have  $P$  and  $N$  rational Cartier, so in particular their nef parts are NQC.

**Proposition 3.3.** *Assume Theorem 3.1. Let  $X$  be a smooth complex quasi-projective variety, projective over a normal quasi-projective variety  $Z$ . If  $K_X$  is pseudo-effective over  $Z$ , it admits an NQC weak Zariski decomposition over  $Z$  with rational Cartier parts.*

*Proof.* Replace  $Z$  by the Stein base of the morphism, which is normal and finite over its original image. The morphism  $f : X \rightarrow Z$  is then surjective with connected fibres. This replacement changes neither the contracted curves nor relative numerical equivalence, nefness or pseudo-effectivity. A very general fibre is smooth and projective by generic smoothness and has pseudo-effective canonical divisor. To see the latter assertion, take a countable sequence of relative effective approximations to  $[K_X]$  and restrict to fibres outside their countably many bad loci. Relative numerical equivalence restricts to numerical equivalence on such a fibre, and  $K_X$  restricts to its canonical divisor. By BDPP, these very general smooth fibres are non-uniruled [14, Theorem 0.2 and Corollary 0.3].

Compactify the projective morphism and resolve away from  $X$  to obtain a projective morphism

$$\bar{f} : \bar{X} \rightarrow \bar{Z}$$

with  $\bar{X}$  smooth projective,  $\bar{Z}$  normal projective, and the original morphism unchanged over  $Z$ . If  $\dim Z = 0$ , the variety  $X$  is already projective and Proposition 3.2 applies. We may therefore assume  $\dim Z > 0$ .

Let  $r : \tilde{Z} \rightarrow \bar{Z}$  be a smooth projective resolution. Choose a sufficiently high multiple  $H$  of a very ample divisor on  $\bar{Z}$ , and a general branch divisor  $B \in |2H|$ . The pulled-back linear systems on both  $\tilde{Z}$  and  $\bar{X}$  are base point free. Bertini therefore makes  $r^*B$  and  $\bar{f}^*B$  smooth, nonempty reduced divisors. The corresponding double covers

$$Z^\sharp \rightarrow \tilde{Z}, \quad \pi : X^\sharp \rightarrow \bar{X}$$

are smooth integral projective varieties. Integrality follows because a nonempty reduced branch divisor cannot be the divisor of a square. The double-cover formula gives

$$(3.15) \quad \begin{aligned} K_{Z^\sharp} &\sim_{\mathbb{Q}} \pi_{\tilde{Z}}^*(K_{\tilde{Z}} + r^*H), \\ K_{X^\sharp} &\sim_{\mathbb{Q}} \pi^*(K_{\bar{X}} + \bar{f}^*H). \end{aligned}$$

For sufficiently positive  $H$ , the first canonical divisor is big. The rational map  $X^\sharp \dashrightarrow Z^\sharp$  has the same very general fibres as  $f$ . The base is non-uniruled because its canonical divisor is big, and

those fibres are non-uniruled by the preceding paragraph. Hence  $X^\sharp$  is non-uniruled: a covering family of rational curves would either give a covering family on the base or consist of vertical curves covering very general fibres. BDPP now makes  $K_{X^\sharp}$  pseudo-effective.

Apply Proposition 3.2 to  $X^\sharp$ . On a projective birational model its canonical pullback is a sum of a nef rational divisor and an effective rational divisor. By (3.15), the same is true, after changing the nef divisor by a rational principal divisor if necessary, for the pullback of  $\pi^*(K_{\overline{X}} + \overline{f}^*H)$ . Take a common equivariant resolution of this model and its conjugate under the covering involution. Writing  $g : T \rightarrow X^\sharp$  for the resulting morphism and  $\iota$  for the involution, average the decomposition and its conjugate. We obtain

$$(3.16) \quad g^*\pi^*(K_{\overline{X}} + \overline{f}^*H) = \overline{P} + \overline{N}, \quad \iota^*\overline{P} = \overline{P}, \quad \iota^*\overline{N} = \overline{N},$$

where  $\overline{P}$  is nef and  $\overline{N} \geq 0$ , both rational Cartier.

Let  $q : T \rightarrow Y = T/\langle \iota \rangle$  be the finite quotient. The normal projective variety  $Y$  is birational to  $\overline{X}$ . An invariant rational divisor descends as a rational Weil divisor by dividing its coefficient at an upstairs prime by the corresponding ramification index. If the invariant divisor is rational Cartier, its descended divisor is rational Cartier as well. Indeed, after clearing denominators, a Cartier divisor has one local equation on a semilocal neighbourhood of the finite orbit above a downstairs point. The product of its translates is an invariant equation whose divisor is the group order times the original divisor. It descends to a rational local equation for a multiple of the downstairs divisor. This is the finite norm argument, and applies at every point of  $Y$ .

Consequently  $\overline{P} = q^*P$  and  $\overline{N} = q^*N$  for rational Cartier divisors  $P, N$  on  $Y$ . Nefness of  $P$  follows by lifting any curve to  $T$  and dividing its intersection by the positive covering degree; effectivity of  $N$  is immediate from its coefficients. Finite pullback is injective on divisors, so (3.16) descends to a decomposition of the pullback of  $K_{\overline{X}} + \overline{f}^*H$ . Restrict to the inverse image of  $X$ . The additional term pulled from  $Z$  is numerically trivial over  $Z$ , and we obtain the required relative weak Zariski decomposition of  $K_X$ .  $\square$

**3.3. The generalized pair and its exact output.** We apply the results of Tsakanikas–Xie in their relative setting [36]. Their Theorem D, stated as Theorem 5.2, says that relative minimal-model existence for smooth varieties in dimension  $n - 1$  upgrades an NQC weak Zariski decomposition of an  $n$ -dimensional NQC generalized lc pair to a minimal model. Starting with dimension zero, Proposition 3.3 and induction therefore give relative minimal models for all pseudo-effective smooth varieties in every dimension. Their Theorem B, stated as Theorem 5.4, now gives minimal models for all pseudo-effective NQC generalized lc pairs. Their Theorem 2.7 identifies existence in this sense with existence in the Birkar–Shokurov sense.

For the generalized rational pair in Theorem 1.1, the nef data are NQC: a rational nef Cartier divisor, after a positive multiple, is itself one of the permitted nef Cartier summands. If its adjoint is pseudo-effective, the preceding paragraph supplies a Birkar–Shokurov minimal model. If it is not pseudo-effective, the alternative hypothesis of Tsakanikas–Xie’s Theorem A applies directly. In either case, that theorem, stated as Theorem 4.2, gives a terminating MMP with scaling starting on the original  $X$ , without a  $\mathbb{Q}$ -factoriality assumption. To meet its scaling hypothesis, choose an effective sufficiently positive ample rational divisor  $A$  such that

$$(X, B + A + M) \text{ is generalized lc,} \quad K_X + B + A + M \text{ is nef.}$$

Such an  $A$  is a general very ample member divided by a sufficiently large integer. On a fixed log resolution carrying the nef data, its pullback is general in a base point free system and transverse to the log boundary. Small coefficients preserve the condition that all boundary coefficients on this resolution are at most one. Its ample class can nevertheless be chosen large enough to make the adjoint nef. This checks all hypotheses of the terminating-MMP theorem.

Let  $X \dashrightarrow Y$  be the resulting finite birational program. The boundary at each stage is the strict pushforward of  $B$ , and the nef b-divisor is unchanged. Every adjoint is rational Cartier. Indeed,

the construction gives a rational Weil divisor which is real Cartier, and a rational Weil divisor in the real span of Cartier divisors lies in their rational span, by solving the finite system of rational linear equations for its coefficients.

We verify the precise discrepancy inequalities rather than only a numerical minimal-model condition. A birational step is given by a negative extremal contraction  $S \rightarrow T$  and a positive model  $S^+ \rightarrow T$ , whose morphism is small; in the ordinary divisorial case the positive model is  $T$  itself. This description also holds without  $\mathbb{Q}$ -factoriality [36, Remark 2.15]. Denote the adjoints by  $D$  and  $D^+$ . On a common resolution  $p : U \rightarrow S$ ,  $q : U \rightarrow S^+$  over  $T$ , taken to carry the fixed nef data, put

$$(3.17) \quad F = p^*D - q^*D^+.$$

It is  $q$ -exceptional. For a  $q$ -contracted curve, its intersection is that of  $p^*D$ , which is nonpositive because  $D$  is negative on the contracted extremal ray. Thus  $F$  is anti-nef over  $q$ , and negativity gives  $F \geq 0$ .

The coefficient of  $F$  is strictly positive along the transform of each prime divisor on  $S$  contracted in the step. If it were zero along such a prime, choose a general point of its transform outside  $\text{Supp } F$  where  $p$  is an isomorphism. Take a curve  $C$  through this point in its positive-dimensional  $q$ -fibre. It maps to a curve on  $S$  and is not contained in  $\text{Supp } F$ . Effectivity therefore gives  $F \cdot C \geq 0$ , whereas  $q^*D^+ \cdot C = 0$  and  $p^*D \cdot C < 0$ , a contradiction. Since the same nef divisor occurs on  $U$  for both generalized pairs, (3.17) gives, for every divisorial valuation  $u$ ,

$$(3.18) \quad A_{S, B_S + M_S}(u) \leq A_{S^+, B_{S^+} + M_{S^+}}(u).$$

For an original prime contracted during the program, strictness holds at the first step that contracts it, and persists thereafter. Composition therefore gives exactly the unscaled inequalities in Theorem 1.1. Smallness of the positive morphisms makes the inverse of  $X \dashrightarrow Y$  non-extractive. Generalized lc is preserved by (3.18).

The endpoint has nef adjoint, or has a final fibre-type extremal contraction  $h : Y \rightarrow T$ , with connected fibres and normal projective target of smaller dimension, on which  $-D_Y$  is relatively ample. We also justify its numerical rank without a factoriality assumption. All vertical curves span its contracted ray. If a Cartier divisor  $L$  has degree zero on that ray,  $L - D_Y$  is relatively ample. Choose a sufficiently positive ample divisor  $H$  on  $T$  and an effective ample rational divisor

$$A \sim_{\mathbb{Q}} L + h^*H - D_Y$$

with small general coefficients, so that adding  $A$  to the boundary preserves generalized lc as above. The new adjoint  $D_Y + A \sim_{\mathbb{Q}} L + h^*H$  is nef, and hence pseudo-effective, over  $T$ . Tsakanikas–Xie’s Theorem F, stated as Theorem 5.26, applies: its added divisor is effective, real Cartier and relatively ample. It gives an MMP to a good minimal model. Since the adjoint is already nef, that MMP has no negative step, so  $D_Y + A$  is semiample over  $T$ .

A relatively semiample divisor which is numerically trivial over  $T$  is numerically a pullback from  $T$ . In fact, its associated morphism over  $T$  sends each connected projective fibre of  $h$  to a point: a positive-dimensional image would contain a curve with positive degree for the ample divisor on the image. Its normal image is consequently finite and birational over  $T$ , and equals  $T$  by normality and the contraction property. Applied here, this shows that  $L$ , up to its already specified pullback twist, is numerically from  $T$ . A contracted curve defines a rational hyperplane in  $N^1(Y)_{\mathbb{R}}$ ; that hyperplane is therefore precisely  $h^*N^1(T)_{\mathbb{R}}$ . If semiampleness is expressed as a positive real sum of semiample rational Cartier divisors, every summand has nonnegative degree on a vertical curve; the vanishing of their sum forces each degree to vanish, and the same pullback argument applies to each summand. The hyperplane has codimension one, giving

$$\rho(Y/T) = 1, \quad \rho(Y) - \rho(T) = 1.$$

The sign of the original adjoint determines the endpoint. Birational steps preserve pseudo-effectivity of the adjoint: on a common resolution the difference is effective and exceptional as in (3.17), and pushforward gives the reverse implication. A pseudo-effective adjoint cannot be anti-ample on the general positive-dimensional fibre of a contraction. Conversely, a nef endpoint and the effective pullback differences make the original adjoint pseudo-effective. Hence the pseudo-effective case ends with a nef model and the other case with a Mori fibre space. Thus all conclusions of Theorem 1.1 hold over  $\mathbb{C}$ .

### 3.4. Descent to arbitrary characteristic-zero fields.

**Proposition 3.4.** *The conclusion of Theorem 1.1 over  $\mathbb{C}$  implies its conclusion over every algebraically closed field of characteristic zero.*

*Proof.* Let  $k$  be such a field. Descend the given projective varieties, rational divisors and nef data, together with a log resolution carrying those data, to a countable algebraically closed subfield  $F \subset k$ . Embed  $F$  into  $\mathbb{C}$ . The initial properties hold over  $F$  and after extension to  $\mathbb{C}$ : generalized lc is checked on the chosen log resolution, and nefness is preserved under algebraically closed field extension by spreading and specializing curves. Apply the complex result. We will descend its finite MMP to  $F$  step by step; the chosen complex scaling divisor need not descend.

Suppose the current variety  $S$  and adjoint  $D$  are defined over  $F$ , and their complex extensions have been identified with the corresponding stage of the program. Consider its next complex contraction  $f_{\mathbb{C}} : S_{\mathbb{C}} \rightarrow T_{\mathbb{C}}$ . Pull back a very ample line bundle from  $T_{\mathbb{C}}$  and choose generating sections. This bundle and its sections spread on  $S \times_F B$ , for a nonempty integral finite-type  $F$ -scheme  $B$ . After shrinking  $B$ , they generate the bundle on every fibre: failure of generation has closed image under the projection, since  $S$  is projective. Specialize at an  $F$ -point and take the Stein contraction of the resulting map. This gives  $f : S \rightarrow T$  over  $F$ , with normal target and connected fibres.

After extension to  $\mathbb{C}$ , the specialized line bundle is numerically equivalent to the original one. Indeed,  $B$  is geometrically integral because  $F$  is algebraically closed. On its connected complex base change, the degree of the universal bundle restricted to any fixed projective curve on  $S_{\mathbb{C}}$  is constant. Hence the two contractions contract exactly the same curves.

We explain why this identifies their targets, including in the fibre-type case. Each morphism is constant on every connected projective fibre of the other: a positive-dimensional image of a fibre component contains a curve, and hyperplane sections of its projective preimage give a curve mapping onto it, contrary to equality of the contracted curves. The constants agree on intersecting components and hence on the whole connected fibre. The reduced joint image in the product of the two targets therefore has finite projections to both. After normalization, these projections are birational: the function field of a normal contraction target is algebraically closed in  $\mathbb{C}(S)$ , so there is no nontrivial finite intermediate field. Both targets are normal, and finite birational morphisms to them are isomorphisms. Thus the descended contraction becomes the specified complex contraction. Stein contraction and normality are preserved by these algebraically closed extensions.

For a birational step, recover its positive model over  $T$  as

$$(3.19) \quad S^+ = \text{Proj}_T \mathcal{R}, \quad \mathcal{R} = \bigoplus_{m \geq 0} f_* \mathcal{O}_S(mbD),$$

with  $b > 0$  sufficiently divisible. Over  $\mathbb{C}$ , the effective exceptional pullback difference (3.17) identifies these pushforward sheaves with the corresponding sheaves on the positive model, whose adjoint is relatively ample. Hence the relative Proj is exactly that model. This includes a divisorial step for which the positive model is  $T$ : the algebra is then the nonnegative section algebra of the Cartier multiple  $bD_T$ , whose relative Proj is  $T$ .

For clarity, both the sheaf algebra and its finite generation descend. On a normal variety over the algebraically closed field  $F$ , the sheaf of a Weil divisor is the rank-one reflexive extension of its

invertible restriction on a regular open containing all codimension-one points. Field extension is flat with geometrically regular fibres in characteristic zero. The codimension-two complement retains its codimension, and the pullback sheaf retains reflexivity and the  $S_2$  property. It is therefore the same reflexive extension as the sheaf of the extended divisor. Multiplication is also compatible, since it agrees on the regular open and its target is torsion-free. The canonical, boundary and nef-trace divisors defining  $D$  are all defined over  $F$ ; the last assertion is checked by computing the trace on a common resolution.

Choose  $b$  clearing their rational coefficients and making the complex extension of  $bD$  Cartier. The divisorial sheaf  $\mathcal{O}_S(bD)$  becomes invertible after the faithfully flat field extension and so is invertible already over  $F$ . The same  $b$  thus works downstairs. Proper flat base change identifies every graded pushforward in (3.19) and its multiplication after extension. On a finite affine cover of  $T_{\mathbb{C}}$ , finite generation of the complex algebra gives a common bound  $d_0$  for the degrees of generators. Downstairs, form the subalgebra generated by its coherent pieces of degrees at most  $d_0$ . In each degree its inclusion in  $\mathcal{R}$  becomes surjective after faithful flat extension, so its cokernel is zero. Thus these finitely many coherent pieces generate  $\mathcal{R}$ , proving finite generation without presupposing it. Formation of relative Proj commutes with extension, so it supplies the next normal projective model and its birational map over  $F$ . This inductively descends the entire finite diagram, including its final fibre contraction when present.

Finally extend the diagram from  $F$  to  $k$ . Projectivity, normality, birationality, the absence of extraction on the positive side, connected contractions and relative ampleness persist. For a proper contraction, connectedness here is also expressed by the equality of the pushforward structure sheaf with the target structure sheaf, which is preserved by flat base change. Cartierness of the specified multiples has already descended from  $\mathbb{C}$  and is preserved over  $k$ .

We record the numerical facts needed for the endpoint. Every line bundle on an algebraically closed extension of a fixed projective  $F$ -variety is numerically equivalent to one from  $F$ : spread the bundle on a finite-type parameter scheme, specialize to an  $F$ -point, and use degree constancy on its geometrically connected base change, as above. Conversely, numerical triviality and nonnegativity of an  $F$ -divisor on relative curves persist under extension. Spread a vertical projective curve, together with the point of the target containing its image, in a flat projective family over a finite-type  $F$ -scheme. A specialization at an  $F$ -point is an effective vertical curve cycle with the same intersection degrees. The corresponding nonnegativity or vanishing over  $F$  therefore gives the same property for the original curve. The reverse implication is tested on base changes of  $F$ -curves. It follows that nefness and the relative numerical divisor spaces, hence relative Picard ranks, are unchanged. In the nef case the final adjoint remains nef; in the Mori case its negative remains relatively ample and the relative Picard number remains one.

Take common smooth resolutions of the descended finite steps, also dominating the original model carrying the nef divisor, and extend them to  $k$ . Their effective pullback differences remain effective, with the same positive coefficients along original contracted primes. The calculation (3.18) then gives weak increase of every generalized log discrepancy over  $k$ , with strict increase for each prime on the original  $X$  contracted by the composite. In particular the output is generalized lc. Its boundary is the pushforward of the original boundary and its nef part is the trace of the original nef  $b$ -divisor. The same effective pullback differences show that a nef endpoint makes the original adjoint pseudo-effective. Conversely, pseudo-effectivity of the original adjoint persists through the birational steps and rules out an anti-ample restriction to a general positive-dimensional fibre. Thus the sign of the original adjoint selects the stated alternative over  $k$  as well, completing Theorem 1.1.  $\square$

It remains to prove Theorem 3.1 and the local assertions on which its proof depends.

#### 4. EQUIVARIANT CONES AND AN LC PROJECTIVE SLICE

Throughout the local argument the ground field is  $\mathbb{C}$ . If  $\theta$  is a nonzero rational volume form, we write  $A_\theta$  for the log discrepancy of the sub-pair whose boundary on a smooth model is  $-\operatorname{div}(\theta)$ . For an  $s$ -fold pluri-volume form the boundary is  $-s^{-1}\operatorname{div}(\theta)$ . These conventions are homogeneous in valuations and are compatible with separable finite extensions: pulling back the form gives the same discrepancy on an extended valuation, with its actual normalization. This is the characteristic-zero ramification formula. An *lc form* has nonnegative discrepancy at every divisorial valuation under consideration; on a projective model this tests the entire function field. A generator with zero Weil divisor on a normal variety recovers the usual discrepancy at centres on that variety. On real quasi-monomial valuations we use the log-linear extension on log-smooth models.

Consider a sequence of pointed normal Gorenstein klt germs  $(V, o)$  of fixed dimension  $n \geq 2$ , with a cyclic group fixing  $o$  and a semi-invariant generator  $\theta_V$  of zero divisor. We shrink affine neighbourhoods equivariantly when necessary. Constants denoted by  $C$  may change, and a uniform assertion about the sequence is allowed to hold after passage to a subsequence.

**4.1. Rational regrading on the whole local algebra.** The stable degeneration theorem [40, Theorem 1.2(1)–(2)] and uniqueness of the normalized-volume minimizer [39, Theorem 1.1] give a quasi-monomial valuation  $v_*$ , normalized by  $A(v_*) = 1$ , with finitely generated associated graded algebra  $R$ . The affine cone  $C_0 = \operatorname{Spec} R$  is normal and klt, and its induced Reeb valuation minimizes normalized volume. The valuation  $v_*$  is invariant under the cyclic group [39, Corollary 1.2]. Only this first cone will be used.

We first explain why a nearby rational monomial valuation has this same graded algebra with a different grading. Related rational regrading is proved in [32, Lemma 2.10]; here we keep track of the whole local algebra and the cyclic action explicitly. Present  $v_*$  at the generic point of an SNC stratum on a smooth model. After a toroidal subdivision respecting the smallest rational subspace containing its weight vector, we may work on a face with positive rationally independent weights. Write  $\rho$  for its number of monomial coordinates. Discrepancy is linear on this face. Choose regular semi-invariant lifts  $h_1, \dots, h_r$  of homogeneous algebra generators of  $R$ , using character projections for the finite group. Enlarge this list by regular eigenfunctions generating the maximal ideal and, when needed, by centred affine coordinates.

In the completed monomial chart, with coefficients in a residue coefficient field, each  $h_i$  has one least exponent for the original weights. That exponent remains uniquely least for every sufficiently nearby weight vector. To justify a common neighbourhood, exponents far out in  $\mathbb{N}^\rho$  have uniformly large weight when all coordinate weights stay in a compact subset of the positive cone; only finitely many exponents can compete with the chosen one. There are finitely many generators.

Fix one neighbourhood  $U$  of positive weight vectors on which every  $h_i$  retains its unique least exponent  $\alpha_i$  and its coefficient  $c_i \neq 0$ . This same neighbourhood works for products of every degree. Indeed, in  $h^\beta = \prod_i h_i^{\beta_i}$ , every exponent other than  $E_\beta = \sum_i \beta_i \alpha_i$  differs from  $E_\beta$  by a sum of generator exponent gaps. Each nonzero gap pairs positively with every weight in  $U$ , so  $E_\beta$  remains uniquely least, with coefficient  $\prod_i c_i^{\beta_i}$ . No bound on  $\beta$  is needed.

For any regular local function  $g$ , successive subtraction of its initial form expresses it, to arbitrarily high  $v_*$ -order, as a sum of polynomials  $P_j(h)$  homogeneous for the original weights. The process goes to infinite order because the finitely generated monoid of positive weights has only finitely many elements in a bounded interval. In one such polynomial all the exponents  $E_\beta$  coincide: their original scalar weights coincide and the original coordinate weights are rationally independent. Its coefficient sum at this exponent is nonzero, because  $P_j$  was chosen to represent the nonzero initial of the current remainder. The product calculation therefore shows that  $P_j(h)$  retains that exponent and coefficient for every weight in the same  $U$ . Distinct subtraction groups have distinct exponent vectors, so even if their new scalar weights tie, their initial monomials cannot cancel. Finally, on a compact positive subneighbourhood of  $U$ , coordinatewise weight comparison gives  $v_{\text{new}}(r) \geq cv_*(r)$

for every regular subtraction error  $r$ , with one  $c > 0$ . The errors thus tend to infinite order for every such new valuation. Consequently the new associated graded ring is exactly the subring generated by the same initial monomials in the polynomial ring over the residue coefficient field.

This conclusion is equivariant on the entire local algebra. Indeed, relations and leading cancellations split by their exponent vectors, and their equivariance is inherited from the  $v_*$ -filtration. For completeness, the error comparison also holds after every group translation. A finite subtraction error  $r$  is regular at  $(V, o)$ , as is  $\gamma r$  because  $\gamma$  fixes  $o$ . Both pull back to regular functions on the fixed monomial chart, even if  $\gamma$  does not preserve that chart. Their series therefore have nonnegative boundary exponents. For a fixed positive comparison constant  $c$ ,

$$v_{\text{new}}(\gamma r) \geq c v_*(\gamma r) = c v_*(r) \longrightarrow \infty.$$

Only this comparison for regular errors is needed.

Choose sufficiently close positive rational weights and rescale them to a primitive integral valuation  $v_{\text{int}}$ . Since  $V$  is Gorenstein and klt,

$$(4.1) \quad a = A_{\theta_V}(v_{\text{int}}) \in \mathbb{Z}_{>0}, \quad v = a^{-1}v_{\text{int}}, \quad A_{\theta_V}(v) = 1.$$

Integral degree  $I$  in  $R$  thus means normalized degree  $I/a$ . We choose the weights close enough that every positive monomial weight, after normalization, lies between one half and twice its  $v_*$ -weight.

The associated extended Rees family is of finite type. More explicitly, on an affine neighbourhood containing the chosen generators, the ideal of functions of  $v_{\text{int}}$ -order at least  $j > 0$  is generated by monomials in the  $h_i$  having weight at least  $j$ . In the local ring, successive initial subtraction gives this statement modulo errors of arbitrarily high valuation. Izumi's inequality for the divisorial valuation at the klt germ [31, Theorem 3.1], with a constant depending on this fixed germ and valuation, makes these errors arbitrarily deep in the maximal ideal; Krull intersection then gives the ideal equality. Away from  $o$  both ideals are the unit ideal, because the chosen generators cut out  $o$ . The family is therefore generated by the original algebra,  $t$ , and  $t^{-v_{\text{int}}(h_i)}h_i$ . It is flat over the  $t$ -line, has a section through the pointed germs, and has central fibre  $C_0$ . It is normal: away from the central fibre this follows from normality of  $V$ , while the normal Cohen–Macaulay Cartier fibre gives the depth and codimension-one conditions along that fibre.

Order along the central fibre restricts to  $v_{\text{int}}$  and is its Gauss extension with  $w(t) = 1$ . Indeed, when terms involving distinct powers of  $t$  tie, their coefficient initials belong to distinct degrees of  $R$  and cannot cancel. The monomial discrepancy formula on the product with the logarithmic  $t$ -line consequently shows that

$$(4.2) \quad t^{-a}\theta_V \wedge \frac{dt}{t}$$

has exactly a simple pole along the central fibre and no other divisor. Thus the total space has Cartier canonical class, and residue along the smooth locus of the normal central fibre gives a generator  $\theta$  on  $C_0$  with zero Weil divisor and integral weight  $a$ .

This generator is also a character for the original grading torus. The reason is that units of a positive graded affine domain with degree-zero part  $\mathbb{C}$  are constants. The generators produced by different rational regradings therefore differ by scalars. Their normalized canonical weights are all one, so continuity of pairing with the fixed canonical character gives canonical weight one for the original Reeb valuation. This weight equals its discrepancy, as in the Reeb-character formula of [32, Lemma 2.18 and Definition 2.20]. In the present volume-form convention this can also be seen directly: choose homogeneous rational monomials forming a basis of the character lattice, view them as transcendental torus coordinates over the degree-zero field, and divide the form by its character monomial. What remains is a base volume form wedged with logarithmic fibre differentials, to which the weighted Gauss formula applies. The same description proves continuity of discrepancy for these Reeb weights.

## 4.2. Alpha estimates and comparison of valuations.

**Proposition 4.1.** *For the preceding choices, every nonzero homogeneous  $h_I \in R_I$  of positive degree satisfies*

$$(4.3) \quad \text{lct}_o(C_0; h_I) \geq \frac{1}{2n(I/a)}.$$

*Every nonzero regular local function  $g$  on  $V$  and every divisorial or real quasi-monomial valuation  $w$  centred at  $o$  satisfy*

$$(4.4) \quad w(g) \leq 2nA_{\theta_V}(w)v(g).$$

*Moreover, for every valuation  $w$  centred at  $o$ ,*

$$(4.5) \quad w(g) \geq s(w)v(g) \quad (g \in \mathcal{O}_{V,o}), \quad s(w) = \min_i \frac{w(h_i)}{v(h_i)}.$$

*In particular each positive normalized homogeneous degree is at least  $1/(2n)$ .*

*Proof.* First take an eigenfunction  $f$  for the original grading torus, of positive weight. An equivariant log resolution gives an invariant divisorial valuation computing its threshold at the vertex. Normalize it to  $A(w) = 1$  and set  $c = w(f) > 0$ . It is nonnegative on  $R$ . For a nearby rational normalized Reeb grading  $\zeta$  and a positive rational number  $s$ , define

$$u_s(h_\chi) = \zeta(h_\chi) + sw(h_\chi)$$

on torus eigenvectors and take the minimum over characters on sums. This is a valuation with

$$(4.6) \quad A(u_s) = 1 + s.$$

Here is a direct verification of the discrepancy statement. On the product with a logarithmic line of coordinate  $z$ , take the Gauss extension of  $sw$  with  $w'(z) = \gamma > 0$  rational. Choose an integral one-parameter subgroup  $j$  and  $\gamma$  so that  $\zeta(\chi) = \gamma j(\chi)$ , and apply the field automorphism  $h_\chi \mapsto h_\chi z^{j(\chi)}$ , fixing  $z$ . Its restriction has the displayed values  $u_s$ . It is still a Gauss extension: after clearing denominators, coefficients that tie at a power of  $z$  do not cancel across distinct characters, because the  $w$ -filtration is torus invariant. The transformation of  $\theta \wedge d \log z$  adds the canonical character weight 1 to discrepancy. The original Gauss extension has discrepancy  $s$ . Both it and the transformed restriction are divisorial up to rational scaling: the value group is discrete and nonzero, and restriction from the product leaves residue transcendence degree at least  $n - 1$ . The logarithmic Gauss extension has the same discrepancy as its restriction. These observations prove (4.6).

Put  $d_\zeta = \zeta(f)$ . Multiplication by  $f$  shifts the  $u_s$ -filtration exactly, while  $u_s \geq \zeta$  on  $R$ . Counting modulo consecutive powers of  $f$  therefore gives

$$(4.7) \quad \text{vol}(u_s) \leq \text{vol}(\zeta) \frac{d_\zeta}{d_\zeta + sc}.$$

Indeed, at cutoff  $p$  the length is at most the sum, over  $k \geq 0$ , of the dimensions in  $R/(f)$  up to  $\zeta$ -weight  $p - k(d_\zeta + sc)$ . The cumulative Hilbert growth of  $R/(f)$  has leading coefficient  $d_\zeta \text{vol}(\zeta)/(n-1)!$ , by the nonzerodivisor identity for the rational graded Hilbert series, after integral rescaling. Summing this polynomial growth proves (4.7); local lengths at the vertex equal these affine counts.

Use normalized-volume minimization on  $C_0$  and let  $\zeta$  approach its minimizing Reeb valuation. Multiplicative comparison of weights gives continuity of volume. Writing  $d_* = v_*(f)$ , we obtain

$$d_* + sc \leq (1 + s)^n d_*.$$

Letting  $s$  decrease to zero yields  $c \leq nd_*$ . A homogeneous  $h_I$  for the rational regrading may have several components for the original torus. A one-parameter torus degeneration selects a nonzero

extreme component. Semicontinuity of the threshold in the trivial family, equivalently inversion of adjunction, bounds the threshold of the general translate below by that of this component. Its original weight is at most  $2I/a$ . This proves (4.3).

For a nonunit  $g$ , the function  $t^{-v_{\text{int}}(g)}g$  is regular near the special point of the Rees family and restricts to its graded initial. Inversion of adjunction for a normal Cartier divisor in a Gorenstein variety [26] transfers the threshold bound to the general pointed germ. Its different is restriction here, since the total space is smooth along the smooth locus of the central fibre. More explicitly, the pair obtained by adding the central divisor is lc near the special point; away from  $t = 0$  the family of divisors is a product, and the pointed section gives the desired bound at  $o$ . This is exactly (4.4) for divisorial valuations. Approximation on a log-smooth model gives it for real quasi-monomial valuations. Units have value zero and cause no exception.

The monomial generators of valuation ideals proved above give (4.5): each generating monomial of  $v$ -weight at least  $v(g)$  has  $w$ -weight at least  $s(w)v(g)$ . Finally a nonunit at the vertex has threshold at most one, so (4.3) implies the lower bound  $I/a \geq 1/(2n)$ .  $\square$

**4.3. Complements and a root cover.** There is a homogeneous nonzero function

$$(4.8) \quad f_* \in R_{aN}, \quad \left( C_0, \frac{1}{N} \operatorname{div}(f_*) \right) \text{ is lc,}$$

where the positive integer  $N$  is bounded in terms of  $n$ . No finite-group equivariance of  $f_*$  is needed. To prove this, let  $E = \operatorname{Proj} R$  and  $H = \mathcal{O}_E(1)$  as an ample rational class. The quotient carries the standard boundary  $B$ , with coefficient  $1 - 1/e_P$  at a prime of generic isotropy order  $e_P$ . The grading is effective because  $v_{\text{int}}$  is primitive. On a homogeneous principal open, invert a homogeneous section  $b$  and take the slice  $b = 1$ . Its product with the radial torus maps finite étale to that cone open, and its cyclic quotient gives the corresponding Proj open with the stated ramification boundary. The slice is klt, so  $(E, B)$  is klt.

Choose a homogeneous rational element  $s$  of weight one. Then  $k(C_0) = k(E)(s)$ , and write

$$\theta = s^a \omega \wedge d \log s.$$

The slice description identifies the divisor of the logarithmic product form over every prime of  $E$  with the ramification multiple of  $K_E(\omega) + B$ . The rational divisor of the section  $s$ , defined by divisible powers, represents  $H$ . Consequently

$$(4.9) \quad K_E + B \sim_{\mathbb{Q}} -aH.$$

Thus  $E$  is of Fano type, the pair has standard coefficients, and its negative log canonical class is nef. Birkar's bounded complement theorem [8, Theorem 1.7] supplies an lc complement  $B^+ \geq B$  with  $N(K_E + B^+) \sim 0$  for bounded  $N$ . For standard coefficients the complement rounding inequality ensures the stated monotonicity.

Choose a rational  $N$ -pluri-volume form on  $E$  with divisor  $-NB^+$  and take its logarithmic product  $\psi$  with  $(d \log s)^N$ . It is lc on the function field: use the compactification of the torus with both boundary points. The same slice ramification calculation makes its divisor on the punctured cone nonpositive. Therefore  $f_* = \theta^N / \psi$  has no pole there, is homogeneous of degree  $aN$ , and extends across the vertex by normality and  $n \geq 2$ . Its discrepancy identity proves (4.8).

Normalize one component of the cover obtained by adjoining  $u^{aN} = f_*$ . Write its ring as  $R'$ , grade  $u$  in degree one, and set

$$(4.10) \quad \widehat{E} = \operatorname{Proj} R', \quad D = aH_u,$$

where  $H_u$  is the effective rational divisor of  $u$ , defined using divisible powers. A connected grading torus preserves each component, so the chosen component remains graded. The variety  $\widehat{E}$  is

normal and projective, and  $D$  is ample, effective and  $\mathbb{Q}$ -Cartier. Because  $u$  has weight one,  $k(\text{Spec } R') = k(\widehat{E})(u)$ . Define a rational volume form  $\eta$  on  $\widehat{E}$  by the pullback identity

$$(4.11) \quad \theta = u^a \eta \wedge d \log u.$$

The  $N$ -th power of  $\eta \wedge d \log u$  is the pullback of  $\psi$ , so  $\eta$  is lc. On  $\widehat{E} \setminus \text{Supp } D$  it has zero Weil divisor and is klt. Indeed the cone cover is étale where  $u \neq 0$ , and that open is the product of the corresponding base open with a torus.

With  $h = \text{vol}(v)$  one has

$$(4.12) \quad h = a^n H^{n-1}, \quad D^{n-1} \leq Nh.$$

For the first equality, count graded pieces up to integral degree  $ap$ . Divisible degrees have the ample Hilbert polynomial on  $E$ . Every other degree class has the same leading growth: nonzero degrees generate  $\mathbb{Z}$ , so multiplication by fixed homogeneous elements compares each class above and below with divisible classes, with bounded degree shifts. These shifts are fixed for the individual cone and need not be uniform along the sequence. Summing yields the first equality. For the second, the cone cover has degree at most  $aN$ , and its degree on  $\text{Proj}$  is the same because the two homogeneous fraction fields have weight-one torus parameters. Since  $H_u$  pulls back  $H$ ,  $a^{n-1} H_u^{n-1} \leq a^{n-1} (aN) H^{n-1} = Nh$ .

## 5. THE PRODUCT OF INDEPENDENT DEGREE SCALES

Continue with the cone and slice constructed in Section 4. For  $1 \leq j \leq n$ , let  $d_j$  be the first positive normalized degree at which the family of *all* homogeneous functions up to that degree has transcendence rank at least  $j$ . These ranks can also be attained using finite-group eigenfunctions, by choosing character bases in each homogeneous piece.

**Proposition 5.1.** *After passage to a subsequence there is a uniform constant  $C$  such that*

$$(5.1) \quad h d_1 \cdots d_n \leq C, \quad h = \text{vol}(v).$$

The proof cuts the projective slice by homogeneous pencils and compares the volumes of successive geometric generic fibres. The following estimate supplies the discrepancy bound used in those comparisons. Its proof in Section 9 is independent of Proposition 5.1, and its index-one hypothesis concerns the volume form itself.

**Proposition 5.2** (Integral jump). *Fix a positive dimension. Consider a sequence of smooth projective varieties  $F$  over algebraically closed fields isomorphic to  $\mathbb{C}$ , with projective birational morphisms  $F \rightarrow F_0$  to normal varieties. Let  $\eta_F$  be an lc rational volume form, and let  $L$  be the pullback of a big semiample effective  $\mathbb{Q}$ -Cartier divisor on  $F_0$ . Suppose the supports of  $\text{div}(\eta_F)$  and  $L$  on  $F$  have simple normal crossings. Write  $l_w = w(L)$  and assume the following conditions.*

- (i) *For every divisorial valuation  $w$ , the equality  $A_{\eta_F}(w) = 0$  implies  $l_w > 0$ .*
- (ii) *For every prime divisor  $P$  on  $F_0$ ,*

$$A_{\eta_F}(P) = 1 \quad \text{if } l_P = 0, \quad A_{\eta_F}(P) \leq \epsilon l_P \quad \text{if } l_P > 0,$$

*where  $\epsilon \rightarrow 0$  along the sequence.*

- (iii) *For every rational  $0 < b \leq 1/10$ ,  $h^0(F, K_F + \lceil bL \rceil) = 1$ .*
- (iv) *There is a uniform constant  $C$  such that every effective  $\mathbb{Q}$ -divisor  $J \sim_{\mathbb{Q}} L$  satisfies  $w(J) \leq Cl_w$  at every divisorial valuation with  $A_{\eta_F}(w) = 0$ .*

*Then, after passage to a subsequence, there is a uniform constant  $C'$  such that every such  $J$  and every divisorial valuation  $w$  satisfy*

$$(5.2) \quad w(J) \leq C' (A_{\eta_F}(w) + l_w).$$

*It suffices that the hypotheses hold sufficiently far along the sequence.*

**5.1. Degree tiers and geometric generic components.** By Proposition 4.1 and (4.8),

$$\frac{1}{2n} \leq d_1 \leq N.$$

Passing to a subsequence, split the ordered degrees into successive tiers: ratios within a tier are bounded above and below, whereas the ratio from the end of one tier to the beginning of the next tends to infinity. This follows by successively passing to subsequences for the finitely many adjacent ratios. Every degree in the first tier is bounded above and below.

Choose algebraically independent homogeneous functions  $f_1 = f_*, f_2, \dots, f_n$  of normalized degrees  $e_1 = N, e_2, \dots, e_n$ , where  $e_j$  is comparable with  $d_j$ . Choose them so that the functions in each complete prefix of tiers form a transcendence basis for the homogeneous functions before the next tier. To do this, extend the single bounded-degree element  $f_*$  already in the first tier; its degree may exceed that tier's initial cutoff by a bounded amount, but it lies below the next tier sufficiently far along the sequence. Thereafter extend the existing independent family at each tier cutoff. No equivariance of these functions is required. If  $1, \dots, m$  is a complete prefix and  $T = d_{m+1}$  starts the next tier, every homogeneous element of normalized degree strictly less than  $T$  is algebraic over  $k(f_1, \dots, f_m)$ , where  $k$  is the cone's ground field.

For any prefix  $1, \dots, m$ , in particular for a complete prefix of tiers, define rational functions on  $\widehat{E}$  by

$$(5.3) \quad \phi_j = \frac{f_j}{u^{ae_j}}, \quad 2 \leq j \leq m.$$

They are independent, since  $f_1 = u^{aN}$ . Resolve the graph of their map to  $(\mathbb{P}^1)^{m-1}$ , obtaining a smooth projective  $W$ , and let  $F$  be a connected component of its geometric generic fibre. Set  $q = n - m$ . Let  $F_0$  be the corresponding component on the normalized graph; it is normal by localization and geometric base change in characteristic zero. The divisor  $D$  pulls back to divisors  $D_W, D_F$ , and to an effective  $\mathbb{Q}$ -Cartier divisor on  $F_0$ . The divisor  $D_F$  is nef and semiample. It is also big: the geometric generic component meets densely the graph over the original domain of definition, so its map to  $\widehat{E}$  is birational onto its image, and  $D$  is ample there.

Divide  $\eta$  by  $d\phi_2 \wedge \dots \wedge d\phi_m$ , with a fixed choice of sign, to obtain the relative volume form  $\eta_F$ . It is lc, and

$$(5.4) \quad A_{\eta_F}(w) = 0 \implies w(D_F) > 0.$$

Indeed, on an SNC resolution generic smoothness restricts the divisor of the form and its discrepancy formula to the geometric generic fibre. Off  $D$  the form is klt, which gives the implication. The algebraically closed fields of these fibres are isomorphic to  $\mathbb{C}$ : adjoining finitely many transcendental elements and then taking an algebraic closure leaves the transcendence degree over  $\mathbb{Q}$  equal to that of  $\mathbb{C}$ .

We may choose one  $W$  dominating all prefix graphs, with every pencil regular and the supports of  $\eta, D_W$  SNC. Apply generic smoothness also to every stratum. On every intermediate generic fibre the restricted support then has normal crossings and its cuts are reduced. Consequently taking ceilings of rational multiples of  $D_W$  commutes with these restrictions. Components missing the generic fibre contribute nothing. The absolute-to-relative division defining  $\eta_F$  likewise restricts the form divisor over the generic parameters.

## 5.2. Discrepancies on the normal graph.

**Lemma 5.3.** *At every prime divisor  $P$  of  $F_0$ , with its primitive valuation, put  $c_P = P(D_F)$ . Then*

$$(5.5) \quad \begin{aligned} c_P > 0 &\implies A_{\eta_F}(P) \leq \left( \sum_{j \leq m} e_j - 1 \right) c_P, \\ c_P = 0 &\implies A_{\eta_F}(P) = 1. \end{aligned}$$

This includes the case  $m = 1$ .

*Proof.* Spread  $P$  to a horizontal divisor valuation  $P_0$  on the normalized graph over the rational base. A finite algebraic extension of the generic base used to label its component is unramified at this horizontal divisor. By separability on the divisor, the order of the relative differential form is its absolute order before division by the base differentials. Thus we can compute using  $P_0$ .

When  $c_P = 0$ , its generic centre lies off  $D$ . The rational map is regular there, so its normalized graph is isomorphic to that open of  $\widehat{E}$ , where  $\eta$  has zero Weil divisor. This gives discrepancy one. Suppose  $c_P > 0$  and let  $S$  be the centre of  $P_0$  on  $\widehat{E}$ , of codimension  $r \geq 1$ . Among the residues of the  $\phi_j$  there are  $r - 1$  algebraically independent elements over  $k(S)$ . To see this, the residue field of  $P_0$  has transcendence degree  $n - 2$ , and is algebraic over the field generated by  $k(S)$  and the base coordinates: normalization of the graph is finite over the graph. All these base coordinates are units at the horizontal divisor, so a transcendence basis can be chosen from their residues. The difference of transcendence degrees is  $r - 1$ .

Extend  $P_0$  to  $k(\widehat{E})(u)$  by the Gauss valuation  $\nu$  with  $\nu(u) = c_P/a$ , and also restrict  $\nu$  to the original cone field. It is nonnegative on homogeneous regular elements: the divisor of a degree- $I$  section divided by  $u^I$  is bounded below by  $-IH_u$ . Thus  $\nu$  has a centre on the affine cone cover.

The field of this affine centre has transcendence degree  $n - r$  and is algebraically related to  $k(S)(\tau)$  over a common subfield of that transcendence degree. Here  $\tau$  is the residue of a positive-degree section that is a unit for  $\nu$ . More explicitly, choose a very ample Cartier multiple and a section nonvanishing at  $S$ ; its residue is transcendental over the residue field of  $P_0$  by the Gauss construction. Ratios of other sections in that multiple recover the dimensions of  $S$ . For every other homogeneous unit, taking powers and ratios with the chosen section makes its residue algebraic over  $k(S)(\tau)$ . Homogeneous decomposition proves the same assertion for residues of units in the entire affine ring. The finite map to  $\text{Spec } R$  preserves the dimension of this centre.

Choose  $n - r$  regular functions on  $\text{Spec } R$  with independent residues on that centre. The selected  $r - 1$  residues of the  $\phi_j$ , or their positive powers, remain independent over its centre field, because  $\tau$  is Gauss-transcendental over the residue field of  $P_0$ . Together with  $f_1$ , use the corresponding  $r - 1$  functions among the  $f_j$ . After taking powers, their degree-zero ratios with  $f_1$  give the chosen powers of the  $\phi_j$ .

These  $n$  cone functions have a logarithmic differential wedge of discrepancy zero at  $\nu$ . In fact monomial combinations with nonzero determinant give one function of positive order and  $n - 1$  units with independent residues. In characteristic zero their logarithmic wedge has a simple pole along the divisorial valuation. Hence their ordinary differential wedge has discrepancy equal to the sum of their values, which is at most  $\sum_{j \leq m} e_j c_P$ . Dividing that wedge by  $\theta$  gives a regular coefficient on the cone: this holds at every codimension-one point, and normality extends it. On the other hand, (4.11) and the logarithmic Gauss formula give

$$A_\theta(\nu) = A_{\eta_F}(P) + c_P.$$

Nonnegativity of the coefficient's value proves (5.5). If  $m = 1$ , necessarily  $r = 1$  in this calculation; the list of  $\phi_j$  is empty and the same proof applies.  $\square$

### 5.3. A single adjoint section before a degree jump.

**Lemma 5.4.** *Suppose  $q > 0$  and  $T = d_{m+1}$  begins the tier after the chosen complete prefix. Uniformly sufficiently far along that jump,*

$$(5.6) \quad h^0(F, K_F + \lceil bTD_F \rceil) = 1 \quad (0 < b \leq 1/10, b \in \mathbb{Q}).$$

*Proof.* The form  $\eta_F$  provides a section. Its poles are simple and lie over  $D_F$ , so the positive ceiling supplies at least one copy of each pole. Suppose a second section has nonconstant ratio with  $\eta_F$ . After extending scalars to the algebraic closure of the rational base field, regard  $F$  as a component of a general smooth complete intersection of the free pencils on  $W$ . Extend the section by zero on

the other components. Let  $M_j$  be the pole divisor on  $W$  of  $\phi_j$ ; it is a member of the corresponding free pencil.

Successive adjunction lifts this section to

$$K_W + [bTD_W] + \sum_{j=2}^m M_j.$$

To give the restriction step explicitly, on an intermediate smooth component  $Y$  write  $A = bTD_Y$ , let  $C$  be the next general member of its free pencil, and let  $R_Y$  be the sum of the remaining pencil divisors. Adjunction gives the exact sequence

$$\begin{aligned} 0 &\longrightarrow \mathcal{O}_Y(K_Y + [A] + R_Y) \\ &\longrightarrow \mathcal{O}_Y(K_Y + [A] + R_Y + C) \\ &\longrightarrow \mathcal{O}_C(K_C + [A|_C] + R_Y|_C) \longrightarrow 0. \end{aligned}$$

The first term has vanishing  $H^1$  by Kawamata–Viehweg vanishing [19, Section 3.1]. Indeed,  $B_Y = [A] - A$  has SNC support and coefficients in  $[0, 1)$ , and the difference of the displayed integral divisor and  $K_Y + B_Y$  is  $A + R_Y$ , which is big and nef. Bigness on each intermediate component follows, as for  $F$ , from its image on the original domain of the graph. All supports were resolved before taking the cuts, and ceilings commute with these cuts as explained above. In the finite base chart, adjunction by the equations  $\phi_j - \lambda_j$  identifies the lifted ratio with the required ratio of relative forms. Denote the lift, written as a rational multiple of  $\eta$ , by  $H'\eta$ .

On the base-changed  $\widehat{E}$ , the poles of  $H'$  are bounded by

$$(5.7) \quad \left( bT + N + \sum_{j=2}^m e_j \right) D.$$

Indeed, at a prime with coefficient  $c > 0$  in  $D$ , the case  $m = 1$  of Lemma 5.3 gives  $\text{ord } \eta \leq (N-1)c-1$ . The ceiling excess is at most one, and the pole divisor of each  $\phi_j$  is at most  $e_j D$ . Adding these inequalities proves (5.7). At a prime with  $c = 0$  the form has order zero and these divisors have no pole.

The selected root field is cyclic Galois over the original cone field, of degree dividing  $aN$ . Split  $H'$  into its deck-character parts, with the enlarged scalar field fixed. Each part still obeys (5.7), because  $D$  is invariant; the deck group is not required to preserve the selected geometric fibre. For a part of given character choose an integer  $I$  in the cancelling congruence class with

$$bT + N + \sum_{j=2}^m e_j \leq \frac{I}{a} < bT + N + \sum_{j=2}^m e_j + N + \frac{1}{a}.$$

Multiplication by  $u^I$  makes it an invariant homogeneous regular function on the cone cover. Regularity follows in codimension one from the pole inequality and the divisor  $H_u$ ; the vertex has codimension at least two, so normality completes the check. This invariant is therefore in the scalar extension of  $R_I$ .

The preceding degrees are negligible compared with  $T$ , and  $a \geq 1$ . Since  $bT \leq T/10$ , one index in the sequence sufficiently far along the jump makes  $I/a < T$  for every rational  $b$  in the asserted range. Every ground-field basis element  $h \in R_I$  is then algebraic over  $k(f_1, \dots, f_m)$ . Its quotient  $h/u^I$  is algebraic over  $k(\phi_2, \dots, \phi_m)$ . To check the latter assertion, after adjoining  $u$  the original algebraic relation is a relation over  $k(\phi_2, \dots, \phi_m)(u)$ ; but  $h/u^I$  belongs to  $k(\widehat{E})$  and  $u$  is transcendental over that field. Expanding a relation in powers of  $u$  gives a nonzero relation over the base field alone. Such a rational function is constant on each connected geometric generic component. This remains true for scalar combinations of the basis elements. Thus every deck-character part of  $H'$  is constant on  $F$ , contradicting the chosen nonconstant ratio.  $\square$

**5.4. Jets, image sheaves and clipping on the first tier.** Take the complete bounded first tier, of length  $m$ ; allow  $m = n$  if there is only one tier. We prove

$$(5.8) \quad h \leq CD_F^q, \quad D_F^q \leq Ch.$$

For  $q = 0$ , the degree on the geometric generic component is one. If  $q > 0$ , we also prove that, for every divisorial valuation  $P$  over  $F$  with  $A_{\eta_F}(P) = 0$ , every effective  $J \sim_{\mathbb{Q}} D_F$  satisfies

$$(5.9) \quad P(J) \leq Cc_P, \quad c_P = P(D_F) > 0.$$

The upper volume bound follows from nef intersections on  $W$ . Both  $D_W$  and the pencil divisors  $M_j$  are nef, and  $e_j D_W - M_j$  is effective. Therefore

$$D_F^q \leq D_W^q \prod_{j=2}^m M_j \leq \left( \prod_{j=2}^m e_j \right) D_W^{n-1} \leq Ch$$

by (4.12) and boundedness of the first tier. If a fibre has several components, the first intersection is their sum with positive multiplicities, so it still bounds the chosen component.

For the other direction test the full vector space  $R_{\leq ap} = \bigoplus_{i \leq ap} R_i$ , with  $p$  large and divisible. Write one of its elements as  $H = \sum_{i \leq ap} h_i^\circ$ , where  $h_i^\circ \in R_i$ . On  $\bar{W} \times \mathbb{G}_m$ , with torus coordinate  $s$ , it becomes

$$(5.10) \quad H_s = \sum_{i \leq ap} s^i \frac{h_i^\circ}{u^i}.$$

It is a rational section with pole bound  $pD_W$ . For every rational divisorial valuation  $w$  centred on this product and every nonzero  $H$ ,

$$(5.11) \quad w(H_s) \leq Cp(A_{\eta \wedge d \log s}(w) + w(D_W)).$$

To prove this estimate, add a Gauss parameter  $z$  with positive rational weight  $\gamma > w(D_W)/a$ , and use the generically finite map to the cone times the  $z$ -line given by dilation  $s \mapsto sz$ . Every positive-degree homogeneous cone function now has positive value, so the valuation is centred at the vertex times zero. Let  $I$  be the largest occurring degree. If  $I = 0$  the assertion is immediate. Otherwise test the single function  $\sum_i z^{I-i} h_i^\circ$  on the target. At  $z = 0$  it is  $h_I^\circ$ , whose threshold is at least  $1/(2np)$  by (4.3). Inversion of adjunction, with the central divisor added, gives

$$w(H_s) + I\gamma \leq 2np(A_{\eta \wedge d \log s}(w) + a\gamma).$$

The discrepancy on the right follows by pulling back  $\theta \wedge dz/z$  and using (4.11); the torus coordinate  $s$  is a unit. Let  $\gamma$  decrease to  $w(D_W)/a$  and discard the nonnegative term on the left to obtain (5.11). The calculation also works after a generically finite base change, by restricting valuations and pulling back the same forms and divisors.

We now give the jet count, retaining the exact section space at a thickened divisor. For the divisor test, first make  $P$  primitive. Spread its extraction after a finite generically étale base change of an open of the pencil base, labelling the selected component  $F$ . All geometric generic data descend over a finite extension. Resolve and shrink so that the family is projective, the required relative SNC data are smooth, and the chosen components and divisor descend geometrically integrally. Include  $s$  as an unchanged extra base parameter. At the generic point of the specialized divisor on a sufficiently general closed fibre, take monomial weights one on the  $m$  base parameters, including  $s - s_0$ , and weight  $1/c_P$  on the equation of the spread divisor. The relative form has a simple pole there and no other support component passes through this generic point. The resulting rational divisorial valuation satisfies

$$(5.12) \quad w(D_W) = 1, \quad A_{\eta \wedge d \log s}(w) = m.$$

For the whole-component test use instead the generic point of a component of a general closed fibre and only the  $m$  base parameters, all of weight one. Then the divisor value is zero and discrepancy

is  $m$ . Commensurability makes both valuations divisorial up to scaling. The indicated parameters form a regular system of parameters at the respective generic points. The nonzero functions being tested remain nonzero by dominance.

By (5.11), the order of each nonzero test section as a local section of  $\mathcal{O}(pD_W)$  is at most  $C_1 p$  for either valuation. The constant is uniform in  $p$ ,  $P$  and the sufficiently general closed parameters, although those parameters may be chosen separately for each  $p, P$ .

Write  $\pi : \mathcal{X} \rightarrow B_0$  for the spreading component family and  $\mathcal{P}$  for the spread divisor. In the whole-component test use  $\pi_* \mathcal{O}_{\mathcal{X}}(pD_W)$ . In the divisor test use the *image sheaf*

$$\mathcal{E}_{p,j} = \text{im}(\pi_* \mathcal{O}_{\mathcal{X}}(pD_W) \longrightarrow \pi_* \mathcal{O}_{j\mathcal{P}}(pD_W)), \quad j = \lceil Kpc_P \rceil,$$

where a fixed rational  $K > C_1$  is chosen once. Generic base change and shrinking make these sheaves vector bundles of respective ranks

$$(5.13) \quad \begin{aligned} & h^0(F, pD_F), \\ & h^0(F, pD_F) - h^0(F, pD_F - jP). \end{aligned}$$

In the second line we work on the fixed extraction. The difference comes from the kernel of restriction of the original section space; it does not require surjectivity onto all sections of the thickening.

The test space injects into the jets of order  $\lceil Kp \rceil$  of this vector bundle at the chosen closed point of the base. Indeed, jets mean reduction modulo the  $\lceil Kp \rceil$ -th power of the maximal ideal  $\mathfrak{m}$  of the base point. Vanishing in them, followed by evaluation upstairs, puts the local section in

$$\mathfrak{m}^{\lceil Kp \rceil} \mathcal{O}_{\mathcal{X}}(pD_W) + \mathcal{I}_{\mathcal{P}}^j \mathcal{O}_{\mathcal{X}}(pD_W)$$

in the divisor test, and in just the first summand in the component test. With the chosen weights both summands have order at least  $Kp$ , since  $j/c_P \geq Kp$ . This contradicts the bound  $C_1 p$ . Jets in  $m$  smooth base parameters have dimension at most  $C_2 p^m$  times the bundle rank, with  $C_2$  depending only on  $m, K$ .

Now  $\dim R_{\leq ap} \sim hp^n/n!$ . Passing to volumes in the first test gives the missing inequality in (5.8); when  $q = 0$ , its rank is one and the same calculation applies. In the second test, for  $q > 0$ , it gives

$$(5.14) \quad h \leq C \left[ D_F^q - \text{vol}(D_F - Kc_P P) \right].$$

Here  $P$  and its extraction are fixed before taking the large divisible  $p$  limit. Rounding  $Kpc_P$  does not change the leading term: one can squeeze it between the corresponding rational coefficients differing by an arbitrarily small positive amount and use continuity of volume. The constants are uniform because they came from the jet estimate; no uniform convergence in  $P$  is required.

Combining (5.14) with the upper bound in (5.8) proves (5.9). In detail, if  $t = P(J) > Kc_P$ , set  $\lambda = Kc_P/t$ . On the extraction,

$$D_F - Kc_P P \sim_{\mathbb{Q}} (1 - \lambda)D_F + \lambda J - Kc_P P,$$

and the last two terms together form an effective divisor. Monotonicity and homogeneity of volume imply

$$\text{vol}(D_F - Kc_P P) \geq (1 - \lambda)^q D_F^q.$$

The two preceding bounds force  $1 \leq C[1 - (1 - \lambda)^q]$ , and hence force a uniform positive lower bound on  $\lambda$ . This is the claimed bound on  $t/c_P$ . If  $t \leq Kc_P$  it holds immediately. There is no divisorial clipping claim to check on a zero-dimensional component.

**5.5. Transfer across every later tier.** Suppose clipping has been proved on a prefix component  $F$  of positive dimension, and the next scale is  $T$ . Set  $L = TD_F$ . The bound (5.9) scales to hypothesis (iv) of Proposition 5.2. Its other hypotheses follow from (5.4), Lemma 5.4, and Lemma 5.3. More explicitly, at the primes of the normal graph we can take

$$\epsilon = \frac{\sum_{j \leq m} e_j - 1}{T} \longrightarrow 0,$$

because every preceding degree is negligible compared with  $T$ . All divisors and form supports were made SNC. Therefore (5.2) holds for  $(F, \eta_F, L)$ , after a subsequence.

Cut by the  $k_1$  pencils of the next tier, all with degrees at most  $CT$ , and choose a geometric generic component  $F^+$  of dimension  $q - k_1$ . Components and resolutions can be chosen compatibly by taking successive algebraic closures of the rational base fields. The next relative form is obtained by dividing by the new coordinate differentials. We claim

$$(5.15) \quad L^q \leq C(L|_{F^+})^{q-k_1}, \quad (L|_{F^+})^{q-k_1} \leq CL^q,$$

and, if  $\dim F^+ > 0$ , clipping for  $L|_{F^+}$ .

Apply the preceding jet argument on  $F$  itself, testing all sections of  $pL$  and using only the  $k_1$  new base parameters. No torus parameter is needed. A nonzero section has effective divisor  $pJ$  with  $J \sim_{\mathbb{Q}} L$ , so its local section order is bounded by

$$pC'(A_{\eta_F}(w) + w(L))$$

by (5.2). Spread any chosen geometric generic divisor after finite base change and take sufficiently general closed points avoiding ramification. The estimates therefore remain valid on this spreading family. For the whole-component test the weights one on the new base parameters give discrepancy  $k_1$  and divisor value zero. At an lc place  $P^+$  over  $F^+$ , give its equation weight  $1/P^+(L)$ . The relative form has a simple pole, so discrepancy is again  $k_1$ , and now  $w(L) = 1$ .

Use the pushforward bundle for the first test, and in the second use the image of restriction to the thickening with order  $\lceil KpP^+(L) \rceil$ . The same weighted-order contradiction injects the test space into jets in  $k_1$  parameters. Passing to volume gives the first inequality of (5.15) and the volume-deficit estimate with source volume  $L^q$ . Nef intersection with the new pencils, each bounded by  $CL$ , gives the second inequality. Combining them with the effective-representative argument just given proves clipping on  $F^+$  for  $L|_{F^+} = TD_{F^+}$ , equivalently clipping for  $D_{F^+}$ . Higher resolutions do not alter these estimates: effective representatives of a pulled-back  $\mathbb{Q}$ -linear system descend and pull back exactly. If  $F^+$  is a point, the first test has rank and degree one and gives the required volume comparison; the divisor test is unnecessary.

This proves the transfer with constants uniform after subsequence. There are only finitely many tiers, so finitely many subsequence choices suffice. To conclude, (5.8) starts the induction

$$h \prod_{j \leq m} d_j \leq CD_F^{n-m},$$

because the first-tier degrees are bounded. Across a tier of length  $k_1$ , the first inequality of (5.15) says

$$T^q D_F^q \leq CT^{q-k_1} D_{F^+}^{q-k_1}, \quad \text{hence} \quad T^{k_1} D_F^q \leq CD_{F^+}^{q-k_1}.$$

Every new  $d_j$  is comparable with  $T$ , giving exactly the new factors in the product. The terminal component is a point of degree one. This proves Proposition 5.1.

## 6. VALUATIVE OPTIMIZATION AND FINITE WINDOWS

We now prepare the proof of Theorem 3.1. Throughout this and the next two sections, put  $k_0 = \mathbb{C}$ ,  $L = k_0(V)$ , and  $\eta = \theta_V$ . We use an invariant affine representative of the pointed klt Gorenstein germ, on which  $\eta$  is a semi-invariant canonical generator. The cyclic group is denoted

by  $G$ . All valuations in these sections are trivial on  $k_0$ . Constants described as uniform are uniform along the subsequence already chosen in Proposition 5.1.

Let  $v$  be the invariant rational divisorial valuation of Proposition 4.1, normalized by  $A_\eta(v) = 1$ . Choose centred regular eigenfunctions  $h_i$  as in that proposition, including centred affine generators, and put  $v_i = v(h_i) > 0$ . Thus

$$(6.1) \quad s(w) := \min_i \frac{w(h_i)}{v_i}, \quad w(g) \geq s(w)v(g), \quad w(g) \leq 2nA_\eta(w)v(g)$$

for nonzero regular  $g$  and centred quasi-monomial  $w$ . Positivity on the centred affine generators ensures that the centre is  $o$ . Choose centred regular eigenfunctions  $f_1^0, \dots, f_n^0$  whose  $v$ -initials are algebraically independent and attain the successive homogeneous transcendence ranks. Set  $d_j = v(f_j^0)$ , in nondecreasing order. The preceding propositions give

$$(6.2) \quad d_j \geq \delta > 0, \quad d_1 \leq C, \quad \text{vol}(v) \prod_{j=1}^n d_j \leq C.$$

The functions may be chosen as germs and then regarded as rational functions; shrinking the affine representative accommodates any finite list subsequently used.

### 6.1. A bound on every finite window.

**Lemma 6.1.** *Put  $d_{n+1} = +\infty$ . There is a uniform constant  $C_W$  such that*

$$(6.3) \quad \mathcal{H}(T) := \dim_{k_0} \mathcal{O}_{V,o}/(v > T) \leq C_W \prod_{j \leq m} \frac{T}{d_j} \quad (d_m \leq T < d_{m+1}, \ 1 \leq m \leq n).$$

*Proof.* The restriction of  $v$  to  $k_0(f_1^0, \dots, f_n^0)$  is the weighted Gauss valuation. The weight-zero Laurent monomials for a basis of the kernel of  $\mathbb{Z}^n \rightarrow \mathbb{Q}$ ,  $\alpha \mapsto \sum \alpha_j d_j$ , have algebraically independent residues. Refine  $v$  by a full-rank monomial valuation on these residues, and extend this refinement to the finite extension of residue fields upstairs. We obtain a composite valuation  $\tau$  with residue field  $k_0$ . Using  $\tau(f_j^0)$  as coordinates, its value group is a lattice  $\Gamma \subset \mathbb{Q}^n$  containing  $\mathbb{Z}^n$  with finite index, and the original real value is the pairing with  $(d_1, \dots, d_n)$ .

Suppose centred regular functions  $l_1, \dots, l_n$  have independent  $\tau$ -values generating a sublattice  $\Lambda$ . Every class of  $\Gamma/\Lambda$  has a representative which is the value of a regular function: the regular value semigroup generates  $\Gamma$ , and its image in a finite group is a subgroup. Fix one such representative per class. Multiplying by the monomials  $l^\nu$ ,  $\nu \in \mathbb{Z}_{\geq 0}^n$ , gives distinct  $\tau$ -values. Those of  $v$ -value at most  $T$  are independent modulo  $(v > T)$ . Comparing the leading coefficients as  $T \rightarrow \infty$  gives

$$(6.4) \quad [\Gamma : \Lambda] \leq \text{vol}(v) \prod_{i=1}^n v(l_i).$$

The fixed representatives contribute only bounded shifts in this asymptotic comparison; no uniform bound for those shifts is needed.

Applying (6.4) to  $f^0$  gives  $[\Gamma : \mathbb{Z}^n] \leq C$ . If the  $j$ th coordinate  $\tau(g)_j$  of a regular nonunit  $g$  is nonzero, replace  $f_j^0$  by  $g$ . The determinant of the new lattice basis is  $|\tau(g)_j|$ , so

$$[\Gamma : \mathbb{Z}^n] |\tau(g)_j| \leq \text{vol}(v) v(g) \prod_{i \neq j} d_i.$$

Consequently

$$(6.5) \quad |\tau(g)_j| \leq C \frac{v(g)}{d_j}.$$

The inequality is automatic when that coordinate is zero. Units have  $\tau$ -value zero.

If  $v(g) \leq T < d_{m+1}$ , the initial of  $g$  is algebraic over the graded fraction field generated by the initials of  $f_1^0, \dots, f_m^0$ . Take a homogeneous polynomial relation. On refining its values by  $\tau$ , at least two lowest terms tie, and they involve distinct powers of the initial of  $g$ : coefficients belong to the graded field generated by an independent tuple, where their nonzero initials cannot cancel. Thus  $\tau(g)$  lies in the rational span of the first  $m$  coordinate vectors. The number of lattice points in this subspace satisfying (6.5) and  $v(g) \leq T$  is at most  $C \prod_{j \leq m} (1 + T/d_j) \leq C' \prod_{j \leq m} (T/d_j)$ . Here one can place  $\Gamma$  in  $\frac{1}{e}\mathbb{Z}^n$  with  $e = [\Gamma : \mathbb{Z}^n] \leq C$ . Finally  $\tau$  has one-dimensional leaves. The bounded box has only finitely many lattice points, so successively taking initials in the finite jet quotient bounds its dimension by this count. This proves (6.3).  $\square$

**6.2. Gauss valuations and continuation.** We say that a valuation is *Gauss on a tuple* when the tuple has algebraically independent homogeneous initials; equivalently, every nonzero polynomial in the tuple has the minimum of its monomial weights. We will also use this definition when the assigned weights have rational relations.

**Lemma 6.2.** *The following facts hold for an  $n$ -dimensional function field in characteristic zero.*

- (i) *If  $w$  is Gauss on a full tuple  $t_1, \dots, t_n$ , then it is quasi-monomial, and*

$$(6.6) \quad A_\eta(w) = w \left( \frac{\eta}{d \log t_1 \wedge \dots \wedge d \log t_n} \right).$$

*This identity is compatible with finite field extension and pullback of the form.*

- (ii) *Suppose  $w$  is quasi-monomial and Gauss on a partial tuple  $f$ . For nearby assignments of real weights to  $f$ , there are quasi-monomial Gauss extensions whose discrepancies and values on any prescribed finite subset of  $L^*$  tend to those of  $w$ . If  $w$  is centred at  $o$ , the extensions can be kept centred there.*
- (iii) *On the finite cone complex of a smooth proper SNC model, the evaluation of a fixed rational function is continuous, homogeneous, and rational piecewise linear, including on faces.*

*Proof.* For (i), make a unimodular monomial change in the tuple so that the first  $r_0$  weights are rationally independent and the remaining weights are zero. Their residues are algebraically independent. Resolve the supports of the first functions and the rational maps to  $\mathbb{P}^1$  defined by the others on a proper smooth model. At the centre, the latter functions are units with independent restrictions, so the codimension is at most  $r_0$ . At least  $r_0$  boundary components are needed to express the independent nonzero weights. Thus the centre is a generic stratum of codimension  $r_0$ , its boundary parameter weights are independent, and the valuation is the corresponding monomial valuation. Indeed different parameter exponents cannot tie, and the valuation of sufficiently high powers of the maximal ideal tends to infinity. The vertical exponent matrix of the first functions has nonzero determinant, and the horizontal residue differentials of the remaining functions are independent. Their log wedge is therefore a unit log volume at the stratum, proving (6.6). Applying the same calculation upstairs gives the finite-extension assertion, consistently with the ramification formula for form-discrepancies.

For (ii), complete  $f$  to a full homogeneous-independent tuple  $t$ . This is possible because the graded field of a quasi-monomial valuation has transcendence degree  $n$ , by the Abhyankar equality. Change the prescribed weights of  $f$ , keep the other tuple weights fixed, and use Gauss valuations on  $k_0(t)$ . The field  $L$  is finite over this rational field. For an algebraic element, its possible extension values are the negatives of Newton-polygon slopes of its minimal polynomial, with multiplicities. These form a finite multiset depending continuously on the coefficient values. One may see this without completeness by extending to an algebraic closure, factoring the polynomial, and using multiplicativity of Newton polygons.

Several specified values must be tracked by one extension. For a finite list of elements, choose integer powers so that the value of their product separates all distinct numerical tuples in the finite

Cartesian product of their candidate value sets at the starting point. Choose a root of the product's minimal polynomial with value tending to the prescribed product value. The resulting embedding of the product field into the algebraic closure extends to  $L$ . Under this single embedding every individual value belongs to its candidate set. Separation forces all those values to converge to the prescribed ones. Include in the list the coefficient of  $\eta$  in the full log frame, and the functions  $h_i$ . Formula (6.6) gives discrepancy convergence, and positivity of the  $h_i$  preserves the centre. This argument asserts no uniform size of the neighbourhood.

For (iii), at a generic stratum write numerator and denominator in the completed regular local ring with a coefficient field and the boundary parameters. The minimum of the occurring exponent weights is determined by finitely many exponents, by the Noetherian property of monomial ideals. This proves rational piecewise linearity. Setting some weights to zero gives the same evaluation at the containing stratum. More explicitly, monomial cutoff ideals in the remaining parameters are primary for the prime generated by those parameters, and localizing to that stratum and contracting leaves them unchanged. The same assertion is visible in the power-series completion. This proves continuity along every face.  $\square$

The following form-boundary version uses the transverse-divisor and toroidal-retraction argument of Jonsson–Mustață [24, Proposition 5.1, Lemma 5.3 and Corollary 5.4]. We include the argument for the precise strictness needed here.

**Lemma 6.3.** *Let  $(Q, D)$  be a smooth proper model with reduced SNC boundary containing  $\text{Supp div}_Q(\eta)$ . Retraction of a quasi-monomial valuation  $w$  to the boundary cone complex decreases  $A_\eta$  by  $A_{Q,D}(w) \geq 0$ , and the decrease is strict unless  $w$  already belongs to that complex.*

*Proof.* The discrepancy formula on a log-smooth model separates into the boundary-linear contribution and  $A_{Q,D}(w)$ . Retraction retains the former. For the equality statement, subdivide the orthant fan near the centre by rational hyperplanes respecting the smallest rational subspace containing the boundary-weight vector, and then take a regular toric refinement. Such subdivisions are realized locally by toroidal modifications: boundary equations extend to smooth, or étale, coordinates, so the construction pulls back from coordinate-axis charts. The reduced pair remains crepant.

The lifted centre lies on the open toric stratum specified by the relative-interior cone of the weight vector. Weight-zero chart monomials are units there. The positive normal weights are rationally independent, since their cone lies in the smallest rational subspace just chosen. If the centre is not the generic point of that stratum, choose, near its generic point, a smooth divisor through the centre transverse to the boundary. Adding it to the boundary subtracts a strictly positive value from the discrepancy, while the resulting smooth SNC pair is lc. Thus  $A_{Q,D}(w) > 0$ . If the centre is generic, independent positive parameter weights force the monomial valuation, as in Lemma 6.2.

The monomial valuations on the refined charts are precisely those on the subdivided original complex. Indeed the new normal parameters are toric monomials, and the weight-zero ratios of old boundary monomials restrict to characters of the torus stratum, with only the monomial lattice relations at its generic point. The monomial calculation therefore pulls back unchanged from the orthant charts. This identifies the equality case on the original complex.  $\square$

**6.3. A selected optimal vertex.** Fix centred regular eigenfunctions  $f = (f_1, \dots, f_r)$  and positive weights  $b = (b_1, \dots, b_r)$  which are linearly independent over  $\mathbb{Q}$ . Suppose a centred quasi-monomial extension assigning these weights exists. Independent weights make every such extension Gauss on  $f$ . We minimize

$$(6.7) \quad \Phi(w) = \frac{A_\eta(w)}{s(w)}$$

over these extensions.

**Proposition 6.4.** *The minimum is attained on a finite rational polyhedral subdivision of the SNC cone complex of a fixed smooth model. There are finitely many invariant regular functions, called detectors, such that maximizing the sum of their values among the minimizers permits the choice of a vertex. At this selected vertex,*

$$(6.8) \quad w(h) = m_h \cdot b \quad (h \in L^*), \quad A_\eta(w) = e \cdot b, \quad s(w) = g_s \cdot b,$$

where all coefficient vectors are rational and have bounded denominators for this fixed vertex. Its rational rank is  $r$ . For an active generator, meaning  $w(h_i) = s(w)v_i$ , one has  $m_{h_i} = v_i g_s$ .

For fixed germ and tuple, selection is described by finitely many piecewise semialgebraic branches as  $b$  varies. On a segment starting at independent weights, these branches have a finite partition into intervals on which the selected ratio is rational-smooth. At dependent weights the polyhedral prescription is initially relaxed, without an extra Gauss condition. A compact limit of bounded-ratio selected vertices from independent positive weights remains a centred quasi-monomial Gauss extension at the limiting weights.

*Proof.* Resolve the supports of  $h_i, f_j, \eta$  on a smooth proper model  $(Q, D)$ . Retraction preserves the values of these functions and  $s$ . It therefore remains in the prescribed fibre, and preserves the Gauss condition because  $b$  is independent. By Lemma 6.3, an attained minimizer must lie on the complex.

To construct detectors, take local rational parameter equations for the boundary components at the generic stratum of each cone. Each parameter is algebraic over  $L^G$ . Take a polynomial equation for it and include invariant regular numerators and denominators for all its nonzero coefficients. Such presentations exist: an invariant rational function acquires an invariant regular denominator by multiplying a denominator by its other group translates. Once the detector values are fixed, Newton root values allow only finitely many values for each parameter. Since there are finitely many cones, simultaneous detector fibres on the complex are finite.

Subdivide so that the evaluations in question and  $s$  are linear. On the closed cones cut out by  $w(h_i) \geq 0$ , every nonzero point has  $A_\eta > 0$ . To check this, first use rational rays, which are divisorial and have affine centres because the  $h_i$  include affine generators; klt gives positivity. Linearity on the finitely many rational cones then gives positivity and properness on each cone. For  $s > 0$ , (6.1) gives

$$(6.9) \quad s \leq \min_j \frac{b_j}{v(f_j)}.$$

Thus a bounded sublevel of  $\Phi$  has bounded  $A_\eta$ . A limit cannot have  $s = 0$ : boundedness of  $\Phi$  would force  $A_\eta = 0$ , hence the zero valuation, contrary to  $w(f) = b \neq 0$ . The sublevel is therefore compact, and both the minimum and the secondary maximum are attained.

On each cut polyhedron the minimum locus is the compact face  $A_\eta - \Phi_{\min}s = 0$ , disjoint from  $s = 0$ . Maximizing the linear detector sum on that face allows a vertex of the original cut polyhedron. Its coordinates solve rational linear equations with right side linear in  $b$ . Hence they are rational-linear in  $b$ . All function values are integral combinations of finitely many parameter weights at its stratum; this proves the common denominator assertion. Since the values of the  $f_j$  are the independent  $b_j$ , the rational rank is exactly  $r$ . The active generator identity follows by independence of  $b$ .

There are only finitely many vertex formulas and polyhedral feasibility conditions, including  $s > 0$ . Comparisons of their ratios and then detector sums are semialgebraic. Restriction to a segment therefore has a finite interval partition. On a valid branch the denominator is positive, so its ratio is rational-smooth. For the last assertion, take a convergent subsequence in the fixed finite complex. The compactness argument keeps  $s > 0$ . For every fixed polynomial  $P$  in the tuple, each

independent-weight valuation satisfies

$$w(P(f)) = \min_{\alpha: c_\alpha \neq 0} \alpha \cdot b, \quad P = \sum_{\alpha} c_\alpha X^\alpha.$$

Both sides are continuous on the complex by Lemma 6.2. The single limit valuation consequently satisfies this equality for every polynomial, even when monomial weights tie. No further subsequence depending on  $P$  is needed. It is therefore Gauss, and admits the approximate continuation of Lemma 6.2.  $\square$

## 7. CARTIER SELECTION ON THE LEADING FIELD

Fix one germ, an independent assignment  $b$ , and the selected vertex  $w$  of Proposition 6.4. Write  $\phi = \Phi(w) > 0$  and retain the vectors  $e, g_s, m_h$  of (6.8). The constructions in this section are made for these fixed characteristic-zero data. Reduction characteristics will be arbitrarily large, but their required size is not asserted to be uniform among germs or choices of tuples.

Fix any finite list of characteristic-zero regular eigenfunctions  $t_\nu$  whose  $w$ -initials are algebraic over the graded tuple field of  $f$ , and any finite list of real cutoffs  $T > 0$ . In the reductions constructed below,  $w$  denotes the valuation obtained by keeping the parameter weights on a spread SNC chart and restricting to the reduced germ. Write  $w(a) = m_a \cdot b$  there; independence of  $b$  makes the rational vector  $m_a$  unique. The finite tracked values and characters are preserved. We write  $w_{\text{orig}}$  when distinguishing the characteristic-zero valuation.

**Proposition 7.1.** *There is a spread of the algebraic data, retaining the given weights and cutoffs, such that, for all sufficiently large reductions of characteristic  $p$ , there is a single product  $H$  of regular eigenfunctions with*

$$\sigma = m_H = p\phi g_s + O(1).$$

*The error depends on the fixed characteristic-zero data but not on  $p$ . For this  $H$  and every vector of nonnegative integers  $(j_\nu)$ , there are a regular nonzero eigenfunction  $R$  and integers  $0 \leq k_i < p$  with*

$$(7.1) \quad pm_R = \sigma + \sum_{\nu} j_\nu m_{t_\nu} + k - (p-1)e,$$

$$(7.2) \quad p\chi_R = \chi_H + \sum_{\nu} j_\nu \chi_{t_\nu} + \chi_{f^k} - (p-1)\chi_\eta.$$

*Characters are written additively. For every preassigned cutoff  $T$ , any collection of these reduced regular eigenfunctions with distinct pairs (value, character) and weights at most  $s(w_{\text{orig}})T$  has size at most the characteristic-zero number  $\mathcal{H}(T)$  of Lemma 6.1.*

We first adjoin roots of the tuple and work on the residue field of an extended valuation. Optimality gives a positivity inequality for its residual form, and logarithmic Cartier turns that inequality into a nonzero leading coefficient. The same coefficient will work for every power vector  $(j_\nu)$ .

**7.1. The residue field and its form.** Choose a positive integer  $N_0$  such that the coefficient lattice of the value group is contained in  $\frac{1}{N_0}\mathbb{Z}^r$ . Adjoin roots  $x_j^{N_0} = f_j$ , and extend  $w$  to  $\tilde{w}$  on  $\tilde{L} = L(x_1, \dots, x_r)$ . Its coefficient lattice is exactly  $\frac{1}{N_0}\mathbb{Z}^r$ . Put

$$(7.3) \quad K_0 = \kappa(\tilde{w}), \quad q = n - r, \quad a' = \text{res}_{\tilde{w}}(ax^{-N_0 m_a}) \quad (a \in \tilde{L}^*).$$

The field  $K_0$  is finitely generated of transcendence degree  $q$  over  $k_0$ , and is generated by the  $a'$  for regular eigenfunctions  $a \in L^*$ .

Here are the field-theoretic details, which also account for the possible failure of  $w$  to be  $G$ -invariant. Let a finite abelian group act on a field with a nontrivial real valuation, and let  $D$  be its decomposition subgroup. Distinct orbit valuations are inequivalent: a positive proportionality

factor between valuations in a finite orbit must be one. Given a  $D$ -invariant homogeneous initial  $\xi$  of value  $\gamma$ , take a lift  $a$ . By weak approximation for the finitely many inequivalent rank-one valuations, choose  $h$  with

$$w(h - 1) > 0, \quad u(h) > \gamma - u(a) \quad \text{at every other orbit valuation } u.$$

Then  $ha$  has initial  $\xi$  at  $w$  and value greater than  $\gamma$  at all the other orbit valuations. Averaging  $ha$  over  $D$  preserves both assertions, since  $D$  permutes those other valuations. Taking the sum over cosets of  $D$  gives a group-invariant element with initial  $\xi$ . Equivalently one uses the full group sum divided by  $|D|$ . These strict inequalities work equally when  $\gamma < 0$ .

Every homogeneous initial splits into  $D$ -eigenvectors. Characters of a subgroup of a finite abelian group extend to the group. For the Kummer extension  $\tilde{L}/L$ , they are realized by monomials in  $x$ ; divide an eigen-initial by the corresponding monomial initial and apply the invariant lifting just proved. Thus the graded field upstairs is generated by those monomials and initials from  $L$ . This proves that its value lattice is exactly  $\frac{1}{N_0}\mathbb{Z}^r$ . Applying the same argument to  $L/L^G$ , rational global eigenfunctions supply all subgroup characters. Their existence follows, after quotienting by the kernel of the action, from a normal basis for this finite Galois extension. A rational eigenfunction is a quotient of regular eigenfunctions: multiply a regular denominator by its group translates to obtain an invariant regular denominator. Normalization by  $x$  now proves the claimed generation of  $K_0$ . The finite extension is still Abhyankar, because graded independence survives finite extension. The stratum description in Lemma 6.2 identifies its residue with a stratum function field, proving finite generation.

Choose valuation units  $y_1, \dots, y_q \in \tilde{L}$  whose residues  $\bar{y}$  form a transcendence basis of  $K_0/k_0$ . Then  $x, y$  are a full Gauss tuple. Write the pulled-back form as

$$(7.4) \quad \begin{aligned} \eta &= a \, d \log x_1 \wedge \cdots \wedge d \log x_r \wedge d \log y_1 \wedge \cdots \wedge d \log y_q, \\ \rho &= a' \, d \log \bar{y}_1 \wedge \cdots \wedge d \log \bar{y}_q. \end{aligned}$$

By (6.6),  $m_a = e$ . The nonzero rational volume form  $\rho$  is independent of the chosen unit lifts and residual basis. To check this, resolve both choices on a common proper smooth model. At the generic centre stratum of codimension  $r$ , the vertical log exponent matrix of  $x$  is invertible. The residue of the horizontal determinant changing the log frame is exactly the determinant changing the restricted differentials on the stratum. The ratio of the two frames is a unit with this residue, while normalization by  $x^{-N_0 e}$  is the same in both. This proves the assertion and permits resolving any additional finite list of rational supports. If  $q = 0$ , the wedge in  $\bar{y}$  is 1 and  $\rho \in k_0^*$ .

**7.2. Strict positivity over an affine residue model.** Choose a finitely generated  $k_0$ -algebra  $A_0 \subset K_0$ , with fraction field  $K_0$ , generated by normalized initials of regular eigenfunctions. Include the normalized initials of every  $h_i$  and every detector. Let  $J$  be the nonempty set of active generators, and, for a divisorial valuation  $z$  of  $K_0/k_0$ , put

$$(7.5) \quad l(z) = - \min_{i \in J} \frac{z(h'_i)}{v_i}.$$

**Lemma 7.2.** *For every divisorial  $z$  one has*

$$(7.6) \quad A_\rho(z) + \phi l(z) \geq 0.$$

*The inequality is strict when  $z$  is nonnegative on  $A_0$ .*

*Proof.* Both assertions are homogeneous in  $z$ , so we may take  $z$  primitive. Consider the lexicographic composite valuation  $(\tilde{w}, z)$ , whose second value on  $a \in \tilde{L}^*$  is  $z(a')$ . Adapt the residual tuple so that  $\bar{y}_1$  is a parameter for  $z$  and the other  $\bar{y}_j$  are units with independent residues at its generic centre. The tuple  $x, y$  is Gauss for its two rows of lexicographic weights. On its rational function field replace those rows by the first row plus  $t > 0$  times the second row.

We require extensions which track exactly, and simultaneously, the projected lexicographic values of finitely many elements. For each test element, take its polynomial over the full tuple field. For sufficiently small  $t$ , each coefficient value is the projection of its lexicographic value. Pairwise equalities of term values give a finite set of lexicographic candidates for the element's value. Choose integer powers of the test elements so that their product value separates all distinct tuples in the Cartesian product of these finite candidate sets; include all candidates, whether or not they are lowest values. Its actual lexicographic value is a Newton root value of its minimal polynomial. The finitely many comparisons determining the Newton polygon are preserved after projection for all sufficiently small  $t$ . Hence the projected product value is the value of a root over an algebraic closure of the projected valued tuple field. This uses the Newton-polygon fact that a value where the minimum of the term values is attained at least twice is a root value for a polynomial with nonzero constant and leading coefficients, as follows by factoring over a valued algebraic closure. The embedding of the product field sending it to this root extends to the whole finite extension  $\tilde{L}$ . The values of all test elements under that one embedding belong to their projected candidate sets. Distinct separating product values remain distinct for small  $t$ , so this forces the entire desired tuple of values.

Include the coefficient of  $\eta$  in the adapted log frame, all  $h_i$ , and every detector. The parameter  $y_1$  already has prescribed weight  $t$  on the tuple field; it can also be included in the list. The resulting extensions, and their restrictions  $w_t$  to  $L$ , are quasi-monomial. Indeed full graded transcendence degree is preserved by finite extension and restriction, and the full-tuple argument of Lemma 6.2 applies. Discrepancies of pulled-back forms agree with those on restriction. Exact tracking therefore gives, for all sufficiently small  $t > 0$ ,

$$(7.7) \quad A_\eta(w_t) = A_\eta(w) + tA_\rho(z), \quad s(w_t) = s(w) - tl(z).$$

In the second equality inactive generators remain inactive for small  $t$ . In the first, the coefficient in the adapted residual log frame has order  $A_\rho(z)$ . The  $h_i$  retain positive values, so  $w_t$  stays centred; the  $f_j = x_j^{N_0}$  retain their weights  $b_j$ . Minimality of  $\phi$  now gives (7.6).

Suppose equality holds and  $z \geq 0$  on  $A_0$ . Equations (7.7) imply  $\Phi(w_t) = \phi$ . Strict retraction places every  $w_t$  on the original cone complex. All detector values are nondecreasing, since their normalized initials lie in  $A_0$ . Their sum was maximal, so every detector value is constant. The joint detector fibre is finite. Each of its valuations on  $L$  has only finitely many extensions to the fixed finite field  $\tilde{L}$ . This contradicts the actual variation  $\tilde{w}_t(y_1) = t$ . Thus equality cannot hold over the affine part. When  $q = 0$  there are no divisorial  $z$ , and both assertions are vacuous.  $\square$

**7.3. A projective model with controlled top sections.** Choose a positive integer  $M$  with  $M/v_i \in \mathbb{Z}_{>0}$  for all  $i \in J$ , and set  $u_i = (h'_i)^{M/v_i}$ . Resolve the rational map defined by these functions on a smooth projective model. They are generating sections of a Cartier divisor  $D_0$  in its chosen rational trivialization, and, after pullback,

$$(7.8) \quad z(D_0) = -\min_{i \in J} z(u_i) = Ml(z).$$

Neither the functions in this trivialization nor  $D_0$  need be effective.

We can choose such a smooth projective model  $Z$ , a reduced SNC divisor  $B_Z$ , and an ample integral divisor  $P$  supported on  $B_Z$  with the following properties. The model maps to a normal compactification of the normalization of  $\text{Spec } A_0$ ; the divisor  $B_Z$  contains the supports of  $D_0$  and  $\text{div}(\rho)$ ; and

$$(7.9) \quad P_T \geq \max\{0, (D_0)_T\} \quad \text{for every component } T \text{ over infinity,}$$

$$(7.10) \quad \beta/\rho \in A_0 \quad \text{for } \beta \in H^0(Z, \Omega_Z^q(\log B_Z)(P)),$$

$$(7.11) \quad H^a(Z, \Omega_Z^i(\log B_Z)(P + jD_0)) = 0 \quad \text{for } a > 0, 0 \leq i \leq q, -|J| \leq j \leq 0.$$

We spell out the construction because negative coefficients of  $P$  on the finite part are useful. Normalize a projective closure of  $\text{Spec } A_0$ , giving ample effective Cartier infinity. Let  $Z_0$  be a smooth projective model resolving the map by the  $u_i$ , with an effective ample divisor  $P_0$ . Choose a nonzero conductor  $c_0 \in A_0$  from its affine normalization into  $A_0$ . Choose  $0 \neq g \in A_0$  vanishing on every finite height-one prime of that normalization lying under the supports of  $D_0, P_0, \text{div}(\rho)$ , or  $\text{div}(c_0)$ . Resolve these supports, those of  $g$ , and the pullback of infinity by a sequence of blowups  $\mu : Z \rightarrow Z_0$ . Include the exceptional divisors in  $B_Z$  and retain the notation  $D_0$  for its pullback. Choose an effective exceptional divisor  $F$  with  $-F$  relatively ample. For  $m$  large,  $m\mu^*P_0 - F$  is ample. For  $c \gg m$  and then  $h \gg c$ , take a sufficiently large positive multiple of

$$(7.12) \quad m\mu^*P_0 - F - c\text{div}(g) + hH_\infty.$$

The principal term does not change ampleness, and the infinity term is nef as a pullback of an ample divisor. Its coefficients can dominate all required infinity coefficients. At the generic points of finite height-one primes on the normal affine base the birational map is an isomorphism. There the coefficients of  $P$  are nonpositive and are as negative as needed at the finitely many bad primes and finite primes in  $B_Z$ . A top log section  $\beta$  therefore has  $\beta/(\rho c_0)$  regular in codimension one on the affine normalization. Normality and the conductor give (7.10). Finally, taking a large multiple in (7.12), Serre vanishing gives all the finitely many conditions (7.11). For  $q = 0$ , take  $Z$  to be a point; the divisors and supports are empty, and the same assertions hold.

**7.4. Logarithmic Cartier nonvanishing.** Spread this finite collection of data over a finite-type integral subring of  $k_0$ , and take geometric fibres in arbitrarily large positive characteristics  $p$ . We keep the same notation on these fibres. After shrinking the spread, smoothness, the relative SNC condition, global generation by the  $u_i$ , ampleness, and the finite cohomological vanishings persist. Spread also a basis of  $H^0(Z, \Omega_Z^q(\log B_Z)(P))$  and polynomial expressions for its ratios to  $\rho$  in the chosen generators  $a'_\lambda$  of  $A_0$ . Cohomology and base change make this a basis after reduction. The parameter base can be chosen smooth over an open of  $\text{Spec } \mathbb{Z}$ . Its geometric points lift to length-two Witt vectors by smoothness, so the reduced smooth SNC pair has a length-two Witt lift.

The finite range of vanishings suffices for every nonnegative twist. To see this explicitly, let  $E = \Omega_Z^i(\log B_Z)(P)$  and  $s_J = |J|$ . The exact generating Koszul complex of the  $u_i$  ends, after twisting, in

$$0 \longrightarrow E((d - s_J)D_0) \longrightarrow \cdots \longrightarrow E((d - 1)D_0)^{\oplus s_J} \longrightarrow E(dD_0) \longrightarrow 0,$$

with the intervening exterior-power multiplicities understood. For  $d \geq 1$  every preceding shift lies between  $-s_J$  and  $d - 1$ . Starting from (7.11), induction makes them all acyclic in positive degree. Break the resolution into short exact sequences of successive images, beginning at the left. The long exact sequences show that every image is acyclic. The final kernel has  $H^1 = 0$ , so the last map is surjective on global sections and the last sheaf is acyclic. Iteration gives

$$(7.13) \quad H^a(Z, E(dD_0)) = 0 \ (a > 0, d \geq 0), \quad H^0(Z, E(dD_0)) = \sum_{|\alpha|=d} u^\alpha H^0(Z, E).$$

This concerns the globally generated line bundle  $\mathcal{O}_Z(D_0)$ ; it makes no effectiveness assumption on its rational trivialization.

Put

$$(7.14) \quad k_p = \lfloor p\phi/M \rfloor, \quad H_p = k_p D_0 + P, \quad A_p = \lfloor H_p/p \rfloor.$$

For all sufficiently large  $p$ ,

$$(7.15) \quad 0 \neq \rho \in H^0(Z, \Omega_Z^q(\log B_Z)(A_p)).$$

At a boundary prime  $T$ , membership is equivalent to  $A_\rho(T) + (H_p)_T/p \geq 0$ , since  $A_\rho(T)$  is integral. Writing  $a_p = p\phi/M - k_p \in [0, 1)$ , this expression is

$$A_\rho(T) + \frac{\phi}{M}(D_0)_T + \frac{P_T - a_p(D_0)_T}{p}.$$

On the finite part the limiting expression is strictly positive by Lemma 7.2; the fixed, possibly negative, coefficient of  $P$  is harmless for large  $p$ . At infinity the limiting expression is nonnegative and the error is nonnegative by (7.9). Only finitely many coefficients are being checked, and their orders have been preserved in the spread.

There is a global top section in pole twist  $H_p$  whose Cartier image on the function field is nonzero. Here is the cohomological argument, including the effect of signs in  $H_p$ . Use relative Frobenius and suppress the perfect-field twists of its target. The meromorphic log de Rham complexes have an inclusion

$$(7.16) \quad \Omega_Z^\bullet(\log B_Z)(pA_p) \longrightarrow \Omega_Z^\bullet(\log B_Z)(H_p),$$

with the ordinary meromorphic differential. After Frobenius pushforward it is a quasi-isomorphism. In an étale chart adapted to the log axes, write an axis coefficient of  $H_p$  as  $h = pa + r$ ,  $0 \leq r < p$ , also when  $h < 0$ . The smaller coefficient module allows exponents at least  $-pa$ , and the larger one allows exponents at least  $-h$ . The extra exponents have nonzero residue modulo  $p$ . Split the Frobenius module by these residues. A summand with nonzero residue in a log direction is contracted by the log Euler field divided by that residue. The residue-zero summands coincide. With several axes use any direction of nonzero residue; ordinary coordinates do not affect the contraction. Relative Frobenius commutes with étale charts, proving the claim.

By the projection formula, the pushed-forward smaller complex is the Frobenius log de Rham complex tensored with  $\mathcal{O}_Z(A_p)$ . For  $p > q$ , logarithmic Deligne–Illusie splitting [16, Section 4.2.3], applied to the smooth SNC length-two lift, gives a surjection from its degree- $q$  hypercohomology to  $H^0(Z, \Omega_Z^q(\log B_Z)(A_p))$  by Cartier. For clarity, the logarithmic splitting can be constructed in the same way as the usual one. Local lifted Frobenius maps carrying each axis to its  $p$ th power times a unit give divided pullbacks of closed one-forms, including divided pullback on  $d \log$  of an axis. These lift inverse Cartier. On overlaps, divided differences of the lifted maps give homotopies; for a log axis use the ratio of the lifted pullbacks minus one, divided by  $p$ . The ratio is a unit congruent to one. Difference and ratio identities make the homotopies additive on triple overlaps. They define the degree-one map in the Čech resolution of the pushed-forward complex. Exterior products and antisymmetrization then give the maps in all degrees, since  $p > q$  makes the factorials invertible. The coordinate Cartier formula identifies their cohomology maps with inverse Cartier. Together with Frobenius in degree zero they give the required derived splitting. The pushed-forward form bundles are flat because relative Frobenius of a smooth variety is finite flat. Tensoring the splitting with  $\mathcal{O}_Z(A_p)$  requires no lift of this last divisor.

By (7.13), every term of the larger complex in (7.16) has vanishing positive sheaf cohomology. Its degree- $q$  hypercohomology is consequently represented by global top sections. Choose a class mapping to the nonzero section  $\rho$  in (7.15) and represent it by such a top section. At the generic point the edge map is ordinary field Cartier  $C$ , in the sense of Cartier’s original operation and its logarithmic coordinate formulation [15][25, Theorem 7.2 and its proof]. Since  $\rho$  is nonzero generically, the representative has nonzero field Cartier image.

Expand that representative using (7.13) and the fixed polynomial expressions for the top-section ratios. Cartier is additive and satisfies  $C(c\omega) = c^{1/p}C(\omega)$  over the perfect constant field. Thus at least one rational monomial term has nonzero image:

$$(7.17) \quad U_p = u^\alpha \prod_{\lambda} (a'_\lambda)^{I_\lambda}, \quad |\alpha| = k_p, \quad C(U_p \rho) \neq 0.$$

The multi-index  $I$  belongs to a fixed finite set. The individual monomial need not separately be a global section; only its field Cartier image is used. Replace  $u_i$  by  $h_i^{M/v_i}$  and  $a'_\lambda$  by  $a_\lambda$ . This gives a product  $H$ , depending on  $p$ , of regular eigenfunctions on the reduced germ, with

$$(7.18) \quad \sigma := m_H = k_p M g_s + O(1) = p \phi g_s + O(1).$$

The implied constants depend on the fixed characteristic-zero data but not on  $p$ . In dimension  $q = 0$ ,  $Z$  is a point and Cartier in degree zero is inverse Frobenius on a nonzero scalar; the same selection and formula apply.

**7.5. Spreading the initials and proving the output.** The test functions  $t_\nu$  fixed at the start of the section have normalized initials satisfying

$$(7.19) \quad t'_\nu \in k_0^*.$$

Indeed after adjoining the initial roots  $x$ , they have degree zero and are algebraic over the graded field of independent variable weights with degree-zero field  $k_0$ . A homogeneous relation makes them algebraic over  $k_0$ , which is algebraically closed.

All these data can be spread simultaneously with the valuation calculation. Resolve on  $\tilde{L}$  at  $\tilde{w}$  the supports of  $x, y$ , of the normalized form coefficient, and of normalized representatives for all initials used. Include the  $h_i, a_\lambda, t_\nu$ , the detectors, and any further prescribed nonzero functions, in particular the ideal generators below. At the resulting codimension- $r$  stratum the parameter weights are independent and generate  $\frac{1}{N_0} \mathbb{Z}^r$ . Hence the exponent matrix of  $x$  in these parameters is unimodular. Normalized representatives of order vector zero are actual units, and restriction computes their residues. Spread these identities on open charts, the identification of the stratum field with  $k(Z)$ , a separating residual tuple  $\bar{y}$ , and the inclusions in the Kummer diagram. Retain geometrically integral strata and models where they were so in characteristic zero. After shrinking, the finite field extensions remain separable. The same real parameter weights define monomial valuations in every such reduced chart; their restriction to the reduced germ is again denoted by  $w$ . The coefficient lattice, the finite tracked values, and the normalized residues are unchanged. In particular (7.19) remains a nonzero constant identity, and the centred generators ensure centre at the reduced closed point. Spread also the group action, its characters with identified roots of unity, normality, and the canonical generator on the smooth locus with complement of codimension at least two. Exclude primes dividing  $N_0|G|$ . Every construction here involves only finitely many data over a finitely generated subring of  $k_0$ .

*Proof of Proposition 7.1.* Set  $Q_p = H \prod_\nu t_\nu^{j_\nu} f^k$ . Choose  $k$  in the indicated box so that  $m_{Q_p} + e \in p(\frac{1}{N_0} \mathbb{Z}^r)$ . This is possible because  $p \nmid N_0$ . Define

$$R = \frac{C(Q_p \eta)}{\eta}.$$

Ordinary Cartier sends regular top forms to regular top forms on the smooth locus. There  $\eta$  is a canonical generator, so  $R$  is regular. Normality extends it across the complement of codimension at least two. Once nonvanishing is known, semilinearity and compatibility with the group action give (7.2).

Compute the value upstairs, where Cartier commutes with separable pullback, as follows also from its field  $p$ -basis formula. The tuple  $x, y$  is an absolute  $p$ -basis over the perfect constants. Indeed its monomials with exponents in  $[0, p)$  are independent over  $p$ th powers already in the graded field: use the basis of the value lattice for  $x$  and the separating residual  $p$ -basis for  $\bar{y}$ . There are  $p^n$  of them, so they form a field basis. In the log frame of (7.4), write the coefficient of  $Q_p \eta$  as

$$c = \sum_{I, J} c_{IJ}^p x^I y^J, \quad 0 \leq I_i, J_j < p.$$

Lowest terms cannot cancel by that graded independence. Since  $m_c = m_{Q_p} + e$  lies in  $p(\frac{1}{N_0}\mathbb{Z}^r)$ , lowest terms have  $I = 0$ . Normalize by  $x^{-N_0 m_c}$ , a  $p$ th power. The residue times  $d \log \bar{y}$  is

$$U_p \prod_{\nu} (t'_{\nu})^{j_{\nu}} \rho.$$

Its residual Cartier image is nonzero by (7.17) and (7.19). Thus the term with  $I = J = 0$  occurs at the lowest value, and  $m_{c_{00}} = m_c/p$ . Log-frame Cartier extracts precisely  $c_{00}$  times that frame. Division by  $\eta$  gives  $m_R = (m_{Q_p} + e)/p - e$ , proving (7.1) and nonvanishing. The selected  $H$  depends on  $p$  but is unchanged for every power vector: multiplication by any product of the nonzero constants  $t'_{\nu}$  cannot kill its residual Cartier image. Thus no infinite list of products needs to be spread.

For the final assertion, fix a cutoff  $T$  in characteristic zero. The ideal  $I_T = (v > T)$  is primary to the maximal ideal, since sufficiently high powers of its finite set of generators have  $v$ -value greater than  $T$ . Choose finite generators of  $I_T$ , identities expressing containment of a high maximal-ideal power, and finite identities for  $G$ -stability. Spread these, keeping any local denominators invertible. After shrinking, the finite quotient has constant length  $\mathcal{H}(T)$ . Track the chosen generators in the monomial chart just described. By (6.1), each has characteristic-zero value strictly greater than  $s(w_{\text{orig}})T$ , and its value is unchanged on the reduction. Nonnegativity on the reduced local ring therefore gives

$$(7.20) \quad I_{T,p} \subset (w > s(w_{\text{orig}})T).$$

There is no assertion that the whole valuation  $v$  is spread. If reduced eigenfunctions with distinct (value, character) pairs of weights at most this cutoff had a dependence modulo  $I_{T,p}$ , split it into characters using stability and  $p \nmid |G|$ . Within a character the distinct lowest value cannot cancel, contradicting (7.20). Their number is therefore at most the unchanged quotient length.  $\square$

The order of choices in Proposition 7.1 will be used repeatedly: first fix the germ, its selected vertex, the finite characteristic-zero test list, and all cutoffs needed for that application; then spread those data and let  $p$  be arbitrarily large. The estimates obtained from Lemma 6.1 may be uniform along the original sequence even though the required reduction characteristic is not.

## 8. COMPLETING TUPLES AND ROUNDING THEIR PHASES

Return to the sequence of germs, with the tier decomposition of Proposition 5.1. Degrees within a tier have uniformly bounded ratios; between consecutive tiers their ratios tend to infinity. Let  $m$  be the end of a tier and  $m_-$  the end of the preceding tier, with  $m_- = 0$  for the first one. Write  $d = d_m$ . Consider a tuple and an independent positive weight assignment as in Proposition 6.4, satisfying

$$(8.1) \quad \max\{1, m_-\} \leq r \leq m, \quad f_1 = f_1^0, \quad \frac{1}{2} \leq \frac{b_1}{d_1} \leq 2, \quad b_j \leq B_* d_j.$$

Assume it has a quasi-monomial Gauss extension of ratio at most  $M_*$ . The bounds  $B_*, M_*$  are prescribed before this application. For its selected optimal vertex, (6.1) gives

$$(8.2) \quad c \leq s(w) \leq 2, \quad A_{\eta}(w) \leq 2M_*, \quad c = \frac{1}{4nM_*}.$$

Indeed the upper bound for  $s$  follows by testing  $f_1$ , while  $b_1 \leq 2nA_{\eta}(w)d_1 \leq 2nM_*s(w)d_1$  gives the lower bound.

For reference, the notation and the stages of its use are as follows.

| Notation                            | Role and scope                                                                                                                                               |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $v, f^0, d_i$                       | Reference valuation and degree scales from (6.2); the stated lower, first-degree and product bounds do not bound every $d_i$ above.                          |
| $f, b, w, b^*$                      | Independent input weights and an optimal Gauss extension at selection; the exact target $b^*$ may be dependent and is reached by continuation, as in (8.13). |
| $s, \Phi$                           | The comparison factor (6.1) and the ratio (6.7), both depending on the valuation.                                                                            |
| $m_h, e, g_s$                       | Rational coefficient vectors at a selected branch, defined by (6.8); their denominator bound there belongs to that fixed vertex.                             |
| $B_*, M_*; S, D_{\text{fin}}, \ell$ | Prescribed numerical budgets, then the common rounding and clearing integers chosen below; see (8.17) and (8.20).                                            |

The denominator of a fixed selected vertex and the required reduction characteristic may depend on the fixed germ, finite tests and cutoffs; no uniform bound on their complexity or number of Newton breakpoints is asserted. The budgets and the later  $S, D_{\text{fin}}, \ell$  have exactly the uniformity along the retained subsequence proved below, whereas the final cutoff  $T$  may depend on the function and germ.

In the following estimates, constants depend only on the prescribed bounds, dimension, tier comparability, and the constants in Proposition 4.1 and Lemma 6.1. For every fixed cutoff multiplier  $K \geq 1$  occurring below, omit a finite initial part of the sequence so that  $Kd$  is below the next tier. With

$$(8.3) \quad \mathcal{D}_j = \prod_{i \leq j} \frac{d}{d_i},$$

Lemma 6.1 and comparability within the current tier give

$$(8.4) \quad \mathcal{H}(Kd) \leq C_W K^m \mathcal{D}_m, \quad \mathcal{D}_m \leq C_2 \mathcal{D}_r.$$

When  $m = n$ , no restriction from a next tier is needed.

**8.1. Three uniform counting estimates.** Suppose the tests in Proposition 7.1 have powers with

$$(8.5) \quad \sum_{\nu} \frac{j_{\nu}}{p} w(t_{\nu}) \leq d.$$

For sufficiently large reductions its output satisfies

$$(8.6) \quad w(R) \leq (2 + rB_*)d.$$

In fact (7.18) and (7.1) have leading contribution  $(\phi g_s - e) \cdot b = 0$ ; the fixed-data error divided by  $p$  tends to zero, and  $0 \leq k_i/p < 1$ . The box of functions

$$(8.7) \quad Rf^{\alpha}, \quad 0 \leq \alpha_i \leq d/d_i, \quad \alpha_i \in \mathbb{Z},$$

has weight at most  $(2 + 2rB_*)d$ . Fix a uniform  $K$  large enough that this is at most  $s(w)Kd$ , using (8.2). There are at least  $\mathcal{D}_r$  distinct exponent-character pairs in the box, since  $b$  is independent. Each product is a regular eigenfunction. The cutoff inclusion (7.20) shows that products with distinct (value, character) pairs have linearly independent classes modulo  $I_{Kd,p}$ : split a putative dependence into characters; within one character its unique lowest value is at most  $sKd$ , contrary to the inclusion. Thus their number is at most  $\mathcal{H}(Kd)$ . By (8.4), at most a uniform number  $C_3$  of such boxes can be pairwise disjoint in their exponent-character pairs. All cutoffs invoked here are fixed before taking the reduction characteristic large.

**Lemma 8.1.** *For regular eigenfunctions  $t$  whose  $w$ -initial is algebraic over the graded tuple field of  $f$ , put*

$$\Lambda_f = \langle (\mathbf{e}_i, \chi_{f_i}) : 1 \leq i \leq r \rangle_{\mathbb{Z}} \subset \mathbb{Q}^r \times \text{Hom}(G, k_0^*),$$

where  $\mathbf{e}_i = m_{f_i}$  is the  $i$ th standard vector. Then

$$(8.8) \quad \#\{(m_t, \chi_t) \bmod \Lambda_f\} \leq C_3,$$

$$(8.9) \quad |m_{t,i}| \leq C_4 \frac{w(t)}{d_i} \quad (1 \leq i \leq r).$$

*Proof.* For (8.8), take any finite list of functions representing distinct input classes. Use them as the tests in Proposition 7.1. For each test take its power one and all other powers zero, using the same  $H$  for these outputs. For arbitrarily large  $p$ , (8.5) holds. If two output classes were equal, subtracting (7.1) and (7.2) would show that the corresponding input classes are equal: the  $k$  terms belong to  $\Lambda_f$ , and multiplying an element of  $\Lambda_f$  by  $p$  remains in  $\Lambda_f$ . No division by  $p$  in a character group is used. Thus these outputs give disjoint boxes (8.7), and the size of every finite list is at most  $C_3$ .

If  $w(t) = 0$ , independence of  $b$  gives  $m_t = 0$ . Otherwise apply the same proposition to the powers of this one test with  $0 \leq j/p \leq d/w(t)$ . Its output's  $i$ th exponent coordinate is a common translate plus  $(j/p)m_{t,i}$  and an error in  $[0, 1)$ . The common  $o(1)$  error from (7.18) is independent of  $j$ . If  $|m_{t,i}|d/w(t)$  exceeded a sufficiently large fixed multiple of  $d/d_i$ , choose more than  $C_3$  evenly spaced values of  $j/p$  in this interval. For large  $p$  they can be rounded to the  $1/p$  grid with arbitrarily small additional error. Since  $d/d_i \geq 1$ , their output coordinates can be made more than  $d/d_i$  apart. The corresponding boxes are disjoint, contradicting their uniform bound. This proves (8.9).  $\square$

**Lemma 8.2.** *If  $r < m$ , there is a centred regular eigenfunction  $t$  with  $w(t) \leq C_5 d$  whose initial is transcendental over the graded tuple field. Adjoining  $t$  preserves the Gauss property at  $w$ . The enlarged tuple can then be continued to arbitrarily close independent assignments, with ratio increased by at most one.*

*Proof.* Suppose all regular eigenfunctions of  $w$ -weight at most  $L_* d$  have algebraic initial. At any fixed pair  $(m_t, \chi_t)$  the initials span at most one dimension: after adjoining  $x$  and normalizing, (7.19) puts them in  $k_0$ . By (8.8) there are at most  $C_3$  classes, and within one class the shift in  $\Lambda_f$  is determined by its integral exponent vector. By (8.9), the number of possible pairs is at most

$$(8.10) \quad C(1 + L_*)^r \mathcal{D}_r.$$

Now use the original tuple, not the changing optimized one. For a fixed large  $D \geq 1$ , consider the span of

$$(f_1^0)^{\alpha_1} \cdots (f_m^0)^{\alpha_m}, \quad 0 \leq \alpha_j \leq Dd/d_j.$$

Its dimension is at least  $D^m \mathcal{D}_m$ . Every nonzero combination has  $v$ -value at most  $mDd$ , because the original initials are independent. By (6.1) and (8.2), its  $w$ -value is at most  $C'Dd$ . Decompose this finite-dimensional space into character subspaces and filter each separately by  $w$ . This does not require the  $w$ -filtration itself to be group-invariant. A valuation filtration on a finite-dimensional vector space has a basis adapted to its finitely many occurring values, so the dimensions of its initial spaces sum to its dimension. If all these eigenfunction initials were algebraic, (8.10) with  $L_* = C'D$  would bound that dimension by  $C(1 + C'D)^r \mathcal{D}_r$ . Since  $r < m$  and  $\mathcal{D}_m \geq \mathcal{D}_r$ , a sufficiently large fixed  $D$  makes this impossible. This supplies  $t$  with the asserted uniform bound.

The function is a nonunit, because the initial of a unit centred at the closed point is a nonzero scalar. Its transcendental initial enlarges the Gauss tuple. The degree  $d_{r+1}$  belongs to the current tier, so  $d/d_{r+1}$  is uniformly bounded. Finally apply Lemma 6.2, retaining the  $h_i$ , to move to independent weights as close as desired. Discrepancy and  $s > 0$  are continuous in that continuation, so one can reserve at most one unit of increase in  $\Phi$ .  $\square$

**Lemma 8.3.** *When  $r = m$ , the selected vertex satisfies*

$$(8.11) \quad \left| \frac{(e - \phi g_s)_i}{s} \right| \leq C_6 \frac{d}{d_i} \quad (1 \leq i \leq m).$$

Consequently, on a selected ratio branch along a segment of weights, at its independent parameters,

$$(8.12) \quad \left| \frac{d\Phi}{dt} \right| \leq C_6 \sum_{i \leq m} \frac{d}{d_i} |\dot{b}_i|.$$

*Proof.* Apply Proposition 7.1 with no  $t_\nu$ . Its output has

$$m_R = \phi g_s - e + k/p + o(1), \quad w(R) \leq (2 + rB_*)d.$$

Fix a sufficiently large integer  $D$  and consider, in the reduction, all boxes

$$R^j f^\alpha, \quad 0 \leq j \leq D, \quad 0 \leq \alpha_i \leq Dd/d_i.$$

Their weights are at most  $sKd$  after increasing the uniform  $K$ . If  $|m_{R,i}| > Dd/d_i$  for some  $i$ , the boxes have disjoint exponent coordinates in that direction. Within each box the independence of  $b$  gives distinct values. There would be at least  $D^{m+1}\mathcal{D}_m$  distinct values. The same cutoff-inclusion argument applies to the regular eigenfunction products  $R^j f^\alpha$ , and bounds their number by  $C_W K^m D^m \mathcal{D}_m$ . Choose  $D > C_W K^m$ , fix this cutoff and all data, and then let  $p$  be arbitrarily large. It follows that  $|(\phi g_s - e)_i| \leq (D+1)d/d_i$ , with harmless adjustment of the constant for the  $o(1)$  term. Divide by the lower bound for  $s$  in (8.2) to obtain (8.11).

On a fixed branch,  $A_\eta = e \cdot b$  and  $s = g_s \cdot b$  are rational-linear formulas. The derivative of their ratio is  $((e - \phi g_s)/s) \cdot \dot{b}$ . At independent parameters the coefficient vectors are exactly those of the selected vertex, so (8.12) follows.  $\square$

**8.2. Uniform constants and dependent endpoints.** We explain the order of constants before constructing the rounded tuples. At most  $n$  functions can be adjoined and at most  $n$  tier rounds occur. There are at most a fixed multiple of  $n$  approximate continuations. Reserve one unit of ratio increase for every possible continuation and every tier round, together with the one-unit margin for the stopping argument below. Choose a single numerical bound  $M_*$  larger than the initial ratio one plus this entire reserve.

Upper weight bounds are fixed by a finite numerical recursion. Start, for example, with  $B_0 = 2$ . At each possible enlargement, apply the already proved estimate  $C_5(B_j, M_*)$ , multiply it by the uniform tier-comparability bound for  $d/d_{r+1}$ , and choose  $B_{j+1}$  above that number and  $B_j$ . Add a further fixed movement allowance at each possible small move or round. The lower bound  $d_i \geq \delta$  makes a bounded absolute coordinate movement fit this allowance. This recursion has predetermined finite depth; its definition does not assume that any future tuple has already been constructed. Evaluate the derivative constants and cutoff multipliers for all these prospective bounds and retain their maxima. Only after these choices will we select a large uniform positive integer  $S$ . Finally omit a finite prefix of the sequence to put all the finitely many fixed cutoffs below the next tier.

For the chosen element  $\gamma \in G$ , let  $\vartheta_f = \text{angle}(\chi_f(\gamma)) \in \mathbb{R}/\mathbb{Z}$ . Targets will satisfy

$$(8.13) \quad Sb_j^* \equiv \vartheta_{f_j} \pmod{\mathbb{Z}}.$$

For each germ use the movement tolerance

$$(8.14) \quad \varepsilon = \frac{d_1}{Sd_n}$$

in every coordinate of an approximate continuation; those movements can always be chosen arbitrarily smaller.

Here is the construction at a tier endpoint  $m$ . For the first tier start with its full original prefix  $f_j^0$ . The valuation  $v$  is Gauss with  $b_j = d_j$ ,  $s = 1$ , and  $\Phi = 1$ . Continue to very close independent weights, allowing one unit of ratio increase. At a later tier start from the previous completed tuple through  $m_-$  at its exact target, with the Gauss extension produced by the previous round, and

again continue to independent weights. While  $r < m$ , use Lemma 8.2 and then make a continuation within the tolerance (8.14). Each step uses the prospectively fixed bounds.

At  $r = m$ , every old coordinate  $j \leq m_-$  is within  $(n+1)\varepsilon$  of its previous target. Keep exactly that old target, and choose for each new coordinate the nearest target satisfying (8.13). Let  $b^*$  be the result and move along the straight segment from  $b$  to  $b^*$ . Its weighted length obeys

$$(8.15) \quad \sum_{i \leq m} \frac{d}{d_i} |b_i^* - b_i| \leq \frac{C_7}{S}.$$

For old coordinates this follows from  $d \leq d_n$ ,  $d_1 \leq d_i$ , and (8.14); there are at most  $n$  of them. For new coordinates, the distance to the nearest target is at most  $1/(2S)$  and  $d/d_i$  is uniformly bounded within their tier.

All required positivity and anchor conditions hold for a sufficiently large common  $S$ . The anchor  $f_1 = f_1^0$  begins arbitrarily close to  $d_1$ ; after its first round its exact target never changes, and subsequent accumulated errors before rounding are bounded as above. Thus  $1/2 \leq b_1/d_1 \leq 2$  throughout. Every newly selected function is nonconstant, and before rounding its coordinate is at least  $sv(f_i) \geq cd_1$  by (6.1). The uniform lower bound for  $d_1$  keeps all targets and segments positive for large  $S$ . The coordinate movements are  $O(n/S)$ , so the prospective upper bounds also remain valid.

**Lemma 8.4.** *For the bounds fixed above, choose  $S$  large enough that  $C_6C_7/S < 1$ . If a completed prefix at the start of the segment has a Gauss extension with an available ratio bound  $M$ , its target endpoint has a centred quasi-monomial Gauss extension with ratio at most  $M+1$ .*

*Proof.* A constant segment requires no change. Otherwise independent parameters are dense along the segment: for each nonzero rational vector a rational dependence is an affine equation in the segment parameter, not identically zero because the starting weights are independent. Apply the finite interval partition of Proposition 6.4. Initially approximate continuation supplies feasible independent assignments with limiting optimal ratio at most  $M$ .

On each open interval where a selected branch is feasible, integrate (8.12) as long as the ratio stays within the reserved bound  $M+1$ . The estimate holds at the dense independent parameters and hence everywhere on the interval by smoothness of the rational branch formula. At a terminal parameter of such a bounded interval, selected vertices at independent parameters have a compact convergent subsequence on the fixed complex:  $s \leq 2$ ,  $A_\eta$  is bounded, discrepancy is proper on the relevant cones, and (8.2) keeps  $s$  positive. Its limit is Gauss on the tuple by Proposition 6.4, including all polynomial identities at these possibly dependent weights. Lemma 6.2 continues this valuation to nearby independent assignments to the right with limiting ratio no larger than the endpoint value. Thus there is neither loss of feasibility nor an upward jump at a branch endpoint.

Proceed in order through the finitely many intervals of the partition. On an interval, stop at a putative first excess of the reserved budget; until that time all constants are valid. With the segment parameterized by  $0 \leq t \leq 1$ , integration and (8.15) propagate the bound

$$\Phi \leq M + t \frac{C_6C_7}{S}.$$

The limsup argument propagates it across every endpoint. Since  $C_6C_7/S < 1$ , a first excess of  $M+1$  is impossible. The polyhedral selection may be relaxed at dependent parameters inside intervals; this causes no difficulty, because the selected branch is actual at the dense independent parameters where the derivative is tested, and compact limits provide actual Gauss extensions at endpoints. At the final endpoint take the compact limit itself.  $\square$

The preceding numerical recursion and Lemma 8.4 justify the construction inductively through every tier. There is no circular dependence in the choice of  $S$ : all prospective upper bounds, ratio budgets, derivative constants, movement constants, and cutoff multipliers precede it. Increasing

$S$  once enforces every one of the finitely many conditions. By the final tier we obtain  $n$  centred regular eigenfunctions, positive rational exact targets  $b_i^*$ , and a centred quasi-monomial Gauss extension  $w$  such that

$$(8.16) \quad Sb_i^* \equiv \vartheta_{f_i} \pmod{\mathbb{Z}}, \quad s(w) \geq c, \quad A_\eta(w) \leq C, \quad b_i^* \leq B_* d_i.$$

The target weights are rational because the character angles are rational. A Gauss valuation on a full tuple with positive rational weights is divisorial-scaled: its value group has rational rank one and its residue field has transcendence degree  $n - 1$ . Finite field extension preserves these properties, or one can use the smooth-stratum description of Lemma 6.2. Thus the resulting  $w$  is divisorial-scaled.

### 8.3. All rational semi-invariants.

*Completion of the proof of Theorem 3.1.* It remains to pass from the congruences on the completed tuple to every rational semi-invariant, with one fixed additional integer. Choose a uniform positive integer  $D_{\text{fin}}$  so large that

$$(8.17) \quad (D_{\text{fin}} + 1) \left( \frac{c}{2nB_*} \right)^n > C_W.$$

For any regular nonzero eigenfunction  $h$ , take  $T \geq d_n$  sufficiently large and consider the indexed functions

$$(8.18) \quad h^j f^\alpha, \quad 0 \leq j \leq D_{\text{fin}}, \quad 0 \leq \alpha_i \leq \frac{sT}{2nb_i^*}, \quad \alpha_i \in \mathbb{Z}.$$

The number of indices is at least

$$(D_{\text{fin}} + 1) \left( \frac{c}{2nB_*} \right)^n \prod_{i=1}^n \frac{T}{d_i} > \mathcal{H}(T).$$

Choose  $T$  still larger so that  $D_{\text{fin}} w(h) \leq sT/2$ . Every indexed function then has  $w$ -value at most  $sT$ . There is a nontrivial linear dependence modulo  $(v > T)$ , whether or not some of the indexed functions already coincide. Since  $v$  is invariant, the ideal is stable and the dependence can be chosen within a single character. Its sum either vanishes or has  $w$ -value greater than  $sT$ , by (6.1). Hence the lowest terms must cancel in the  $w$ -graded ring.

This cancellation involves at least two distinct exponents  $j, j'$ . Otherwise, after factoring the nonzero initial of  $h^j$ , a nonzero polynomial in the initials of  $f$  would vanish. This contradicts their Gauss property, even though the target weights have rational relations. Choose two lowest terms with distinct powers. Equality of their weights and equality of their characters, together with (8.16), give

$$(8.19) \quad (j - j')(Sw(h) - \vartheta_h) = 0 \quad \text{in } \mathbb{R}/\mathbb{Z}, \quad 1 \leq |j - j'| \leq D_{\text{fin}}.$$

Set

$$(8.20) \quad \ell = \text{lcm}(1, \dots, D_{\text{fin}}), \quad w' = \ell Sw.$$

Then for every regular nonzero eigenfunction,

$$w'(h) \equiv \ell \text{ angle}(\chi_h(\gamma)) \pmod{\mathbb{Z}}.$$

The cutoff  $T$  may depend on  $h$  and on the germ; the integers  $D_{\text{fin}}, \ell, S$  do not. In particular the same  $\ell$  works for all these functions at once.

For a rational semi-invariant  $h$ , choose a regular denominator  $b$  and replace it by the product of its group translates. This is a nonzero invariant regular denominator  $b_G$ , and  $hb_G$  is regular and has the same character as  $h$ . Subtracting the two already established congruences proves the assertion for  $h$ . On invariant rational functions the resulting value is integral. Finally  $w'$  is centred and

divisorial-scaled, and its discrepancy is bounded by  $\ell SC$  uniformly on the retained subsequence. This is exactly Theorem 3.1.

The only input whose proof has been deferred is Proposition 5.2, used in Proposition 5.1. The remaining sections prove that proposition and thereby complete the dependency chain.  $\square$

## 9. THE INTEGRAL ADJOINT ARGUMENT

We prove Proposition 5.2 using two uniform estimates for forms. The estimates are reduced in Section 10 to Theorem 10.1, whose proof occupies Sections 11–13. All fields in these sections are finitely generated over  $\mathbb{C}$ ; the varieties in Proposition 5.2 are transported to  $\mathbb{C}$  by the stipulated field isomorphisms.

A form of index  $q$  on a function field is a nonzero rational section of the  $q$ th tensor power of its absolute top differential line. Its boundary on a normal model is minus its divisor divided by  $q$ . On a model over a smaller function field, we divide by the  $q$ th power of a base volume form to interpret this boundary relatively. Rational convex combinations of form discrepancies are realized by taking tensor powers and multiplying forms. Indices of these products need not be bounded unless a bound is expressly asserted. A projective variety is of *Fano type* if it admits an effective klt boundary whose negative adjoint is ample.

**Theorem 9.1** (Lifting and Mixing). *Fix positive integers  $d, r$  and a positive real number  $C$ . The following assertions hold.*

- (1) *Let  $P/Q$  be a regular extension with  $\text{trdeg}_{\mathbb{C}} P \leq d$ , having a projective Fano type generic model. Let  $\sigma, \psi$  be forms on  $P$  whose boundaries on this model are effective, with  $\sigma$  of index at most  $r$ . Suppose, on every nonzero divisorial valuation of  $P$ , that*

$$A_{\psi} > 0, \quad 0 \leq A_{\sigma} \leq C A_{\psi}.$$

*For every  $\lambda > 0$  there is  $\delta = \delta(d, r, C, \lambda) > 0$  with the following property. If  $u$  is a nonzero integral-valued divisorial valuation of  $Q$  and*

$$\inf_{w|_Q=u} A_{\psi}(w) < \delta,$$

*then there is an integral-valued divisorial valuation  $v$  of  $P$  whose restriction is a positive multiple of  $u$  and for which  $A_{\psi}(v) < \lambda$ . In relative dimension zero the regular extension is  $P = Q$ , and the same assertion holds.*

- (2) *Let  $P$  have a projective Fano type model over  $\mathbb{C}$  of dimension at most  $d$ . Let  $\sigma, \psi, \chi$  be forms with effective boundaries on that model, with  $\sigma$  of index at most  $r$ , and suppose*

$$A_{\psi} > 0, \quad 0 \leq A_{\sigma} \leq C A_{\psi}.$$

*Suppose in addition that, for some  $c > 0$ ,*

$$A_{\chi}(w) \geq c A_{\psi}(w) \quad \text{whenever } w \neq 0 \text{ and } A_{\sigma}(w) = 0.$$

*There is a rational number  $a \in (0, 1)$  depending only on  $d, r, C, c$  such that*

$$(1 - a)A_{\psi}(w) + aA_{\chi}(w) > 0 \quad \text{for every nonzero divisorial } w.$$

*All infima and homogeneous inequalities allow positive rational scalings of prime divisorial valuations. An integral-valued valuation is not required to be primitive.*

### 9.1. The integral barrier.

*Proof of Proposition 5.2, assuming Theorem 9.1.* Write  $A = A_{\eta_F}$  and  $l_w = w(L)$ . The dimension of  $F$  and the clipping constant in Proposition 5.2 are fixed along the sequence. We begin with a rational  $t > 0$ , which will be chosen only after obtaining constants independent of  $t$ . Use the canonical divisor  $K_F = \text{div}(\eta_F)$  and put

$$(9.1) \quad \Delta = \lceil tL \rceil - tL, \quad E = K_F + \lceil tL \rceil.$$

The boundary  $\Delta$  has coefficients in  $[0, 1)$  and SNC support, so  $(F, \Delta)$  is klt. Since  $\eta_F$  is an lc volume form, every coefficient of  $K_F$  is an integer at least  $-1$ . A coefficient  $-1$  gives  $A(P) = 0$  and hence  $l_P > 0$ . Thus  $E$  is an effective integral divisor.

Choose a sufficiently divisible  $q$  and a general  $H_q \in |qL|$ , and set  $H = H_q/q$ . Semiampleness permits  $H_q$  to be chosen smooth and transverse to the fixed SNC support. Taking  $q \geq 2$ , the pair  $(F, \Delta + 2H)$  is sub-lc. Consequently

$$(9.2) \quad w(H) \leq \frac{1}{2}A_{F,\Delta}(w) \quad \text{for every divisorial } w.$$

Here and below discrepancies carrying a pair subscript are ordinary pair discrepancies. Define forms  $\sigma, \psi$  by

$$(9.3) \quad A_\sigma = A, \quad A_\psi(w) = A(w) + tl_w - w(H).$$

The first form is  $\eta_F$ , of index one; the second exists because  $tL - H \sim_{\mathbb{Q}} 0$ . From

$$A_{F,\Delta}(w) = A(w) + tl_w - w(E)$$

and (9.2), we obtain

$$(9.4) \quad A_\psi(w) \geq \frac{1}{2}(A(w) + tl_w) > 0, \quad A_\sigma(w) \leq 2A_\psi(w).$$

Strict positivity uses the hypothesis  $l_w > 0$  at every nonzero divisorial valuation with  $A(w) = 0$ .

The pair  $(F, \Delta + H)$  is klt, its boundary is big, and its adjoint is rationally linearly equivalent to  $E \geq 0$ . The good minimal model theorem of [11] gives a  $\mathbb{Q}$ -factorial good minimal model  $Y$ . Let  $h : Y \rightarrow S$  be its semiample contraction, with connected fibres. Subscripts denote birational transforms. The rational principal divisor  $tL - H$  is the same on every model, so the ordinary MMP discrepancy inequality gives, on a common resolution,

$$(9.5) \quad E_Y = K_Y + \Delta_Y + tL_Y = h^*N, \quad w(E) \geq w(E_Y) = w(h^*N).$$

Here  $N$  is an effective ample  $\mathbb{Q}$ -Cartier divisor when  $\dim S > 0$ . To obtain equality of divisors, take a globally generated multiple of the effective semiample divisor  $E_Y$ . Its defining section descends to the ample system on  $S$  because  $h_*\mathcal{O}_Y = \mathcal{O}_S$ , and its zero divisor defines that multiple of  $N$ . If  $S$  is a point, the effective divisor  $E_Y$  is rationally linearly trivial and is zero; we then put  $N = 0$ . The effective pair  $(Y, \Delta_Y + H_Y)$  is klt, and

$$(9.6) \quad A_{Y,\Delta_Y+H_Y}(w) = A_\psi(w) - w(h^*N) > 0.$$

We claim that every integral-valued divisorial valuation satisfies

$$(9.7) \quad w(h^*N) > 0 \implies A_\sigma(w) + tl_w \geq 1.$$

If the conclusion fails, the nonnegative integer  $A_\sigma(w)$  is zero. Let  $D$  be the reduced SNC union of the supports of  $K_F$  and  $L$ . If  $k_P$  is the coefficient of  $K_F$  at a component of  $D$ , then

$$0 = A_\sigma(w) = A_{F,D}(w) + \sum_{P \subset D} (1 + k_P)w(P).$$

All summands are nonnegative. Thus every component seen by  $w$  has  $k_P = -1$  and  $l_P > 0$ . Since  $w(P)$  is a positive integer,

$$0 < tl_P \leq tl_w < 1, \quad \text{coeff}_P(E) = -1 + \lceil tl_P \rceil = 0.$$

The support of  $E$  is contained in  $D$ , so  $w(E) = 0$ , contradicting (9.5). This proves (9.7); its unit threshold is exactly where the index-one hypothesis on  $\eta_F$  is used.

**9.2. A uniformly positive base polarization.** Suppose  $\dim S > 0$ , and put  $P = \mathbb{C}(Y)$  and  $Q = \mathbb{C}(S)$ . Connectedness makes  $P/Q$  regular. On the generic fibre of  $h$ , the boundaries of  $\sigma$  and  $\psi$  are respectively

$$\Delta_Y + tL_Y, \quad \Delta_Y + H_Y,$$

restricted to that fibre, because  $h^*N$  restricts to zero. They are effective. The second gives a klt log Calabi–Yau pair with big boundary:  $H_Y$  is big by birational pushforward and its restriction to the generic fibre is big. A klt log Calabi–Yau pair with big boundary is of Fano type. Explicitly, write its boundary as a rational linear equivalence  $B \sim_{\mathbb{Q}} A_0 + D_0$  with  $A_0$  ample and  $D_0 \geq 0$ . For sufficiently small  $e > 0$ ,  $(1 - e)B + eD_0$  is effective klt and its negative adjoint is rationally linearly equivalent to  $eA_0$ . If the generic fibre has dimension zero, regularity gives  $P = Q$ .

Apply Theorem 9.1(1) with reference index one, comparison constant 2, and, for example,  $\lambda = 1/4$ . By (9.4) and (9.7), an integral-valued valuation lying over a positive multiple of a valuation  $u$  with  $u(N) > 0$  has  $A_\psi \geq 1/2$ . Thus there is  $\delta > 0$ , depending only on the fixed dimension and independent of  $t$ , such that

$$(9.8) \quad \bar{A}_\psi(u) := \inf_{w|_Q=u} A_\psi(w) \geq \delta \quad \text{if } u \text{ is integral-valued and } u(N) > 0.$$

In relative dimension zero this follows directly from the barrier; decreasing  $\delta$  accommodates that case as well.

Apply the lc-trivial canonical bundle formula with b-semiample moduli part, [5, Theorems 1.5 and 6.28], to  $h : (Y, \Delta_Y + H_Y) \rightarrow S$ . The morphism is projective and connected; the boundary is effective and klt, and

$$K_Y + \Delta_Y + H_Y \sim_{\mathbb{Q}} h^*N.$$

The discrepancy rank condition is one: on a generic resolution, the rounded discrepancy divisor has no negative coefficients and its positive coefficients occur only on exceptional divisors. Its section space is therefore the constants on the connected generic fibre. These verify generic effectivity, generic sub-log-canonicity, rank one, and the rational adjoint pullback required by the cited theorem.

Its generalized base discrepancy is

$$(9.9) \quad \beta(u) = \bar{A}_\psi(u) - u(N) > 0.$$

Indeed, after putting  $u$  on a smooth base model, the lc threshold over its DVR is computed by resolving the fibre divisor together with the boundary. It is the minimum of the finitely many discrepancy-to-multiplicity ratios of vertical components; the horizontal directions are already klt. Thus the infimum in (9.9) is attained and is positive. On the original  $S$  the discriminant is effective: over the generic point of a prime of  $S$  there is a prime of  $Y$  of positive integral fibre multiplicity, and its ordinary discrepancy is at most one.

On a high smooth model where the moduli divisor is semiample and the crepant discriminant has SNC support, replace the moduli divisor by  $H_0/q_0$ , where  $H_0$  is a general member of its  $q_0$ th multiple and  $q_0 \geq 2$  is sufficiently divisible. The subpair obtained by adding  $H_0$  with coefficient one is sub-lc, so  $u(H_0) \leq \beta(u)$  for all  $u$ . After pushforward this gives an effective ordinary klt boundary  $B_S$  with

$$(9.10) \quad K_S + B_S \sim_{\mathbb{Q}} N, \quad \frac{1}{2}\beta(u) \leq A_{S, B_S}(u) \leq \beta(u).$$

Both the discriminant trace and the pushed-down general member are effective. On every model the replacement changes the adjoint by the same rational principal divisor. In particular  $K_S + B_S$  is  $\mathbb{Q}$ -Cartier and its ordinary crepant pullback is the boundary just constructed. This argument does not require  $K_S$  itself to be  $\mathbb{Q}$ -Cartier.

First take a small  $\mathbb{Q}$ -factorialization  $S_1 \rightarrow S$  of the ordinary klt pair  $(S, B_S)$ . Its crepant boundary is effective and pair discrepancies are unchanged. Now  $K_{S_1}$  is  $\mathbb{Q}$ -Cartier and  $S_1$  is klt, so

underlying-variety discrepancies are defined. Fix

$$0 < \epsilon' < \min\{1, \delta/4\},$$

and extract to a  $\mathbb{Q}$ -factorial model  $S' \rightarrow S_1$  precisely the prime divisors whose underlying log discrepancy is less than  $\epsilon'$ . This is a finite set: on a fixed log resolution, write the discrepancy as the nonnegative integral discrepancy of the full reduced SNC support plus a positive linear combination of its integral component orders. A discrepancy below one forces the integral term to vanish. Such valuations are the monomial lc places of the reduced SNC pair, and the positive coefficients bound all their integral weights. There are finitely many strata and finitely many possible weight vectors. The klt extraction furnished by [11] applies since each desired divisor has pair discrepancy at most its underlying discrepancy, hence less than one. An explicit extraction construction is given in Lemma 10.3 below.

For this extraction  $\pi : S' \rightarrow S_1$  one has

$$K_{S'} + B_{\text{exc}} = \pi^* K_{S_1}, \quad B_{\text{exc}} \geq 0.$$

Thus all remaining underlying discrepancies improve; the extracted primes themselves have underlying discrepancy one on  $S'$ . It follows that  $S'$  is  $\epsilon'$ -lc. The crepant transform  $B_{S'}$  of the ordinary pair boundary is effective. If  $N'$  denotes the pullback of  $N$ , then

$$(9.11) \quad N' \geq 0, \quad N' \text{ is nef and big}, \quad N' - K_{S'} \sim_{\mathbb{Q}} B_{S'} \geq 0.$$

Every positive coefficient of  $N'$  is at least  $\delta/2$ . For an extracted primitive valuation  $u$  with  $u(N) > 0$ , (9.10) gives

$$\beta(u) \leq 2A_{S, B_S}(u) \leq 2A_{S_1}(u) < 2\epsilon'.$$

Together with (9.8) this yields  $u(N) \geq \delta - 2\epsilon' > \delta/2$ . For a prime already on  $S_1$ , use its corresponding prime on  $S$ . Choose a prime of  $Y$  dominating it, and let  $k \geq 1$  be its fibre multiplicity. The coefficient  $ku(N)$  is a positive integer, because  $E_Y$  is the birational pushforward of the integral divisor  $E$ . Effectivity of  $(Y, \Delta_Y + H_Y)$  gives

$$\beta(u) \leq \frac{1}{k} \leq u(N), \quad \delta \leq \bar{A}_\psi(u) = u(N) + \beta(u) \leq 2u(N).$$

This proves the assertion for all primes of  $S'$ .

Birkar's effective birationality theorem [10, Theorem 4.2] now applies. Its underlying variety is  $\epsilon'$ -lc; its polarization  $N'$  is nef and big;  $N' - K_{S'}$  is pseudo-effective; and in its required decomposition into an integral pseudo-effective summand and an effective summand we take  $0 + N'$ . The positive coefficients of the latter are at least  $\delta/2$ . We obtain a positive integer  $m$ , taking a common multiple for the finitely many possible positive dimensions of  $S$ , such that  $\lfloor mN' \rfloor$  is birational and in particular nonconstant. Crucially,  $m$  depends on neither  $t$  nor the denominators of the coefficients of  $N'$ .

**9.3. The strict ceiling inequality and final mixture.** We now fix a rational  $t > 0$  with  $mt < 1/20$ . Pass far enough along the sequence that, writing  $\epsilon$  for the prime estimate in Proposition 5.2,

$$(9.12) \quad c_0 := m(t + \epsilon) < 1/10.$$

We may enlarge  $\epsilon$  slightly to make it rational while retaining this inequality. The section 1 belongs to  $\lfloor mN' \rfloor$ , and nonconstancy gives another section represented by a nonconstant rational function  $g$  on  $S'$ . Push its effective divisor  $\text{div}(g) + mN'$  to  $S$ , pull back by  $h$ , and use (9.5). This gives

$$(9.13) \quad \text{div}(g) + mE \geq 0 \quad \text{on } F.$$

Let  $L_0$  be the effective big semiample  $\mathbb{Q}$ -Cartier divisor on  $F_0$  whose pullback is  $L$ . At a prime of  $F_0$  with  $l_P = 0$  we have  $A(P) = 1$ , so its coefficient in the pushforward of  $E$  is zero. If  $l_P > 0$ , its coefficient is

$$A(P) - 1 + \lceil tl_P \rceil \leq A(P) + tl_P \leq (t + \epsilon)l_P.$$

Pushing (9.13) to  $F_0$  therefore gives the effective  $\mathbb{Q}$ -Cartier bound

$$\operatorname{div}(g) + c_0 L_0 \geq 0.$$

Its pullback controls every prime of  $F$ , including the exceptional ones. Set  $b = 1/10$ . If  $l_P > 0$  and  $n = \operatorname{ord}_P(g) \in \mathbb{Z}$ , then

$$n \geq -c_0 l_P > -b l_P \implies n \geq 1 - \lceil b l_P \rceil.$$

Since  $K_{F,P} \geq -1$ , this proves  $\operatorname{div}(g) + K_F + \lceil b L \rceil \geq 0$  at such a prime. If  $l_P = 0$ , the pullback bound gives  $n \geq 0$ , while  $K_{F,P} \geq 0$ : otherwise  $K_{F,P} = -1$  would be an lc place with  $l_P = 0$ . The same inequalities with  $g = 1$  hold as well. Thus 1 and the nonconstant  $g$  are two independent sections of  $K_F + \lceil b L \rceil$ , contrary to the assumed dimension one. The positive-dimensional case is impossible, and for this fixed  $t$  we have  $N = 0$  sufficiently far along the sequence.

Now  $(Y, \Delta_Y + H_Y)$  is a klt log Calabi–Yau pair with big boundary, so  $Y$  itself is of Fano type. Let  $J \geq 0$  with  $J \sim_{\mathbb{Q}} L$  and define a form  $\chi_0$  by

$$A_{\chi_0}(w) = A(w) + t l_w - t w(J).$$

Its boundary on  $Y$  is the effective divisor  $\Delta_Y + t J_Y$ . Write  $C_0$  for the fixed clipping constant on the set  $A = 0$ . There, (9.4) and the hypothesis on  $J$  give

$$A_{\chi_0}(w) \geq -(C_0 - 1)_+ t l_w \geq -2(C_0 - 1)_+ A_\psi(w),$$

where  $(x)_+ = \max\{x, 0\}$ . Choose a rational  $\theta \in (0, 1)$ , depending only on  $C_0$ , such that

$$1 - \theta(1 + 2(C_0 - 1)_+) \geq \frac{1}{2}.$$

The form with discrepancy  $A_\chi = (1 - \theta)A_\psi + \theta A_{\chi_0}$  has effective boundary on  $Y$  and satisfies  $A_\chi \geq A_\psi/2$  on  $A_\sigma = 0$ . Theorem 9.1(2), with  $r = 1$ ,  $C = 2$  and  $c = 1/2$ , gives a uniform rational  $a \in (0, 1)$  such that, putting  $a_0 = a\theta > 0$ ,

$$\begin{aligned} 0 &< (1 - a)A_\psi(w) + a A_\chi(w) \\ &= A(w) + t l_w - (1 - a_0)w(H) - a_0 t w(J). \end{aligned}$$

Since  $H$  is effective, we conclude

$$w(J) \leq \frac{A(w) + t l_w}{a_0 t} \leq \frac{\max\{1, t\}}{a_0 t} (A(w) + l_w).$$

The final constant is uniform because  $t$  was fixed once and for all. This is the asserted integral jump estimate.  $\square$

## 10. REDUCTION TO BOUNDED FANO FIBRES

We reduce Theorem 9.1 to an estimate on geometrically bounded-singularity Fano generic fibres. The reference form retains a bounded index through the reduction; no index bound is needed for the other forms.

**10.1. Valuations, forms, and the bounded-fibre statement.** For a form  $\omega$  of index  $q$  and a homogeneous divisorial valuation  $v = s \operatorname{ord}_E$ , our convention is

$$(10.1) \quad A_\omega(v) = s \left( 1 + \frac{\operatorname{ord}_E(\omega)}{q} \right),$$

where the order is taken in the  $q$ th pluricanonical sheaf. We allow  $s \in \mathbb{Q}_{>0}$ ; for the trivial valuation we put all form discrepancies equal to zero.

The restriction of a divisorial valuation to a function subfield is divisorial or trivial. Indeed, if the restriction to  $K \subset L$  is nontrivial, its value group is a nonzero subgroup of the discrete value group upstairs. The relative Abhyankar inequality and  $\operatorname{trdeg}_{\mathbb{C}} \kappa(v) = \operatorname{trdeg}_{\mathbb{C}} L - 1$  give

$$\operatorname{trdeg}_{\mathbb{C}} \kappa(v|_K) \geq \operatorname{trdeg}_{\mathbb{C}} K - 1.$$

The opposite inequality is the absolute Abhyankar inequality. Equality and discreteness characterize geometric rank-one valuations, proving the assertion.

For a finite extension, the discrepancy of the pulled-back form at an upstairs valuation equals its discrepancy at the restricted valuation, with the restriction's scale included, as proved in Section 2. In positive relative dimension, dividing by a base differential frame gives the same boundary and discrepancies on the generic fibre for valuations trivial on the base field. To see this, spread a resolution over a smooth base open and use the determinant of the exact sequence of differentials along horizontal primes. These facts also justify extending constants when studying the generic varieties below.

**Theorem 10.1** (Bounded-fibre estimate). *Let  $L/K$  be a regular extension of finitely generated fields over  $\mathbb{C}$ , with  $\text{trdeg}_{\mathbb{C}} L \leq d$ . Suppose it has a positive-dimensional projective generic model  $G/K$  that is geometrically  $\epsilon_0$ -lc Fano, where  $\epsilon_0 > 0$  is fixed:  $-K_G$  is ample and  $\mathbb{Q}$ -Cartier, and the geometric underlying variety is  $\epsilon_0$ -lc. Let  $\sigma$  be a form on  $L$  of index at most  $r$ , with effective boundary on  $G$  and with  $A_\sigma \geq 0$  on all absolute divisorial valuations of  $L$ . For every nonzero divisorial valuation  $u$  of  $K$ , set*

$$\bar{A}_\sigma(u) = \inf_{w|_K=u} A_\sigma(w),$$

and use the same notation for other forms.

- (1) *There is a form  $\sigma_K$  on  $K$ , with index bounded only in terms of  $d, \epsilon_0, r$ , such that*

$$\frac{1}{2}\bar{A}_\sigma(u) \leq A_{\sigma_K}(u) \leq \bar{A}_\sigma(u) \quad \text{for every nonzero divisorial } u.$$

- (2) *Let  $\psi$  be a form on  $L$  with effective boundary on  $G$ ,  $A_\psi > 0$  on all nonzero absolute divisorial valuations, and  $A_\sigma \leq CA_\psi$ . For every  $\lambda > 0$ , there is  $\delta > 0$ , depending only on  $d, \epsilon_0, r, C, \lambda$ , such that an integral-valued nonzero divisorial valuation  $u$  of  $K$  with  $A_\psi(u) < \delta$  has an integral-valued divisorial lift  $v$  on  $L$  satisfying*

$$v|_K \in \mathbb{Q}_{>0}u, \quad A_\psi(v) < \lambda.$$

- (3) *Let  $\chi, \psi$  be forms with effective boundaries on  $G$ ; neither is required to be absolutely lc in this part. Suppose that, for a fixed  $c > 0$ ,*

$$A_\chi(w) \geq cA_\psi(w) \quad \text{on all nonzero divisorial } w \text{ with } A_\sigma(w) = 0.$$

*There is  $D \geq 0$ , depending only on  $d, \epsilon_0, r, c$ , such that*

$$(10.2) \quad A_\chi(w) \geq cA_\psi(w) - DA_\sigma(w)$$

*whenever  $w$  is trivial on  $K$ , or  $u = w|_K$  is nonzero and  $\bar{A}_\sigma(u) = 0$ .*

*The boundaries on  $G$  are interpreted by dividing out a base volume form as above. All comparisons allow positive rational scalings, and integral-valued valuations need not be primitive. If  $K = \mathbb{C}$ , the first part has no nonzero valuations to test and a constant form of index one suffices; the second part is vacuous, while the third part includes every nonzero divisorial valuation of  $L$ .*

Theorem 10.1 is proved in Sections 11–13. The present section proves that it implies Theorem 9.1.

**10.2. Ordinary forms on the base.** We first record a base substitution that does not bound the index. It will be used for the positive anchor and the forms being mixed, while Theorem 10.1(1) supplies the reference form.

**Lemma 10.2** (Ordinary base substitution). *Let  $L/K$  be a regular extension as above, and let  $G/K$  be a normal geometrically integral projective generic model. Let  $\omega$  be a form whose boundary on  $G$  is effective and sub-lc. Then the threshold canonical bundle formula gives a discriminant  $b$ -divisor and a  $b$ -Cartier  $b$ -semiample moduli  $b$ -divisor with generalized discrepancy  $\bar{A}_\omega$ . Each defining infimum for a nonzero divisorial base valuation is finite and attained.*

Suppose another such form  $\psi$  has  $\bar{A}_\psi(u) > 0$  for every nonzero divisorial  $u$  of  $K$ . For every rational  $\gamma > 0$ , there is a form  $\omega_K$  on  $K$  such that

$$(10.3) \quad \bar{A}_\omega(u) - \gamma \bar{A}_\psi(u) \leq A_{\omega_K}(u) \leq \bar{A}_\omega(u).$$

No index bound is asserted. The boundary of  $\omega$  need not be sub-lc at vertical valuations.

*Proof.* Choose a smooth projective model  $T$  of  $K$  and spread out  $G$ , taking a dominant projective closure and its normalization. This gives a projective morphism  $f : X \rightarrow T$  with generic model  $G$ . Its Stein factor is finite birational over  $T$  because  $K$  is algebraically closed in  $L$ ; normality of  $T$  makes it an isomorphism. Hence  $f$  has connected fibres.

For the boundary  $B_\omega$  defined by  $\omega$ , we have

$$K_X + B_\omega \sim_{\mathbb{Q}} 0 = f^*0, \quad A_{X, B_\omega} = A_\omega.$$

The boundary is effective and sub-lc over the generic point of  $T$ . We check the discrepancy rank condition, including the lc case. On a resolution  $\mu : G' \rightarrow G$  of the generic pair, let  $K_{G'} + B_{G'} = \mu^*(K_G + B_G)$ . The modified discrepancy divisor omits the components having non-log discrepancy  $-1$ , or equivalently boundary coefficient one. At every remaining component its coefficient is the negative of the coefficient of  $B_{G'}$ . All coefficients of  $B_{G'}$  are at most one, and nonexceptional coefficients lie in  $[0, 1]$ . Its modified rounded discrepancy divisor is therefore effective and exceptional. If this divisor is denoted by  $D$ , normality gives  $\mu_*\mathcal{O}_{G'}(D) = \mathcal{O}_G$ : sections with poles only at exceptional divisors have no poles at codimension-one points of  $G$ . Consequently

$$H^0(G', \mathcal{O}_{G'}(D)) = H^0(G, \mathcal{O}_G) = K.$$

This verifies rank one. We have thus checked generic effectivity, generic sub-log-canonicity, rank one, and the rational adjoint pullback in the lc-trivial canonical bundle formula. The b-semiampleness and the identification with the threshold moduli part follow from [5, Theorems 1.5 and 6.28]. On a sufficiently high smooth base, write this formula as

$$(10.4) \quad K_T + B_{\omega, T} + M_{\omega, T} \sim_{\mathbb{Q}} 0.$$

The moduli b-divisor descends to  $T$  and its trace is semiample.

To identify the generalized discrepancy and prove attainment, put a primitive base valuation  $\text{ord}_P$  on a smooth base and work over the DVR at its generic point. Resolve the boundary and the fibre divisor. On this resolution write

$$K_Z + B_Z = \mu^*(K_X + B_\omega), \quad (f\mu)^*P = \sum_j m_j E_j,$$

where  $m_j > 0$  and the combined support is SNC. If  $b_j$  is the coefficient of  $E_j$  in  $B_Z$ , the threshold is

$$(10.5) \quad t_P = \sup\{t \in \mathbb{R} : (X, B_\omega + tf^*P) \text{ is sub-lc over } \eta_P\} = \min_j \frac{1 - b_j}{m_j}.$$

The supremum is over all real  $t$ , since vertical discrepancies may initially be negative. Horizontal coefficients are at most one by the generic hypothesis. At the displayed minimum all coefficients of the resolved SNC pair are at most one; increasing  $t$  violates this at a minimizing component. Moreover,  $m_j^{-1} \text{ord}_{E_j}$  restricts to  $\text{ord}_P$ . Sub-log-canonicity at  $t_P$  gives the converse lower bound for every valuation above  $\text{ord}_P$ . Hence

$$\bar{A}_\omega(\text{ord}_P) = t_P, \quad \text{coeff}_P(B_{\omega, T}) = 1 - t_P.$$

This proves both attainment and the claimed generalized discrepancy identity; homogeneity treats nonprimitive base valuations. In particular absolute positivity of  $A_\omega$  implies strict positivity of  $\bar{A}_\omega$ , because the latter is an attained minimum.

Perform the same construction for the anchor  $\psi$ . Take a common high smooth projective base on which both moduli b-divisors descend and on which the discriminant of  $\psi$  has SNC support. Since its moduli part pulls back to every higher model, the ordinary discrepancy of this discriminant

subpair is exactly  $\bar{A}_\psi$  on all valuations. Choose a sufficiently divisible  $q$  with  $1/q \leq \gamma$  and a general member

$$H \in |qM_{\omega,T}|.$$

If this system is trivial, take  $H = 0$ . Otherwise  $H$  is reduced and transverse to all strata of the anchor's SNC boundary. Adding it with coefficient one leaves that anchor subpair sub-lc, even if some of its boundary coefficients are negative. Thus

$$0 \leq u(H) \leq \bar{A}_\psi(u).$$

The ordinary boundary  $B'_T = B_{\omega,T} + H/q$  has discrepancy

$$A_{T,B'_T}(u) = \bar{A}_\omega(u) - \frac{1}{q}u(H),$$

which proves (10.3). Equation (10.4) gives  $K_T + B'_T \sim_{\mathbb{Q}} 0$ . After clearing denominators, write  $m(K_T + B'_T) = \text{div}(g)$  and choose a volume form  $\eta_T$  with divisor  $K_T$ . The form  $\eta_T^{\otimes m}/g$  has boundary  $B'_T$ , as required. For  $\omega = \psi$  and  $\gamma = 1/2$ , this construction gives

$$(10.6) \quad \frac{1}{2}\bar{A}_\psi \leq A_{\psi_K} \leq \bar{A}_\psi.$$

□

Two details of this construction will be used repeatedly. First, if a carried boundary is effective on a specified source model over a base  $U$ , then the discriminant trace on  $U$  is effective. For a prime  $P$  of  $U$ , take a source prime  $E$  dominating it, of fibre multiplicity  $k \geq 1$  and boundary coefficient  $b_E \geq 0$ . Its threshold satisfies

$$(10.7) \quad \bar{A}_\omega(\text{ord}_P) \leq \frac{1 - b_E}{k} \leq 1.$$

Any substituted form whose discrepancy is at most  $\bar{A}_\omega$  therefore has effective trace on  $U$ .

Second, ordinary adjoint descent is an equality of pullbacks. For an adjoint pulled back from a  $\mathbb{Q}$ -Cartier divisor  $N$  on a normal base  $S$ , choose the representatives on a high  $\pi : T \rightarrow S$  so that  $K_T + B_T + M_T = \pi^*N$ . If  $H = qM_T + \text{div}(g)$  and  $B'_S = \pi_*(B_T + H/q)$ , then

$$(10.8) \quad \begin{aligned} K_S + B'_S &= N + \frac{1}{q} \text{div}_S(g), \\ K_T + B_T + \frac{1}{q}H &= \pi^*(K_S + B'_S). \end{aligned}$$

Thus the ordinary adjoint on  $S$  is  $\mathbb{Q}$ -Cartier and its crepant boundary is precisely the constructed boundary on  $T$ . No separate  $\mathbb{Q}$ -Cartier hypothesis on  $K_S$  is used.

### 10.3. The geometric tower step.

**Lemma 10.3.** *Let  $P/Q$  be a regular extension of positive transcendence degree having a projective Fano type model over  $Q$ . Suppose finitely many forms on  $P$  have effective boundaries on this model. There is a projective Mori contraction  $H \rightarrow U$  over  $Q$ , with  $Q(H) = P$  and  $R = Q(U)$ , such that:*

- (i)  $P/R$  and  $R/Q$  are regular, and  $\text{trdeg}_Q R < \text{trdeg}_Q P$ ;
- (ii) the generic fibre of  $H \rightarrow U$  is positive-dimensional and geometrically 1/10-lc Fano;
- (iii)  $H$  and  $U$  are of Fano type over  $Q$ , and all the carried form boundaries remain effective on  $H$ .

*In addition, any base substitutions with discrepancy at most the appropriate barred discrepancy have effective traces on  $U$ .*

*Proof.* Choose an effective klt log Fano boundary on the initial model. First take a small  $\mathbb{Q}$ -factorialization of this ordinary pair. Pair discrepancies and codimension-one boundary coefficients are preserved, and the negative adjoint becomes nef and big. Only after this step do we use the separately  $\mathbb{Q}$ -Cartier underlying canonical class. Its underlying variety is klt because its effective pair boundary is klt.

Put  $\epsilon_0 = 1/10$  and consider the geometric prime valuations whose underlying log discrepancy is less than  $\epsilon_0$ . This is a finite set. On a fixed geometric log resolution write  $K_V + B_V = \rho^* K_X$  and let  $D$  be the full reduced SNC support. Every component has  $a_i = 1 - \text{coeff}_{D_i} B_V > 0$ . For a primitive prime valuation  $v$ ,

$$(10.9) \quad A_X(v) = A_{V,D}(v) + \sum_i a_i v(D_i).$$

The first summand is a nonnegative integer. If  $A_X(v) < 1$ , it is zero, so  $v$  is a monomial lc place of the reduced SNC pair at a stratum. Each nonzero  $v(D_i)$  is a positive integer, bounded by  $1/a_i$ . The finitely many strata and weight vectors give finiteness. The same argument works with any fixed threshold strictly below one, as used in Section 9.

The finite set is invariant under the Galois action and can be extracted over  $Q$ . To make both the extraction and its effectivity properties explicit, take a high resolution containing the desired divisors and use the chosen effective klt log Fano boundary. At each divisor to be kept, its crepant coefficient is positive and less than one: its pair discrepancy is at most the underlying discrepancy  $< \epsilon_0$ . Keep these coefficients and the strict transform of the boundary. At every other exceptional divisor choose a rational coefficient below one but above both zero and its crepant coefficient. The resulting pair is klt, and its adjoint differs from the pullback of the old adjoint by an effective exceptional divisor  $D_0$  whose support does not contain any divisor to be kept.

Run a klt MMP over the old variety to a nef model, using [11]; the morphism is birational, so the boundary is big over that variety. Its relative adjoint is  $D_0$ . Negative steps are isomorphisms at the generic points of the divisors to be kept, which lie outside its support. On the final model its transform is effective, exceptional and relatively nef; the negativity lemma makes this transform zero. The resulting  $\mathbb{Q}$ -factorial extraction has precisely the required exceptional divisors and carries the crepant effective klt boundary. This extraction construction uses only the effective klt pair and the discrepancy bound at the retained divisors; the log Fano assumption is used for the following positivity. Its negative adjoint is the pullback of the old nef and big negative adjoint, hence nef and big. It is still of Fano type: if  $-(K + B)$  is nef and big, write it as  $A + D$  up to rational linear equivalence, with  $A$  ample and  $D \geq 0$ ; a sufficiently small addition  $eD$  preserves klt and has negative adjoint rationally linearly equivalent to  $(1 - e)(-(K + B)) + eA$ , which is ample.

For the underlying varieties, the same extraction satisfies

$$K_{X'} + B_{\text{exc}} = \pi^* K_X, \quad B_{\text{exc}} \geq 0.$$

All nonextracted underlying discrepancies therefore improve, whereas each extracted prime has underlying discrepancy one on  $X'$ . This proves geometric  $\epsilon_0$ -log-canonicity. The comparison may be made with the entire effective  $\mathbb{Q}$ -Cartier exceptional combination  $B_{\text{exc}}$ , so it does not require its individual geometric components to be defined over  $Q$ . A carried effective form boundary likewise remains effective: at an extracted divisor its discrepancy is at most the old underlying discrepancy, and its new coefficient is one minus that discrepancy.

Here are the descent and field-of-definition details. A geometric divisor and the resolution containing it are defined over a finite extension of  $Q$ . Passing to a finite separable constant extension at a valuation trivial on  $Q$  does not introduce ramification or change the primitive discrepancy. One can thus choose the finite set and its resolution as Galois orbits and carry out the preceding construction over  $Q$ . Alternatively, spread the resolution, the boundary and the effective exceptional difference over a sufficiently small base with function field  $Q$ , and apply the relative form of [11]; its generic restriction is the same extraction. A small  $\mathbb{Q}$ -factorialization on the spread preserves the crepant identities and the geometric generic discrepancies.

Run a  $K_{X'}$ -MMP with scaling over  $Q$ . Since  $X'$  is of Fano type,  $K_{X'}$  is not pseudo-effective: its intersection with a sufficiently high power of an ample divisor is negative after writing  $-K_{X'}$  as an ample class plus an effective boundary. The klt Mori fibre space theorem of [11] gives a Mori

contraction  $H \rightarrow U$ . The underlying discrepancies improve along this program, also after extending constants, so  $H$  remains geometrically  $\epsilon_0$ -lc. Every carried effective boundary stays effective by pushforward through the non-extractive map.

The variety  $H$  is still of Fano type. Indeed the log Fano structure on  $X'$  gives an effective klt log Calabi–Yau boundary that is big, by adding a sufficiently general small-coefficient representative of the ample negative adjoint. Push this boundary through the MMP. Its adjoint is crepant: on a common resolution the difference is exceptional and rationally linearly trivial over the common base, so the negativity lemma makes it zero. Its boundary remains big by pushforward of sections, and the big-boundary perturbation used in Section 9 makes  $H$  of Fano type. Over the function field  $Q$ , the Mori program can also be obtained by spreading the klt log Fano data over a small open and applying the relative theorem. The canonical class is not pseudo-effective relative to that open, as is seen on its generic fibre. Restriction of the resulting program to the generic fibre preserves all the discrepancy comparisons and yields the stated Mori contraction.

The generic fibre of  $H \rightarrow U$  is positive-dimensional and has ample negative canonical class. Its discrepancies are computed by primes horizontal over  $U$ ; after extending the constants of  $R = Q(U)$ , the preceding geometric comparisons make it  $\epsilon_0$ -lc. Thus it is geometrically  $\epsilon_0$ -lc Fano. Connectedness of the Mori contraction and characteristic zero make  $P/R$  regular. As a subfield of the regular extension  $P/Q$ , the extension  $R/Q$  is regular as well. The relative dimension of the Mori contraction is positive, so the transcendence degree strictly decreases.

For completeness,  $U$  is of Fano type by the following canonical bundle formula argument. Over an algebraic closure of  $Q$ , let  $B_H$  be an effective klt boundary with  $A_H = -(K_H + B_H)$  ample. Fix an ample divisor  $A_U$  on  $U$ . For sufficiently small rational  $e > 0$ ,  $A_H - eh^*A_U$  is ample. Choose a general effective rational representative  $D_H$  of this class with small coefficients so that  $(H, B_H + D_H)$  is klt. Then

$$K_H + B_H + D_H \sim_{\mathbb{Q}} -eh^*A_U.$$

This is a projective connected klt-trivial fibration with effective generic boundary; the rank-one verification in Lemma 10.2 applies. Apply the canonical bundle formula and the moduli-replacement construction from that lemma’s proof to this adjoint pullback. Its threshold discrepancy  $\bar{A}$  is positive at every nonzero divisorial base valuation: the DVR minimum in (10.5) is attained, and all its numerators are positive because the source pair is klt.

On a high smooth  $\pi : T \rightarrow U$ , let the moduli part  $M_T$  descend and be semiample, and let the discriminant  $B_T$  have SNC support. Choose compatible representatives as in (10.8), so that

$$K_T + B_T + M_T = \pi^*(-eA_U).$$

For sufficiently divisible  $q \geq 2$ , choose a general member  $H_T \in |qM_T|$ , taking  $H_T = 0$  if the system is trivial. The reduced transverse addition  $H_T$  leaves the discriminant subpair sub-lc, including when  $B_T$  has negative coefficients. Thus  $0 \leq u(H_T) \leq \bar{A}(u)$  for every divisorial  $u$ , and the discrepancy of  $B_T + H_T/q$  lies between  $\bar{A}/2$  and  $\bar{A}$ . Its pushforward  $B_U$  is effective: the discriminant trace is effective by (10.7), and the added divisor is effective. Formula (10.8), with  $N = -eA_U$ , gives the  $\mathbb{Q}$ -Cartier adjoint and its crepant pullback. Consequently  $(U, B_U)$  is klt and  $K_U + B_U \sim_{\mathbb{Q}} -eA_U$ , proving that  $U$  is of Fano type. The data descend to a finite Galois extension of  $Q$ ; averaging its conjugate boundaries preserves effectivity, klt and ampleness of the negative adjoint and descends the Fano type structure to  $Q$ . The algebraic characteristic-zero canonical bundle formula may be applied here after identifying the algebraic closure of the finitely generated extension of  $\mathbb{C}$  with  $\mathbb{C}$  as an abstract field and then descending these finite data.

Finally, for every prime of  $U$ , a prime of  $H$  over its generic point has effective carried boundary. Formula (10.7) gives the final assertion about the effectivity of all substituted forms on this model.  $\square$

#### 10.4. Induction for Lifting and Mixing.

*Proof of Theorem 9.1, assuming Theorem 10.1.* For Lifting, induct on  $n = \text{trdeg}_Q P$ . If  $n = 0$ , regularity implies  $P = Q$ ; take  $v = u$  and, for example,  $\delta = \lambda$ . If there are no nonzero valuations of  $Q$ , the assertion is vacuous.

Suppose  $n > 0$  and perform Lemma 10.3. Write  $R = Q(U)$  and use  $\epsilon_0 = 1/10$ . Theorem 10.1(1) and Lemma 10.2 give forms  $\sigma_R, \psi_R$  such that

$$(10.10) \quad \frac{1}{2}\bar{A}_\sigma \leq A_{\sigma_R} \leq \bar{A}_\sigma, \quad \frac{1}{2}\bar{A}_\psi \leq A_{\psi_R} \leq \bar{A}_\psi.$$

Their traces on  $U$  are effective,  $\sigma_R$  has a uniformly bounded index, and  $A_{\psi_R} > 0$ . Taking infima in  $A_\sigma \leq CA_\psi$  and using (10.10), we have

$$(10.11) \quad 0 \leq A_{\sigma_R} \leq 2CA_{\psi_R}.$$

Let  $\delta_B > 0$  be the threshold in Theorem 10.1(2) for the generic fibre  $P/R$  and target  $\lambda$ . If  $R = Q$ , that assertion directly proves the desired result. Otherwise apply the induction hypothesis to  $R/Q$ , its bounded-index reference  $\sigma_R$ , comparison constant  $2C$ , and target  $\delta_B/2$ . It supplies a uniform threshold  $\delta_I$ . For a given nonzero integral-valued  $u$  of  $Q$  we have

$$(10.12) \quad \inf_{z|_Q=u} A_{\psi_R}(z) \leq \inf_{w|_Q=u} A_\psi(w).$$

Indeed  $z = w|_R$  is nonzero divisorial, and  $A_{\psi_R}(z) \leq \bar{A}_\psi(z) \leq A_\psi(w)$ . If the right side is below  $\delta_I$ , induction gives an integral-valued  $z$  of  $R$  restricting to a positive multiple of  $u$ , with  $A_{\psi_R}(z) < \delta_B/2$ . Then (10.10) gives  $\bar{A}_\psi(z) < \delta_B$ , and the bounded-fibre lifting assertion produces the desired integral-valued  $v$  of  $P$ . Its restriction to  $Q$  is again a positive multiple of  $u$ . Each tower step lowers  $n$ , updates the reference index by a function of already bounded data, and doubles the comparison constant. There are at most  $d$  steps. The recursively chosen thresholds therefore depend only on  $d, r, C, \lambda$ , proving Lifting with its stated uniformity.

For Mixing, induct on  $\text{trdeg}_{\mathbb{C}} P$ . Dimension zero has no nonzero divisorial valuations and is vacuous. Apply Lemma 10.3 over  $\mathbb{C}$ , and write  $R = \mathbb{C}(U)$ . Theorem 10.1(3) gives

$$A_\chi \geq cA_\psi - DA_\sigma$$

on valuations trivial on  $R$  and on those restricting to a nonzero  $u$  with  $\bar{A}_\sigma(u) = 0$ . Together with  $A_\sigma \leq CA_\psi$ , this gives  $A_\chi \geq (c - DC)A_\psi$  there. Choose a rational  $b \in (0, 1)$ , depending only on the stated constants, with  $b(1 + DC) \leq 1/2$ , and define

$$A_{\chi_+} = (1 - b)A_\psi + bA_\chi.$$

It has effective boundary on  $H$ , and on the indicated valuations

$$(10.13) \quad A_{\chi_+} \geq (1 - b(1 + DC - c))A_\psi \geq \frac{1}{2}A_\psi > 0.$$

If  $R = \mathbb{C}$ , this treats every valuation and proves Mixing with  $a = b$ .

Otherwise (10.13) makes  $\chi_+$  klt on the generic model over  $R$ , since it applies to all valuations trivial on  $R$ . Its boundary there is effective, so base substitution applies even if some vertical discrepancies are negative. Obtain  $\sigma_R, \psi_R$  as in (10.10), and apply Lemma 10.2 to  $\chi_+$  with the anchor  $\psi$  and  $\gamma = 1/4$ . It gives  $\chi_R$  with

$$(10.14) \quad \bar{A}_{\chi_+} - \frac{1}{4}\bar{A}_\psi \leq A_{\chi_R} \leq \bar{A}_{\chi_+}.$$

All three traces on  $U$  are effective. The zero set of  $A_{\sigma_R}$  is exactly that of  $\bar{A}_\sigma$ , by (10.10). At a valuation in this set, taking infima in (10.13) gives  $\bar{A}_{\chi_+} \geq \bar{A}_\psi/2$  and hence

$$A_{\chi_R} \geq \frac{1}{4}\bar{A}_\psi \geq \frac{1}{4}A_{\psi_R}.$$

The lower-dimensional Mixing assertion on the Fano type model  $U$  applies with comparison constant  $2C$  and zero-set constant  $1/4$ . Thus a uniform rational  $a_R \in (0, 1)$  satisfies

$$(1 - a_R)A_{\psi_R}(u) + a_RA_{\chi_R}(u) > 0 \quad \text{for all nonzero divisorial } u \text{ of } R.$$

For a valuation  $w$  of  $P$  with nonzero restriction  $u$ , we have

$$A_\psi(w) \geq A_{\psi_R}(u), \quad A_{\chi_+}(w) \geq A_{\chi_R}(u),$$

by the barred definitions and the upper bounds in (10.10) and (10.14). The same convex mixture is therefore strictly positive upstairs. For valuations trivial on  $R$ , positivity follows from (10.13). Finally

$$(1 - a_R)A_\psi + a_R A_{\chi_+} = (1 - a_R b)A_\psi + a_R b A_\chi.$$

The product  $a = a_R b$  is positive, rational and uniform. The dimension decreases at every step, and only the controlled reference index and the displayed constants enter the induction. This proves Mixing and completes the reduction to Theorem 10.1.  $\square$

## 11. BOUNDED PRESENTATIONS AND RELATIVE LOGARITHMIC FRAMES

We begin the proof of Theorem 10.1. Throughout this section and the next two, its dimension, singularity and reference-index bounds are fixed. A constant called uniform depends only on these bounds, and on an additional numerical input when this is stated explicitly. Put  $n = \text{trdeg}_K L > 0$ .

**11.1. Finite extensions and projective presentations.** We first explain precisely how bounded finite extensions of  $K$  can be used. Since  $L/K$  is regular,  $L \otimes_K K'$  is a field for a finite extension  $K'/K$ ; we denote it by  $LK'$ . Let  $u'$  prolong a divisorial valuation  $u$  of  $K$ , with its scale chosen so that  $u'|_K = u$ . Every divisorial  $w$  of  $L$  over  $u$  has a compatible prolongation to  $LK'$ . Indeed, tensor the completion at  $w$  with the completion at  $u'$  over the completion at  $u$ , choose a field factor, and use its extended discrete valuation with the original scale. Its restriction to  $LK'$  has the required restrictions. It is divisorial, and the finite-extension discrepancy formula from Section 10 leaves the discrepancies of pulled-back forms unchanged. Consequently, for each prolongation separately,

$$(11.1) \quad \inf_{w'|_{K'}=u'} A_{\sigma_{LK'}}(w') = \inf_{w|_K=u} A_\sigma(w).$$

The two inequalities follow respectively by restriction and by the compatible prolongation just constructed. Prolonging any given valuation also proves that homogeneous discrepancy inequalities descend, including inequalities at valuations trivial on the base.

If  $u$  is integral-valued, its prolonged value group has denominator dividing a ramification index at most  $[K' : K]$ . Multiplication by that index restores integrality, at a bounded cost in a smallness threshold. For the first assertion of Theorem 10.1, an upstairs form descends by its norm: multiply its conjugate pluri-top forms in a normal closure. Dividing its divisor by the resulting tensor index gives the average of the conjugate normalized divisors. Formula (11.1) therefore preserves both discrepancy bounds, and the tensor index is multiplied by a bounded degree. One may equivalently use the usual norm over the given finite separable extension.

By BAB boundedness [9, Theorem 1.1], geometric  $\epsilon_0$ -lc Fano varieties of the relevant dimensions form a bounded family. Hence  $G$  admits a very ample polarization of uniformly bounded degree. It can be defined over a bounded extension of  $K$ . Here is a descent argument that does not presume a polarization over  $K$ . The geometric Picard group is finitely generated of bounded rank: klt Fano vanishing gives  $H^1(G_{\bar{K}}, \mathcal{O}) = 0$ , and the exponential sequence, after identifying the algebraically closed field with  $\mathbb{C}$ , bounds its rank by the Betti ranks in a bounded complex family. It is torsion-free. In fact, if a line bundle  $T$  is torsion, polynomiality of Euler characteristic gives  $\chi(T) = \chi(\mathcal{O}) = 1$ , and klt Fano vanishing applies to  $T$  as well. Thus  $T$  has a nonzero section; an effective numerically trivial divisor is zero, so  $T$  is trivial.

The continuous Galois action on this free group has finite image of uniformly bounded order, since finite subgroups of  $\text{GL}_q(\mathbb{Z})$  inject under reduction modulo 3. After an extension killing that action, the polarization class is invariant. Its complete linear system descends to a possibly twisted projective space: the isomorphisms of the polarization differ only by scalars, so their projectivizations satisfy descent. The dimension of this Severi–Brauer variety is bounded by the

degree and dimension of  $G$ . A splitting extension of degree at most the degree of its central simple algebra is therefore bounded. We may and do use an embedding over the resulting field, with  $A = \mathcal{O}_G(1)$ , whose ambient dimension and degree are bounded.

Choose a nonzero rational absolute top differential  $\zeta$  on  $K$ . The differential determinant for  $L/K/\mathbb{C}$  writes the reference form as

$$(11.2) \quad \sigma = \eta \otimes \zeta^{\otimes r_0}, \quad 1 \leq r_0 \leq r,$$

where  $\eta$  is a rational relative  $r_0$ -pluri-top form. Its divisor on  $G$  is nonpositive, and its total absolute degree is uniformly bounded. Indeed,

$$\deg_A \left( -\frac{1}{r_0} \operatorname{div}_G \eta \right) = -K_G \cdot A^{n-1};$$

adjunction to a general complete-intersection curve in the smooth locus bounds this intersection in the bounded embedded family. For  $n = 1$  the normal curve itself is smooth and the same formula applies.

Take a finite linear projection with affine coordinates  $z_1^0, \dots, z_n^0$ . The differential  $dz_1^0 \wedge \dots \wedge dz_n^0$  has poles bounded by a fixed multiple of the infinity hyperplane, as is seen by differentiating at each codimension-one point. Its zeros have bounded degree as well, by the preceding canonical-degree calculation. Hence

$$(11.3) \quad \eta = (dz_1^0 \wedge \dots \wedge dz_n^0)^{\otimes r_0} \frac{P}{Q},$$

where  $P, Q$  are restrictions of homogeneous polynomials of the same uniformly bounded degree.

For completeness, a rational function with pole divisor of degree at most  $a$  on one of these normal embedded varieties is a quotient of polynomial restrictions of degree at most  $C(1+a)$ . Cut each pole prime by a hypersurface containing it but not the variety, of degree at most the degree of the prime; a general finite linear projection gives such a hypersurface. Multiply the equations with the pole multiplicities. Normality makes the resulting numerator a regular section of the corresponding twist. Uniform regularity in the bounded projective Hilbert families lifts it to an ambient polynomial after a uniform further twist. Applying this argument to (11.2) gives (11.3). Applying it to a ratio of relative forms gives a second bound which will be used repeatedly: if another form has effective boundary on  $G$ , and the forms are compared at a common tensor index  $p$ , then their ratio has a polynomial presentation of degree at most  $Cp$ , independently of the index of that other form.

**11.2. Preparation valid at every parameter.** Hilbert schemes and the coefficients in (11.3) give finitely many algebraic families of varieties with rational relative forms. We retain the scalar coefficient of each form, so the form itself, and not just its divisor, specializes correctly. Their parameter spaces may be taken to be dense opens of projective varieties over  $\mathbb{C}$ . We will repeatedly compactify dominantly, resolve, and shrink these opens.

These operations can be arranged to cover every field-valued parameter under consideration. To see this, take reduced closures of the valid parameter points. On a dense open of each irreducible component, flatness and the open geometric loci give normal, geometrically integral fibres. Resolve the generic fibre together with the relative form divisor and spread this resolution out. After shrinking, its relative divisor has fixed simple-normal-crossings coefficients and specializes as the divisor of the fibre form; the exceptional and divisorial images are also constant in the required sense. Effectivity of the boundary and log canonicity are thus tested on an open. Density of the valid points forces these properties at the generic point, so this open contains all the desired properties. This argument uses no bounded index for the underlying  $K_G$ : the adjoint with the form boundary is already  $\mathbb{Q}$ -principal.

We also spread all inverse rational maps, their identities on dense opens, the exact differential and form identities, and their nonzero denominators. Shrinking makes these identities valid for

every field-valued point of the open, including a point whose map to the parameter space is not dominant. The finitely many irreducible closed complements have smaller dimension. Noetherian induction applied to those complements produces finitely many such packages. In particular, any later shrinking is accompanied by this induction; no parameter is discarded.

**Proposition 11.1** (Relative logarithmic preparation). *The finitely many packages can be chosen with projective morphisms  $f : Y \rightarrow B$  having the following properties. The base is smooth and strict toroidal; the source is strict simplicial toroidal; and every source cone maps onto a base cone. The good parameter open is disjoint from the base boundary. For each field-valued point used in a package,  $Y_K$  maps projectively and birationally to the prescribed  $G$ , and the differential and form identities specialize. The divisor of  $\eta$ , regarded as a relative logarithmic pluri-top form, is toroidal. Arbitrary prescribed proper closed subsets can be included in the boundary during this construction.*

*Proof.* We use weak toroidalization [2, Theorem 1.1] for projective dominant morphisms in characteristic zero, with prescribed closed subsets, and prove the additional assertion about the logarithmic form. Only projective birational modifications of the source over the resulting base change are needed.

*Induction and the relative form.* In relative dimension zero, mark the zeros and poles of the relative form function. A dominant generically finite toroidal morphism has invertible logarithmic differential determinant in characteristic zero. This follows in toric charts from the full rank of the map of torus lattices, including the unit directions. More generally that map has full target rank for a dominant toroidal morphism: otherwise the map on logarithmic differentials in the full toric frames would be generically rank-deficient even after completion, contrary to dominance and separability. Faithful flatness of completion transfers this rank test to the original chart.

For the inductive step, choose independent relative rational functions and resolve their graph to factor the morphism projectively through an intermediate base of relative dimension one less. The top arrow then has relative dimension one. In the determinant factorization write  $\eta = \eta_1 \otimes \eta_{\text{low}}$ , with the appropriate common tensor power understood. Apply weak toroidalization to the top arrow, marking the prescribed sets and the closures of the support of the divisor of  $\eta_1$  on its original normal proper generic curve. A birational normal proper model of that generic curve is the same curve. Consequently every component of the relative logarithmic divisor of  $\eta_1$  which is not already in the boundary is vertical.

Refine this top toroidal arrow so that its base  $B_1$  is smooth strict toroidal, its source is strict simplicial toroidal, and all cone maps are onto cones. Mark on  $B_1$  the images of those finitely many bad vertical components and the entire boundary of  $B_1$ . Apply the induction hypothesis to the lower arrow  $B_1 \rightarrow B_{\text{old}}$ . It gives an arrow  $B_2 \rightarrow B_{\text{new}}$ , mapping to the old lower arrow, with the logarithmic form property for  $\eta_{\text{low}}$ .

*Normalized main pullback.* We verify the normalized main pullback of the top arrow under  $B_2 \rightarrow B_1$ . This map need not be toroidal, or dominant on every subsequent parameter. The marked inverse images nevertheless ensure that the old boundary equations pull back to monomials times units in complete toroidal charts. One can see this using the log structure of functions invertible off the boundary. Alternatively, on a strict simplicial chart some multiple of the pull divisor is the divisor of a character monomial. Normality makes the corresponding power of the function that monomial times a unit. At a geometric closed point the unit has a formal root in characteristic zero. The first term for generic positive character weights shows that the monomial exponent itself is divisible by that power, proving the assertion for the original function.

At such a point of the old top arrow, write the boundary coordinates of the smooth base as monomials  $z^{p_j}$  times units in the source monoid. The  $p_j$  are linearly independent. The smooth base parameters transverse to its boundary can be made part of the smooth source parameter system: the stratum map is smooth, as follows from the full-rank torus map in characteristic zero. First absorb the old source units by a unit-valued character on the source lattice, taking formal roots

when its finite lattice index requires them. Scaling monomials by such units is an automorphism of the complete local ring; its action on the monoid generators in the cotangent space is invertible, and completeness gives an inverse by successive approximation. Combined with the preceding smooth-parameter change, the cotangent matrix is invertible. The logarithmic Jacobian of this old source-coordinate change is a unit. Its relative differential need not be zero, but it changes a logarithmic determinant frame by a unit and hence does not change its divisor.

Now pull back to  $B_2$ . Absorb the remaining unit factors, which come from the new base, along the old source characters. These multipliers have zero relative differential over  $B_2$ . Substitute for the old smooth base parameters, retaining all the smooth parameters of  $B_2$ . The remaining equations are binomial:

$$(11.4) \quad z^{p_j} = (z')^{q_j}.$$

Their normalized main branches are toroidal, with cone the intersection of the product cones with

$$p_*x = q_*y.$$

This intersection meets the relative interior of the product cone: the first cone maps onto the old base cone, and the relative interior of the second maps into the relative interior of its relevant old base cone. Inverting all monomials in (11.4) gives the character-group pushout. Its finite torsion, when present, separates the torus branches. The closure of each such branch is an affine monoid variety; its normalization is the saturation of that monoid in its character lattice. The interior condition makes its monoid sharp and gives a unique point over the monoid origin on each branch. Completion at that point is one normal monoid-series domain, with the free smooth parameters adjoined.

These are the branches of the actual algebraic main pullback. The joint nonboundary open is dense in that main component, so boundary monomials are nonzerodivisors there. Flat completion preserves this property; no minimal completed branch can be supported solely on vanishing boundary monomials. Thus the relevant completed branches are precisely the closures of the torus branches just computed. Excellence gives finite normalization and its compatibility with this completion; geometric generic integrality identifies the prescribed main generic component. These facts also justify performing the binomial computation first in polynomial monoid algebras and then completing. They do not require an arbitrary formal coordinate change to be algebraic.

The chart map to  $B_2$  retains its given character map and smooth parameters. Its cone is onto the  $B_2$  cone, since the old top cone was onto. Characters rationally complementary to the  $p_j$ , together with the remaining relative smooth parameters, give the same relative logarithmic determinant frame after pullback: saturation changes no rational span, and the new unit multipliers contribute only base differentials. It follows that the relative logarithmic determinant pulls back unchanged. The divisor of  $\eta_1$  is now wholly supported in the boundary, because its bad vertical images were marked. Composing with  $B_2 \rightarrow B_{\text{new}}$  gives a toroidal map; in these same charts its base equations are monomials, its log rank is full, and its stratum coordinates are smooth. The relative determinant factorization proves the assertion for  $\eta$ . The models continue to map birationally to the old ones and retain all inverse marks.

*Projective cone refinement.* We carry out these refinements on the finite-type packages over  $\mathbb{C}$ , before evaluating at field-valued parameter points. First take compatible projective toroidal resolutions, so that the source and base are smooth and strict. On the resulting regular source cone complex, assign value 1 to each primitive ray generator and extend linearly on each cone. These functions agree on common faces and define a slicing function  $h$ , positive away from the origins. Projective equidimensionalization [3, Proposition 4.4 and Lemma 4.1] then gives a smooth base and makes every source cone map onto a base cone. Retain  $h$  on the source subdivision: every new cone lies in an old cone, so  $h$  remains linear there and positive away from the origins. Apply [1, Corollary 0.11] to its 1-skeleton with the unchanged triangulation induced by the zero lifting function. This gives

a projective triangulation with no new rays. Since the base is simplicial, each old source ray maps to a base ray or to zero [3, Remark 4.2]; the image of each new simplicial cone is therefore a face of a base cone. The final source is strict and simplicial, and the cone-surjectivity condition is preserved. All these are toroidal modifications, so the inverse marks and the pullback identities for the logarithmic divisors are retained.

*Return to all parameters.* Apply the construction to the compactified families, and use the all-parameter shrinking and induction above. Any loci newly excluded by these refinements undergo the same noetherian restart before the finite package list and its common moduli multiple are fixed. On the good isomorphism open the parameter map already lifts to the modified base; properness then extends a valuation-ring map when required. The retained inverse maps and form identities give the claimed birational and differential assertions on every fibre used.  $\square$

All models, cones and chart functions in Proposition 11.1 belong to finitely many finite-type families. After the indicated shrinking and spreading, the generic models  $Y_K$  are bounded in fixed projective embeddings. Pullback of the polynomial presentations on  $G$  shows that a comparison function at tensor index  $p$  has generic zero and pole divisors on  $Y_K$  of combined degree at most  $Cp$ , counted with multiplicities.

**11.3. The minimum function.** Let  $\phi$  denote the toroidal order function of  $r_0^{-1} \operatorname{div}_{Y/B}^{\log}(\eta)$ . It is rational linear on each source cone. On every horizontal ray, that is, a ray mapping to zero, it is nonnegative by log canonicity of the generic form pair. For a base weight  $b$ , define

$$(11.5) \quad \mu(b) = \min_{f_*x=b} \phi(x),$$

using all source cones over its base cone.

Every cone mapping onto the cone at a given base point can be used over that point, with an appropriate ordinary-fibre component label. Indeed, its smooth stratum map has open dense image in the base stratum. Properness makes its closure meet every point of that stratum. At a deeper point of this closure, the face corresponding to the original stratum still maps onto the base cone. In the formal orbit-closure chart for that face, all base boundary monomials vanish identically, and the remaining base parameters are smooth stratum parameters which can be eliminated. The full-rank Jacobian in these unconstrained stratum coordinates survives passage to the face orbit-closure quotient. The open face stratum is dense in this fibre chart. Thus that fibre has actual points of the desired open source stratum. This last calculation, in addition to properness, supplies the assertion at every point. Points and components may be taken geometrically.

The minimum in (11.5) exists and is rational piecewise linear. On any simplicial cone, allocate each base coordinate to the source rays above its axis and choose a ray of least cost. Horizontal rays have nonnegative cost and can be set to zero. There are only finitely many cones and choices, so taking their minimum gives the claim. The resulting minima are continuous across base faces. A candidate on a face can be approached from every incident deeper base cone: properness specializes its source-stratum closure to a stratum over that deeper cone, and arbitrarily small added weights in the new directions supply the missing base coordinates. Conversely, a limit of minimum candidates from the interior supplies a face candidate; the horizontal nonnegative directions can be omitted. After compatible projective subdivisions, with the refinements in Proposition 11.1, we may therefore assume that  $\mu$  is linear on each base cone. Write

$$(11.6) \quad M_0 \text{ for the toroidal } \mathbb{Q}\text{-divisor on } B \text{ with order function } \mu.$$

The next section compares this geometric minimum with actual divisorial prolongations of a base valuation. That comparison will identify  $M_0$  as the moduli divisor and give the bounded-index base form.

## 12. TRAIT COMPARISON, BASE FORMS AND INTEGRAL LIFTING

Fix a package  $Y \rightarrow B$  from Proposition 11.1, a field-valued parameter with field  $K$ , and a nonzero divisorial valuation  $u$  on  $K$ . The generic parameter lifts on the good isomorphism open, and properness extends it to  $\text{Spec } \mathcal{O}_u \rightarrow B$ . We use  $\delta$  for a source cone, keeping  $\sigma$  for the reference form. Its stratum is  $O_\delta$ , its span lattice is  $N_\delta$ , and its dimension is  $j_\delta$ .

The four labelled formulas organize the argument. Formula (TC) describes the actual normalized main trait chart and its integral lattice. Formula (TE) separates discrepancy into the base term, the toroidal order, and a nonnegative excess; minimizing it gives (T), which identifies the geometric minimum with the barred reference discrepancy. The two estimates in (TV) control retraction and variation along a labelled cell. We then apply these calculations to construct the bounded-index base form and prove integral lifting, the first two assertions of Theorem 10.1. For lifting we return to the bounded parameter and axis extensions; the auxiliary roots used in proving (TV) do not enter its denominator bound.

**12.1. Bounded saturation and the normalized trait chart.** Near the centre on the smooth base take boundary equations  $t_1, \dots, t_a$ . The base cone is  $\mathbb{R}_{\geq 0}^a$ , with lattice  $\mathbb{Z}^a$ . A source cone over this centre maps onto it by  $p_*$ . Its stratum map is smooth: in toroidal charts the map of orbit tori has full target rank, and characteristic zero makes it smooth.

We first arrange, simultaneously on every mapped face span, that these maps are surjective on lattices. Take  $h$ -th roots of the base axes, where the fixed integer  $h$  is divisible by the indices of all the finitely many image lattices in their saturated image spans, and normalize the main pullback. The base lattice is replaced by  $h\mathbb{Z}^a$ , and each source lattice by its preimage under  $p_*$ . Since  $h\mathbb{Z}^a$  is contained in every relevant full image lattice, the new map is onto, including for every face mapping onto the base cone. For faces over smaller base faces the same assertion holds in their image span. Substitution of powers of the new axes, followed by monoid saturation, proves the chart description. Rational cone shapes and their order functions pull back unchanged. So does the relative logarithmic determinant, as is immediate in rational character spaces.

There can be several new strata of the same cone shape. The new source is nevertheless strict and simplicial: locally its different boundary germs lie over different old boundary germs and have the same real cone configuration. We use finitely many such charts along the fixed base strata. The Kummer bases are smooth, and the models remain projective over them. Every old cone option is available over every lifted base point. Indeed, the axis cover is finite flat; the main pullback before normalization meets every point of the fibre product, since the universal generic fibre is geometrically integral. The same conclusion follows directly from the local saturated monoid calculation. Normalization is finite and surjective onto that main pullback.

On an actual parameter this operation requires at most a bounded finite field extension. We prolong valuations with their old scale, as in (11.1). Hence neither the proposed discrepancy minimum nor its pulled-back order formula changes. From now on  $Y, B, t_j$  refer to these new data, and all ordinary-fibre labels are taken *after* this normalization. Set

$$b_j = u(t_j) > 0, \quad \tau = u(\pi) > 0, \quad k_0 = k(u), \quad \ell = b/\tau \in \mathbb{Z}_{>0}^a,$$

where  $\pi$  is a uniformizer of  $\mathcal{O}_u$ . When  $a = 0$ , the axis and relation lists are empty; all the arguments below include this case.

Complete the trait, choose a coefficient section fixing  $\mathbb{C}$ , and extend the residue field to an algebraic closure  $\kappa$ . Thus

$$R = \kappa[[\pi]], \quad K_R = \kappa((\pi)).$$

Let  $s \in B_\kappa$  be the resulting geometric closed point. Over  $R$  take the normalization of the main component of the pullback of  $Y$ , whose generic field is that of the geometrically integral generic model after scalar extension. The following calculation identifies its actual local branches.

At a geometric closed point  $y \in O_\delta$  over  $s$ , put  $M = N_\delta^\vee$  and  $P = \delta^\vee \cap M$ . Boundary coordinates pull back to  $X^{p_j}$  times units, with  $p_j = p^*e_j$ . Surjectivity of the lattice map splits its dual inclusion integrally:

$$(12.1) \quad M = p^*\mathbb{Z}^a \oplus M_{\text{rel}}.$$

Given prescribed unit multipliers on the  $p_j$ , assign value one on a basis of  $M_{\text{rel}}$  and extend multiplicatively. This produces a unit-valued character on  $M$  using only integral powers of units. In particular it does not extract roots of their residues, even at the generic point of a labelled stratum over a nonclosed residue field.

The substitution  $X^m \mapsto U(m)X^m$  preserves all monoid relations and is an automorphism of the complete local ring. On its cotangent monoid generators it is multiplication by nonzero residue scalars; the smooth variables can be fixed. Successive approximation gives surjectivity, and a surjective endomorphism of a Noetherian ring is injective. This also works on a singular toric chart, where monoid atoms are not independent coordinates. The old smooth base parameters can simultaneously replace the appropriate smooth source parameters, since their transverse Jacobian has full rank. The combined cotangent matrix is block triangular and invertible. Make these changes on the source first; after pullback, absorb the remaining trait units using (12.1) again. The latter multipliers come from the coefficient ring of the trait. Eliminating the smooth base parameters now gives precisely

$$X^{p_j} = \pi^{\ell_j} \quad (1 \leq j \leq a),$$

with  $n - j_\delta + a$  free smooth parameters. All these changes preserve the face ideals. If a generic strict toroidal chart is given only by an etale chart, the calculation may first be made after a separable residue extension; the preserved face ideals and the uniqueness below then descend it.

The dual cone and lattice of the normalized main chart are

$$(TC) \quad \begin{aligned} D_\delta &= \{(x, e) \in \delta \times \mathbb{R}_{\geq 0} : p_*x = \ell e\}, \\ \Lambda_\delta &= \{(x, e) \in N_\delta \oplus \mathbb{Z} : p_*x = \ell e\}. \end{aligned}$$

To verify normalization and completion in this assertion, consider the character-group pushout

$$G = (M \oplus \mathbb{Z}) / \langle (p^*c, -\ell \cdot c) : c \in \mathbb{Z}^a \rangle.$$

By (12.1), it is  $M_{\text{rel}} \oplus \mathbb{Z}$ , hence torsion-free. Let  $S$  be the image of  $P \oplus \mathbb{N}$  in  $G$ , and  $\bar{S}$  its saturation. A point  $x_0 \in \text{relint } \delta$  with  $p_*x_0 = \ell$  exists because an onto cone map takes relative interior onto relative interior. The functional  $(x_0, 1)$  is strictly positive on every nonzero image in  $S$ . Thus  $\bar{S}$  is sharp. The relation subspace meets the product interior, so  $D_\delta$  spans  $\Lambda_\delta \otimes \mathbb{R}$ ; no further quotient lattice is needed.

Localizing the binomial algebra at all monomials gives the group algebra of  $G$ , which has one torus component. Its closure is  $\kappa[S]$ , and its finite normalization is  $\kappa[\bar{S}]$ . Other components of the unsaturated binomial equations, if any, lie on the monomial boundary. Sharpness gives one monomial origin. The completion of the normal monoid algebra there, with the free smooth parameters adjoined, is a normal domain: the algebra is excellent, and a positive integral grading gives the domain property for its monoid series. Hence there is one normalized main formal branch at this origin.

This branch is the one in the algebraic trait model. Its generic fibre lies in the good open, is geometrically integral, and is not contained in any boundary component. Boundary monomials are therefore nonzerodivisors on the main model, and flat completion preserves that fact. No completed main branch is supported in the monomial boundary. The preceding torus closure calculation then lists exactly the main completed branches; excellence identifies their finite normalizations with completion of the algebraic normalization. This proves (TC) for the actual model, rather than merely for an abstract formal binomial quotient.

**12.2. Monomial valuations, labels and algebraic ray divisors.** Faces of  $D_\delta$  meeting  $e > 0$  correspond exactly to faces  $\beta \leq \delta$  mapping onto the base cone, and their relative interiors correspond. In the slice  $e = \tau$ , a rational point  $x$  defines a monomial valuation  $w_x$ : take the minimum of the weights  $(x, \tau)$  in a monoid series, giving the free smooth parameters weight zero, and then restrict to the function field. This is a valuation also on a face. The support of a nonzero series generates a monomial ideal in a finitely generated monoid, so finitely many exponents from that support give its minimum against every nonnegative weight. The zero-weight monomials and smooth parameters form the coefficient part; products of two nonzero initial forms remain nonzero in its series domain. The same argument proves that evaluation of every rational function is continuous on the entire closed slice: write it as a quotient of regular series and subtract the two finite minima. It applies to real as well as rational weights. Moreover,

$$(12.2) \quad (mx, m\tau) \in \Lambda_\delta \implies mw_x \text{ is integral-valued.}$$

If  $x \in \text{relint } \beta$ , label this valuation by the incident irreducible component of  $O_\beta \cap Y_s$ , taken geometrically in the *ordinary* parameter fibre. At a deeper point in its closure, the specified face determines its local branch, by strictness. The ordinary orbit-closure fibre germ for that face is irreducible: base boundary equations vanish on the orbit closure, and the smooth base parameters can be eliminated. It meets its open stratum densely.

The corresponding face centre in (TC) has exactly this ordinary-fibre branch. The relevant comparison is integral:

$$(12.3) \quad \Lambda_\delta / \Lambda_\beta \simeq N_\delta / N_\beta, \quad N_\beta = N_\delta \cap \text{span}_{\mathbb{R}} \beta.$$

Indeed, the map  $(n, e) \mapsto n \bmod N_\beta$  has kernel  $\Lambda_\beta$ . Given  $n \in N_\delta$ , the already arranged surjectivity  $p_* N_\beta = \mathbb{Z}^a$  supplies  $b' \in N_\beta$  with  $p_* b' = -p_* n$ . Then  $(n + b', 0)$  lifts its class. The quotient cones agree as well. For  $n \in \delta$ , choose  $e$  large enough that  $\ell e - p_* n$  is in the base cone, and choose  $b' \in \beta$  mapping to this difference. The point  $(n + b', e) \in D_\delta$  projects to the class of  $n$ , and the reverse containment is immediate. For an empty axis list the same proof simply uses the product cone.

Thus the quotient orbit-closure monoid has the original integral lattice, cone and free smooth coordinates. There is no finite orbit cover at the generic ordinary-fibre component. The old unit-character changes preserve face ideals and are formal automorphisms of these quotients; the subsequent trait multipliers reduce to constants on the face centre, where the axes and  $\pi$  vanish. Hence the centre maps dominantly to the stated ordinary branch. At an interior point, where  $\delta = \beta$ , its free smooth coordinates are precisely the coordinates of that branch. Formula (12.3) proves compatibility at every deeper incident point.

We next show that rational  $w_x$  is an intrinsic algebraic ray order with this label. Choose effective Cartier multiples of the boundary primes. These are available on the strict simplicial source and their characters span the dual of the cone over  $\mathbb{Q}$ . For every positive boundary weight, compare an appropriate positive power of its equation with a power of  $\pi$  having the same weight at  $(x, \tau)$ . The corresponding two-generated ideals impose the equal-weight walls. A zero boundary weight already gives a face wall. Together the walls isolate the ray through  $(x, \tau)$ , because the boundary equations give separate axis multiples on the simplicial cone. These ideals are algebraic: use the Cartier divisor ideals and  $(\pi^q)$ ; changing a local generator by a unit does not change either the ideal or its wall. The choices can be used consistently wherever this face and ray occur.

Their normalized blowups are projective vertical modifications, and do not change the generic fibre. In the completed chart, adjoining the blowup ratios and saturating is the toric subdivision by those walls. It extracts the desired ray prime. That prime remains irreducible after completion of the original chart. To see the last point explicitly, in the algebraic toric modification its image is the orbit closure of the supporting old face. The generic fibre over this closure is geometrically integral: the orbit-lattice map is the projection modulo the larger face span, which is onto with torsion-free kernel. The supporting orbit closure is still integral on completion. Flat base change

makes any newly appearing minimal prime contract to the old ray prime. It therefore meets the inverse image of an open where this map of orbit closures is flat, and on that open dominates the completed supporting closure. Geometric integrality of the generic fibre permits only one such component.

The minimum-weight rule for monoid series is the scaled order of this prime. One precise verification is to make a smooth toric refinement retaining the ray. Powers of the ideal of its ray divisor push down to the monomial ideals defined by the corresponding order cutoffs. Flat completion preserves these pushforwards, and membership in the completed ideals is termwise membership. The ray divisor remains a single reduced Cartier divisor there; the excellent completion is a regular map. This proves the order calculation on all series and therefore on rational functions.

Contract the completed ray prime to the algebraic modification over  $R$ . Its height is one: it contains the nonzero vertical parameter, whereas flatness bounds the height of its contraction by the height-one prime above it. Its image dominates the labelled ordinary-fibre component, by the orbit calculation. It is the unique algebraic ray prime with the given label and weights. Indeed, at an interior point of that component the normalized main chart has one branch and the same free stratum coordinates. On its ray modification there is one ray prime dominating that local component. Properness brings every global prime candidate dominating the component above this chosen point. Faithful flatness lifts its generic point to the completed modification; the equal-cut unit conditions and positive vertical order place it in that single ray closure. Its height-one contraction forces all candidates to coincide. Equivalently, one can work at the generic component, using the primitive unit splitting and (12.3) to exclude both unit-root branches and orbit covers.

The orders computed at different interior points, or at incident deeper points with this same label, consequently agree, with the scale determined by  $w_x(\pi) = \tau$ . This is uniqueness of the toric ray order; a valuation with additional positive excess can of course have the same boundary weights. Paths between labelled cells will use only comparable strata with incident ordinary-fibre components. A segment, including its incident endpoints, can be evaluated in the one full chart at the deeper labelled point. All labels thus come from the fixed parameter families.

These identifications also commute with a root of the uniformizer. If  $\pi = \rho^d$ , then

$$(12.4) \quad \Lambda_d = \{(n, e) : p_* n = d\ell e\}, \quad \Lambda_d \longrightarrow \Lambda_\delta, \quad (n, e) \longmapsto (n, de).$$

The weight  $(x, \tau/d)$  maps to  $(x, \tau)$ , so the labelled ray order restricts with its original scale. The quotient calculation (12.3) is independent of  $\ell$ , proving preservation of the ordinary-fibre label.

*Remark 12.1.* The preliminary saturation and the subsequent choice of labels cannot be omitted. On the open set  $y(y^2 + x) \neq 0$ , the map

$$t = x^2(y^2 + x)$$

has boundary lattice map multiplication by two at the ordinary component  $x = 0$ . Its generic fibre is geometrically integral: with  $z = xy$ , its equation is  $z^2 = t - x^3$ . After  $t = \pi^2$ , normalization introduces  $q = \pi/x$ , with

$$q^2 = y^2 + x, \quad \pi = xq.$$

The special fibre over that old component has two primes,  $q = y$  and  $q = -y$ . Both have boundary weights  $w(x) = w(\pi) = 1$ ,  $w(y) = 0$ , but their orders differ on  $q - y$ . The bounded Kummer normalization separates them into different new strata, each with primitive boundary map. Taking the labels on that new model is exactly what makes the uniqueness proved above applicable.

**12.3. Logarithmic coordinates and the excess formula.** Near a chosen stratum point, take equations for effective Cartier multiples of the boundary primes. Select  $j_\delta - a$  of their characters which rationally complement the base axes, and add regular units with independent relative stratum differentials. This gives a list of  $n$  functions  $z = (z_1, \dots, z_n)$ . In the complete toroidal chart the

character block and the stratum-coordinate block are invertible, so  $d \log z_1 \wedge \cdots \wedge d \log z_n$  is a relative logarithmic determinant frame. Finitely many Zariski neighbourhoods along the finitely many strata suffice. Their functions have uniformly bounded rational presentations in the fixed family embeddings: use finitely many affine neighbourhoods and express their regular functions by bounded-degree ratios whose denominators are nonzero there. A chart used by the main trait closure has denominators surviving generically. Hence these are actual rational functions on the corresponding generic model.

The ratio

$$(12.5) \quad F = \frac{\eta}{(d \log z_1 \wedge \cdots \wedge d \log z_n)^{\otimes r_0}}$$

is toroidal locally, of order function  $r_0 \phi$ . Algebraically, a suitable power becomes a unit after dividing by the corresponding boundary equations. The retained fibrewise differential identities and nonzero denominators ensure that this assertion specializes to the prescribed form in  $L$ , including for nondominant parameter maps.

The labelled valuation  $w_x$  is the Gauss valuation on  $K(z)$ , with  $z$ -values linear in  $x$  and the smooth units of weight zero. First suppose  $x$  is rational and interior to the cone. Modulo the relations  $p_j = \ell_j[\pi]$ , the complementary boundary characters and  $[\pi]$  are rationally independent. Their angular residues, obtained by clearing denominators with powers, therefore have  $j_\delta - a$  independent character directions. Their unit factors reduce to units in the stratum parameters. The remaining  $z_i$  reduce to functions of the free smooth parameters with independent differentials. These two blocks give  $n$  algebraically independent angular residues over the base residue field. One can take a root of  $\pi$  to make each rescaled  $z_i$  have weight zero, or take powers of the angular expressions to stay over the original field. This proves Gauss without cancellation. Continuity of series evaluation extends the Gauss formula to faces and real weights in the fixed chart.

In particular, for rational  $x$ , the restriction of  $w_x$  to  $L$  is divisorial. Its value group is discrete, its residue field has the  $n$  independent additional residues just exhibited over the divisorial base residue field, and  $L/K(z)$  is finite. The rank-one Abhyankar criterion gives divisoriality. The same argument applies at a face, using its chart, and when some rounded coefficients in the final construction below vanish.

Conversely, a rational divisorial  $w$  on  $L$  over  $u$  has a centre with rational boundary weights  $x$  in one of these slices. It can be followed on the geometric trait model with its scale unchanged. Indeed,  $\widehat{L}_w$  contains  $\widehat{K}_u$ ; a coefficient field for the former can be chosen extending the chosen one for the latter by lifting a residue transcendence basis and then using separable Hensel lifting. In a Laurent-series presentation, extend this coefficient field to an algebraic closure and thereby embed  $\kappa((\pi))$  compatibly. The scalar-extension function field embeds as well: regularity of  $L/K$ , and the  $n$  independent additional residues from relative Abhyankar equality, ensure that transcendence degree is not lost after the algebraic base-residue extension. The prolongation has the same value scale. Its centre selects a component label; use a closed specialization within that component when choosing a chart, and let  $w_x$  be the resulting retraction. These large residue and completion fields are used to calculate charts and values of functions, never to define absolute form discrepancies.

**Proposition 12.2** (Trait comparison). *For every parameter map allowed above and every nonzero divisorial base valuation  $u$ , the barred reference discrepancy is*

$$(T) \quad \bar{A}_\sigma(u) = A_\zeta(u) + \mu(b).$$

*For a divisorial  $w$  over  $u$ , with labelled monomial retraction  $w_x$ , there is an excess satisfying*

$$(TE) \quad A_\sigma(w) = A_\zeta(u) + \phi(x) + E(w), \quad E(w) \geq 0, \quad E(w_x) = 0.$$

Let  $\omega$  be any other form with effective boundary on  $G$ , and compare it to  $\sigma$  at a common tensor index  $p$ , writing

$$A_\omega(v) = A_\sigma(v) + \frac{v(H)}{p}.$$

There is a uniform constant  $C_2$  such that

$$(TV) \quad \begin{aligned} |w(H) - w_x(H)| &\leq C_2 p E(w), \\ |w_x(H) - w_{x'}(H)| &\leq C_2 p \|x - x'\|. \end{aligned}$$

The second inequality holds along every segment in one labelled slice cell, including incident endpoints, in fixed cone coordinates. The rational monomial valuations in these assertions are the compatible algebraic labelled ray orders constructed above. Formulae (T) and (TE) do not require absolute log canonicity of  $\sigma$ .

*Proof.* First compute the excess using finitely generated fields. After a finite base extension by a root uniformizer  $\tilde{\pi}$ , clear the ratios  $w(z_i)/u(\pi)$  and put  $z'_i = \tilde{\pi}^{-h_i} z_i$ , of value zero. Prolong all valuations with the old scale. Form discrepancies are unchanged under this finite extension. On the smaller rational field in the variables  $z'$  over the extended base field, extract the prolonged base prime and take, near its generic point, a smooth base model times the smooth unit torus with coordinates  $z'_i$ . Denote this model by  $T$ , its reduced base prime pullback by  $P_0$ , and the prolonged restriction of  $w$  by  $v$ . Then

$$(12.6) \quad E(w) = A_{T,P_0}(v).$$

To check the equality, use the finite-extension discrepancy formula for  $L/K(z)$ , the ratio (12.5), and invariance of the relative  $d \log z$  determinant under rescaling by base functions. They reduce the absolute form to  $\zeta \wedge d \log z'_1 \wedge \cdots \wedge d \log z'_n$ . At the base prime, the divisor of  $\zeta$  contributes exactly  $A_\zeta(u)$  after the discrepancy of the reduced-fibre pair is separated off. The contribution of  $F$  is  $\phi(x)$ . This calculation uses the homogeneous scale throughout and proves (TE). The smooth reduced pair  $(T, P_0)$  is lc, so  $E(w) \geq 0$ ; for  $w_x$ , the established Gauss rule is the order of  $P_0$  with its scale, giving  $E(w_x) = 0$ . Every cone option occurs, and the minimum defining  $\mu(b)$  has a rational minimizer with a label. Minimizing (TE) therefore proves (T), including for the universal generic family.

We now prove (TV). The combined degree of the generic zero and pole divisors of  $H$ , with multiplicities, is at most  $C_1 p$ . This remains a uniform bound after the preliminary bounded Kummer preparation. Its normalized models are finite over the old base pullbacks; over the good opens they give the prescribed scalar-extended generic models mapping to  $G$ . Pulling back the bounded polynomial cuts on  $G$  bounds degrees in fixed projective embeddings of these finitely many families. Intersection degrees of fixed line bundles in finite type are bounded after stratifying by flatness. The generic fibres used are geometrically normal, both by preparation and, off the base boundary, by eliminating smooth parameters in the toroidal charts.

For the first inequality take the root uniformizer large enough that  $x/u(\tilde{\pi}) \in N_\delta$ . This auxiliary root degree is allowed to be unbounded. If  $\pi = \rho^d$  is the chosen extension, the new ray has primitive lattice vector

$$(12.7) \quad v_0 = (dx/\tau, 1), \quad \Lambda_d = \mathbb{Z}v_0 \oplus \ker(e).$$

Its last coordinate proves primitivity and the displayed splitting. Thus the open ray chart is a torus over  $\kappa[\rho]$ , with  $\rho$  a regular coordinate vanishing to order one. Including the free smooth parameters gives a regular chart with a single reduced smooth vertical divisor.

Make the algebraic cut-ideal modification over the new discretely valued function field, and follow the prolonged  $w$ . Every equal-cut ratio has value zero; in a blowup chart the ratio or its inverse is regular, so it is a unit at the centre. Imposing all these unit conditions, and removing boundary components outside the supporting face, puts the centre in this open ray locus. This remains true

when smooth parameters vanish additionally at the centre: the open ray chart, with its smooth parameters, is still regular. The vertical prime is the prime with the label already specified, and its scaled order agrees with  $w_x$  by (12.4). The rescaled boundary functions  $z'_i$  are units: a suitable power is a toric character of zero ray order times a local unit, and normality removes that power. The chosen free stratum units remain units as well.

These assertions hold on the algebraic trait, not only on its geometric completion. The map from its excellent characteristic-zero valuation DVR to the completed trait with algebraically closed residue is flat and regular, with the same uniformizer. Regular base change preserves normality. Geometric generic integrality and flatness identify the main component, so finite main normalization commutes with this base change. Blowups commute with flat base change, and the same normality argument applies to their finite normalizations. Smoothness and a reduced smooth special fibre may consequently be checked in the geometric completed charts and descend at the centre. The resulting local vertical divisor downstairs has the same uniformizer-normalized order.

In this regular local ring factor the principal divisor of  $H$  into its vertical order and its horizontal prime factors. The vertical part gives  $w_x(H)$ . Thus  $w(H) - w_x(H)$  is a signed sum of the orders of effective local Cartier horizontal primes  $D$ , with their zero or pole multiplicities. Their total weighted geometric generic degree is still at most  $C_1 p$ . The modifications have changed no generic fibre. Under the possibly unbounded scalar extension the degrees of split components sum, with multiplicities, to the original geometric degree. Rescaling the  $z_i$  by base constants changes coefficients, not their rational presentation degrees. Thus this degree budget has no dependence on the auxiliary root degree.

For each horizontal  $D$ , choose a nonzero polynomial  $q(z)$  vanishing on its generic image with

$$(12.8) \quad \deg q \leq C_3 \deg D.$$

The  $z$ -projection is defined at the generic point of  $D$ , since the rescaled coordinates are regular units at our centre and their chosen presentations have nonvanishing denominators on this neighbourhood. Its image closure has dimension at most  $n - 1$ . Bezout, applied to the bounded-degree presentations of  $z$ , bounds its degree by  $C \deg D$ , even when the image is contracted. A subvariety of that degree in affine  $n$ -space lies in a nonzero hypersurface of at most that degree, by generic linear projection. This proves (12.8) over the actual extended base field. Rewrite  $q$  in  $z'$  and multiply by a base constant so that its minimum coefficient valuation is zero. It is regular at the centre and its reduction is a nonzero polynomial. It still vanishes along  $D$ , whence

$$(12.9) \quad 0 \leq w(D) \leq v(q).$$

On a smooth torus over a characteristic-zero field, a nonzero polynomial of degree  $d_q > 0$  has log canonical threshold at least  $1/d_q$ . One may extend the field algebraically and prove this local assertion at each zero as follows. Choose successive general affine linear sections through the point, ending in a line on which the polynomial is not identically zero. Its vanishing order on that line is at most  $d_q$ , so the line pair with coefficient  $1/d_q$  is lc. Smooth-divisor inversion of adjunction [26], applied successively and then dropping the added section divisor, gives log canonicity in the ambient smooth space near the point. Points where the polynomial is a unit impose no condition. A nonzero constant likewise contributes zero order.

Apply this bound to the reduction of the normalized  $q$ . Inversion of adjunction along the smooth reduced fibre  $P_0$  gives

$$(12.10) \quad v(q) \leq (\deg q) A_{T, P_0}(v) = (\deg q) E(w).$$

It suffices to apply this over the generic point of the extracted base prime, where all centres in use lie. Spreading its residue-field resolution, or spreading the polynomial lc assertion over a base neighbourhood, verifies the hypotheses there. Combining (12.8)–(12.10) and summing with the zero and pole multiplicities proves the first inequality of (TV).

For the second inequality keep one logarithmic coordinate chart and its label along the segment. There is a polynomial relation

$$(12.11) \quad \sum_j a_j(z) H^j = 0, \quad \deg_z a_j \leq C_4 p.$$

Indeed,  $z$  is generically finite by differential independence. Take the equation of its irreducible graph image in  $\mathbb{P}_z^n \times \mathbb{P}_H^1$ , then dehomogenize. Its degree in the  $z$ -block is counted by  $n - 1$  general  $z$ -hyperplanes and one general  $H$ -hyperplane. These intersections can be taken in the rational-map open over which the generic map to the graph image is finite. The  $z$  cuts have bounded degree and the general  $H$  cut has degree at most  $C_1 p$ , bounded by its pole divisor. Bezout in the fixed bounded embedding proves (12.11). The chosen chart-presentation open is dense on the generic fibre and its image contains a dense open of the graph image, so it suffices for this computation. For constant  $H$  the assertion is immediate.

By the Gauss rule over the fixed  $u$ , the value of each nonzero  $a_j(z)$  is a finite minimum of affine functions on the segment, with Lipschitz constant at most  $C_5 p$ . In (12.11) the least term value must be attained at least twice. Hence  $w_x(H)$  lies among the finitely many candidates

$$\frac{w_x(a_i) - w_x(a_j)}{j - i} \quad (i \neq j),$$

each piecewise affine with Lipschitz constant at most  $2C_5 p$ . The actual value is continuous on the closed labelled segment by the monoid-series calculation. Partition the segment at the finitely many affine breakpoints and intersections of these candidates. On each remaining interval a continuous selection follows one of the affine candidates, and therefore has the same Lipschitz bound. Adding lengths gives that bound on the whole segment, including its labelled endpoints. Enlarge  $C_2$  to dominate the constants in the two inequalities. This proves (TV).  $\square$

**12.4. The bounded-index base form.** We prove Theorem 10.1(1). For this construction return to the prepared projective families of Proposition 11.1, before the axis extensions. The minimum identity (T) holds on these families by (11.1) and the unchanged pulled-back order functions. It applies to every allowed field-valued parameter map, including a nondominant one, and to the universal generic family with any absolute base volume form, without requiring absolute log canonicity.

Apply the lc-trivial canonical bundle formula and b-semiampleness [5] to the universal generic form  $\eta \otimes \zeta^{\otimes r_0}$ . Use a projective fibration model carrying the original effective lc generic pair; effectivity of its crepant transform on  $Y$  is unnecessary. The rank-one condition and threshold interpretation are precisely those verified in Section 10. On every smooth higher base model, (T) identifies the discriminant trace as the negative of the canonical divisor represented by  $\zeta$ , minus the pullback of  $M_0$ . The absolute adjoint is  $\mathbb{Q}$ -principal. Thus  $M_0$  represents the actual moduli divisor and is  $\mathbb{Q}$ -semiample. If the base-free multiple is first obtained on a higher model, it descends along the projective birational map: the pushforward of its structure sheaf is  $\mathcal{O}_B$ , and the pullback divisor is the pullback of  $M_0$ . Choose  $m \geq 2$  with  $mM_0$  globally generated. The finitely many packages make this choice uniform.

Resolve the actual parameter map and the divisor supports on a smooth projective model of  $K$ . The pullback  $M_{0,K}$  is defined even for a nondominant map, because its generic point lies off the boundary support. Formula (T) says

$$\bar{A}_\sigma(u) = A_\zeta(u) + u(M_{0,K}) \geq 0.$$

Choose a general member  $H = mM_{0,K} + \text{div}(h)$ , transverse to a log resolution of this lc form boundary. Along every component of  $H$  the old boundary coefficient is zero; the reduced transverse addition therefore preserves log canonicity. In valuation language,  $u(H) \leq \bar{A}_\sigma(u)$ . The absolute

form

$$(12.12) \quad \sigma_K = \frac{\zeta^{\otimes m}}{h} \quad \text{satisfies} \quad A_{\sigma_K}(u) = \bar{A}_\sigma(u) - \frac{u(H)}{m}.$$

Since  $m \geq 2$ , this lies between  $\frac{1}{2}\bar{A}_\sigma(u)$  and  $\bar{A}_\sigma(u)$ . Taking the norm if a bounded preliminary field extension was used proves the required bounded index on the original base. If  $K = \mathbb{C}$ , there are no nonzero base divisorial valuations and a nonzero scalar form of index one gives the same assertion.

**12.5. Return to bounded denominators.** We prove Theorem 10.1(2). Let  $u$  be integral-valued and choose a rational divisorial  $w$  over it with  $A_\psi(w)$  sufficiently small. Perform only the bounded parameter and axis extensions, retaining the old scale. Their degrees are bounded by some fixed integer  $N$ . Set  $D_0 = \text{lcm}(1, \dots, N)$ , enlarging  $N$  to bound the total extension degree. The values  $b, \tau$  then belong to  $D_0^{-1}\mathbb{Z}$ . All data in this paragraph refer to these bounded extensions; the auxiliary roots used to prove (TV) have been discarded.

Equations (T) and (TE) give

$$A_\sigma(w) = \bar{A}_\sigma(u) + (\phi(x) - \mu(b)) + E(w).$$

The first two summands are nonnegative. The hypothesis  $A_\sigma \leq CA_\psi$  therefore yields  $E(w) \leq CA_\psi(w)$ . Applying (TV) to the ratio with  $\psi$  gives

$$(12.13) \quad A_\psi(w_x) \leq C_6 A_\psi(w), \quad C_6 = 1 + CC_2.$$

In fact the difference equals  $-E(w)$  plus  $(w_x(H) - w(H))/p$ ; the displayed larger constant works without a sign restriction on  $C_2 - 1$ . On each closed labelled cell,  $A_\psi(w_x)$  has a uniform Lipschitz constant  $L_0$ : this follows from the second inequality of (TV) and the fixed linear part  $\phi$  in (TE).

Write  $x = \sum_{i=1}^s d_i r_i$ , with  $d_i \geq 0$  and the primitive rays  $r_i$  of its source cone. For the finite list of cones choose uniform bounds

$$S_0 \geq s, \quad P'_0 \geq \sum_i \|p_* r_i\|_\infty, \quad R_0 \geq \sum_i \|r_i\|.$$

Given a small  $\epsilon > 0$ , choose an integer  $Q \geq 2$  with  $Q > P'_0$  and  $Q^{-1} \leq \epsilon$ . Place the  $Q^s + 1$  points

$$j(D_0 d_1, \dots, D_0 d_s) \pmod{\mathbb{Z}^s}, \quad 0 \leq j \leq Q^s,$$

in the  $Q^s$  half-open cubes of side  $Q^{-1}$ . Two lie in the same cube. Subtracting them gives  $1 \leq h \leq Q^s$  and integers  $q_i$  such that, with  $m = D_0 h$ ,

$$(12.14) \quad |md_i - q_i| < Q^{-1}, \quad D_0 \leq m \leq D_0 Q^{S_0}.$$

Since  $md_i \geq 0$  and  $Q^{-1} < 1$ , all the  $q_i$  are nonnegative. The residual vector  $\sum_i q_i p_* r_i - mb$  is integral, and its sup norm is less than  $P'_0/Q < 1$ . Hence it is exactly zero. Thus

$$y' = \sum_i \frac{q_i}{m} r_i \quad \text{satisfies} \quad p_* y' = b, \quad m \|y' - x\| < R_0 \epsilon.$$

It lies in the same closed cell, with the incident label supplied by the fixed chart, even if some  $q_i$  are zero. For a cone without rays the same assertion holds with  $x = y' = 0$  and  $m = D_0$ ; the approximation step is empty.

Now  $(my', m\tau) \in \Lambda_\delta$ , and  $m\tau > 0$ . By (12.2) and the Gauss residue calculation,  $v = mw_{y'}$  is integral-valued and divisorial. Its restriction to the original field remains integral-valued and divisorial, and its base restriction is  $mu$ . Homogeneity, the Lipschitz estimate and (12.13) give

$$(12.15) \quad A_\psi(v) \leq mC_6 A_\psi(w) + L_0 R_0 \epsilon.$$

First choose  $\epsilon$  with  $L_0 R_0 \epsilon < \lambda/2$ , omitting this condition if  $L_0 R_0 = 0$ . Then choose the input threshold smaller than  $\lambda/(2D_0 Q^{S_0} C_6)$ . If  $\bar{A}_\psi(u)$  is below that threshold, the defining infimum supplies the required  $w$ , and (12.15) gives  $A_\psi(v) < \lambda$ . The choices are in the stated order and are uniform. This proves the second assertion of Theorem 10.1.

### 13. LC INCIDENCE AND COMPLETION OF THE BOUNDED-FIBRE ARGUMENT

It remains to prove part (3) of Theorem 10.1. We first consider a divisorial valuation  $w$  whose restriction  $u$  to  $K$  is nonzero and satisfies  $\bar{A}_\sigma(u) = 0$ . Make the bounded parameter and Kummer extensions of Sections 11 and 12, and prolong  $w$  with its scale unchanged. The barred equality and the hypothesis on the zero set of  $A_\sigma$  are preserved under these extensions. We use the notation of Proposition 12.2. In particular  $s$  is the geometric closed point on the parameter base,  $b$  is its vector of boundary orders, and a label is an irreducible component of a stratum in the *ordinary* fibre  $Y_s$ .

Define the nonnegative piecewise linear function

$$(13.1) \quad g(x) = \phi(x) - \mu(p_*x).$$

The subdivisions already made ensure that  $g$  is linear on every source cone. Equations (T) and (TE) give

$$(13.2) \quad A_\sigma(w) = g(x) + E(w), \quad p_*x = b, \quad E(w) \geq 0.$$

Here  $x$  is the boundary-weight vector of  $w$ , and the support cone  $\delta$  of  $x$  carries its label. We will connect this labelled vector to a zero of  $g$  in the same slice, with path length controlled by  $g(x)$ . The next two subsections establish the incidence of zero-gap strata needed for that construction.

**13.1. An effective dlt model for the parameter family.** The incidence argument takes place on the algebraic parameter family  $f : Y \rightarrow B$ , including the local Kummer changes, after extension to the algebraically closed residue field  $\kappa$ . It does not take place on the formal trait. The base is smooth and may be shrunk to a connected affine neighbourhood of  $s$ . The morphism is projective, its generic fibre is geometrically integral, and its fibres are connected by Stein factorization and normality of  $B$ . Its generic fibre maps birationally to the original effective lc form model, which we denote by  $G_{\text{gen}}$ . These properties persist after a connected pointed étale base change.

Let  $\partial Y$  be the reduced toroidal boundary, and pull all the parameter data to the present base. Consider the ordinary subpair

$$(13.3) \quad C^\natural = \partial Y - \frac{1}{r_0} \operatorname{div}_{Y/B}^{\log}(\eta) + f^*M_0.$$

The relative log determinant gives  $K_Y + C^\natural \sim_{\mathbb{Q},B} 0$ . The ordinary log discrepancy of a toroidal ray is  $g$  on that ray, including after subdivision. Thus  $(Y, C^\natural)$  is sub-lc: on a smooth toroidal resolution its boundary is SNC with coefficients at most one. On the generic fibre it is crepant to the effective form boundary on  $G_{\text{gen}}$ .

The construction uses Birkar's very-exceptional-divisor method [6, Section 3 and Theorem 3.4]; the argument below specifies the finite model on which its negativity step is applied.

**Lemma 13.1.** *There is a smooth projective toroidal resolution  $P \rightarrow Y$  and a finite sequence of ordinary  $\mathbb{Q}$ -factorial dlt MMP steps over  $B$  from  $P$  to a model  $P_j$  with an effective dlt boundary  $D_j$  such that*

$$K_{P_j} + D_j \sim_{\mathbb{Q},B} 0,$$

*and  $(P_j, D_j)$  is crepant to  $(Y, C^\natural)$ . The lc centres of the crepant subpair on  $P$  and of  $(P_j, D_j)$  correspond by strict transform, preserving their inclusions and their proper images in  $B$ .*

*Proof.* Choose  $P$  smooth with strict SNC toroidal boundary, and let  $C_P^\natural$  be the crepant transform of  $C^\natural$ . Set

$$(13.4) \quad D = \max\{C_P^\natural, 0\}, \quad N = D - C_P^\natural.$$

Then  $(P, D)$  is effective dlt,  $N \geq 0$ , and

$$(13.5) \quad K_P + D \sim_{\mathbb{Q},B} N, \quad A_{C^\natural}(v) = A_D(v) + v(N).$$

The lc places of the two pairs agree. To check the nontrivial direction, the centres of lc places of the SNC pair  $(P, D)$  are intersections of coefficient-one components, and none is contained in the components truncated in (13.4). Every such place has  $v(N) = 0$ . Conversely  $A_{C^\natural} = 0$  forces both nonnegative terms on the right of (13.5) to vanish. On the generic fibre  $N$  is exceptional over  $G_{\text{gen}}$ , since the form boundary on that model is effective.

We construct an MMP with scaling, give the scaling alternative, and then show that the exceptional error vanishes on a finite model. All the varieties used here are of finite type over  $\kappa$ . If needed for the ordinary complex statements of the MMP theorems, transport the entire family by an abstract isomorphism  $\kappa \simeq \mathbb{C}$ . Such an isomorphism exists because  $\kappa$  is the algebraic closure of a finitely generated extension of  $\mathbb{C}$  and has the same transcendence degree over  $\mathbb{Q}$  as  $\mathbb{C}$ . This identification need not fix the original copy of  $\mathbb{C}$ .

Choose a rational ample divisor  $H$  with  $K_P + D + H$  nef over  $B$ . Write  $P_i, D_i, N_i, H_i$  for successive models and pushforwards, and put  $C_i^\natural = D_i - N_i$ . These subpairs are crepant throughout: the adjoint of  $C^\natural$  is rationally linearly pulled back from the base, so, with compatible canonical divisors, its pullbacks to a common resolution agree. For each negative MMP step, with common resolution maps  $a$  and  $a^+$ , negativity therefore gives

$$(13.6) \quad a^* N_i - (a^+)^* N_{i+1} \geq 0.$$

All  $N_i$  remain effective. In particular  $K_{P_i} + D_i \sim_{\mathbb{Q}, B} N_i$  is relatively pseudo-effective, so a Mori fibre ending is impossible.

*Scaling construction.* Here are the existence and scaling details. For any rational  $0 < t \leq 1$ , choose a sufficiently small rational  $\epsilon > 0$ . The divisor  $tH + \epsilon D$  is ample, and a sufficiently divisible general representative  $T_{t, \epsilon}$  gives an effective big klt boundary

$$(13.7) \quad \Delta_t = (1 - \epsilon)D + T_{t, \epsilon} \sim_{\mathbb{Q}, B} D + tH.$$

The general representative has arbitrarily small coefficients and meets the SNC support transversely;  $(P, (1 - \epsilon)D)$  is klt. If the steps so far have scaling thresholds at least  $t$ , they are nonpositive for  $K_P + \Delta_t$ . Hence its pushed boundary is still klt, big, and rationally linearly equivalent to  $D_i + tH_i$ . Bigness is preserved under these non-extractive birational maps. On any such model a klt big boundary can moreover be represented with a small general ample part. Indeed take an effective big decomposition with the required ample part and mix it, with sufficiently small positive coefficient, into the existing klt representative.

Suppose that  $K_{P_i} + D_i$  is not nef, while  $K_{P_i} + D_i + \lambda_{i-1}H_i$  is nef. Its nef threshold  $\lambda_i$  on this interval is positive. Choose a rational  $t \in (0, \lambda_i)$ . The klt cone theorem applied to a representative at  $t$  with ample part leaves only finitely many negative extremal rays to test. The remaining part of the cone is nonnegative both at  $t$  and at  $\lambda_{i-1}$ , hence throughout their interval. Thus  $\lambda_i$  is a rational maximum of finitely many ray thresholds, and is attained on a ray  $R_i$  with

$$(K_{P_i} + D_i) \cdot R_i < 0, \quad (K_{P_i} + D_i + \lambda_i H_i) \cdot R_i = 0.$$

The klt contraction theorem at  $t$  contracts  $R_i$ . At  $\lambda_i$  the nef divisor is relatively semiample: use its klt representative with an ample part and apply basepoint-freeness to that representative with the ample part removed. It therefore descends through the contraction. If the contraction is small, the klt flip supplied by [11] at  $t$  is also the flip for  $K_{P_i} + D_i$ : over the contraction their classes are positive multiples, because the  $\lambda_i$  combination descends. These are the ordinary dlt MMP steps; they preserve  $\mathbb{Q}$ -factoriality and dlt, and are nonpositive for all the positive-scaling combinations with parameter at most  $\lambda_i$ . This constructs the sequence until nefness, or constructs an infinite sequence with nonincreasing thresholds.

*Limiting threshold.* We next prove that an infinite sequence must have

$$(13.8) \quad \lambda_i \longrightarrow 0.$$

Every divisorial contraction lowers the relative Picard number, so there are only finitely many. Fix an index  $j$  after the last one. If the thresholds had positive limit, choose a rational closed interval bounded away from zero, with upper endpoint at most the previous scaling bound, containing all sufficiently late thresholds. On  $P_j$ , choose klt big representatives of  $D_j + tH_j$  at the two endpoints. They can be chosen to contain small common positive multiples of finitely many general ample divisors spanning  $N^1(P_j/B)$ , as well as a fixed ample part. To obtain these supports, subtract the desired sufficiently small ample sum from the big class, represent the remainder effectively, and mix this representative into a klt one with a small coefficient. The endpoint representatives determine a rational segment of klt boundaries. Allowing small independent changes in the coefficients of the spanning divisors embeds it in a rational polytope of klt boundaries with fixed ample part. The adjoints remain big after shrinking these changes: on the segment their relative classes are  $N_j + tH_j$ , an effective class plus a big class with  $t > 0$ .

BCHM finiteness for this polytope gives finitely many marked ample models over  $B$  [11, Corollary 1.1.5]. Every sufficiently late  $P_i$  is one of them. In fact  $P_j \dashrightarrow P_i$  is small, and the pushes of the chosen divisor classes still span  $N^1(P_i/B)$ . For completeness, a small birational map between  $\mathbb{Q}$ -factorial models identifies these numerical spaces: the pullback difference of a numerically trivial divisor and its transform on a common resolution is exceptional over the other model and numerically trivial on its contracted curves; negativity applied with both signs makes that difference zero. The same argument in reverse proves the assertion. At the parameter  $\lambda_i$  the pushed adjoint is nef. A sufficiently small rational change in the spanning coefficient directions adds a relatively ample class on  $P_i$ , while staying in the polytope, so the resulting adjoint is ample there. Sections of the corresponding divisors agree across the small map, as they are determined by the same inequalities at prime divisors. Thus  $P_i$ , with its marking from  $P_j$ , is the ample model of this perturbed boundary.

Distinct negative steps cannot return to the same marked model. The pullback difference for  $K + D$  is effective at each step and is nonzero at a genuinely negative step. If it were zero, the two pullbacks would agree on a common resolution. A curve dominating a negative contracted curve would then have negative intersection on the old side, whereas its image on the other side is contracted to the same contraction base and has nonnegative intersection there, a contradiction. This gives strict discrepancy improvement at some valuation. Subsequent discrepancies cannot decrease, so repetition of the marked pair is impossible. The finite list of marked ample models therefore rules out a positive limiting threshold, proving (13.8).

*Preservation of lc centres.* The lc centres and their incidences have been retained at every step. Indeed discrepancy monotonicity and crepancy give

$$0 \leq A_{D_i}(v) \leq A_{C^{\sharp}}(v), \quad A_{C^{\sharp}}(v) = A_{D_i}(v) + v(N_i).$$

Together with the equality of initial lc places, this shows that the lc places are exactly the same and have  $v(N_i) = 0$ . Their centres are consequently not contained in  $\text{Supp } N_i$ . Every negative contracted curve is contained in  $\text{Supp } N_i$ , since an effective  $\mathbb{Q}$ -Cartier divisor has nonnegative intersection with a curve not contained in its support. The contraction is therefore an isomorphism over a target neighbourhood of the image of each lc centre's generic point. To see this, a positive-dimensional projective fibre through a point outside  $\text{Supp } N_i$  would contain a contracted curve through that point. A zero-dimensional point of a connected fibre cannot be a separate component of a larger fibre. Properness and normality of the target then give the asserted isomorphism neighbourhood. The flip agrees over the same neighbourhood. For an inclusion of lc centres, test it at the generic point of the smaller centre; this lies in that common isomorphism locus. Inclusions are thus preserved in both directions, and properness preserves their images in  $B$ . Only finite compositions will be used below.

*Movable representatives.* Fix  $j$  after the last divisorial contraction in a putative infinite sequence, or take its final nef model in the finite case. Fix also a relatively ample Cartier divisor  $A$  on  $P_j$ .

We claim that for every rational  $\delta > 0$  and every finite list of primes on  $P_j$  there is an effective divisor avoiding that list with

$$(13.9) \quad E_\delta \sim_{\mathbb{Q},B} N_j + \delta A.$$

In the finite case  $N_j$  is nef and the sum is ample, so general representatives on the affine base give the claim. In the infinite case choose  $i$  first, sufficiently large that  $\delta A - \lambda_i H_j$  is ample on the fixed model  $P_j$ . All maps from  $P_j$  to  $P_i$  are small. Choose an ample Cartier divisor  $A_i^+$  on  $P_i$ . Once  $i$  and this divisor have been fixed, openness of the ample cone permits a positive rational  $\alpha$  such that

$$(13.10) \quad \delta A - \lambda_i H_j - \alpha(A_i^+)_{P_j} \quad \text{is relatively ample.}$$

The divisor  $N_i + \lambda_i H_i + \alpha A_i^+$  is also ample, being nef plus ample. On the affine base choose effective representatives of it and of (13.10), avoiding the corresponding finite lists of primes. Transform the first back to  $P_j$  and add the second. Smallness identifies the primes and yields (13.9). The choice of  $\alpha$  is made *after*  $i$ ; no bound on the size of  $(A_i^+)_{P_j}$  is required.

*Vanishing of the exceptional error.* We show first that  $N_j$  is vertical over  $B$ . On the generic fibre take a common resolution of  $P$ ,  $P_j$ , and  $G_{\text{gen}}$ . The effective Cartier pullbacks of  $N_j$  are bounded above by those of  $N$  by (13.6). Since  $N$  is exceptional over  $G_{\text{gen}}$ , the order of  $N_j$  at every prime of  $G_{\text{gen}}$  is therefore zero. This assertion concerns Cartier pullback data on a common resolution; it does not assert a morphism  $P_j \rightarrow G_{\text{gen}}$ .

Suppose a horizontal component  $F$  of  $N_j$  has coefficient  $n_F > 0$ . Apply (13.9) with  $F$  in the avoided list. On the generic fibre relative linear equivalence becomes ordinary linear equivalence, so, for a rational function  $h_\delta$  and a positive integer  $m_\delta$ ,

$$E_\delta = N_j + \delta A + \frac{1}{m_\delta} \text{div}(h_\delta).$$

Taking its coefficient at  $F$  gives, with  $a_F = \text{ord}_F(A)$ ,

$$(13.11) \quad \frac{\text{ord}_F(h_\delta)}{m_\delta} = -n_F - \delta a_F.$$

Pull this effective divisor to the common resolution and take its trace on  $G_{\text{gen}}$ . The zero trace of  $N_j$  gives

$$(13.12) \quad \frac{1}{m_\delta} \text{div}_{G_{\text{gen}}}(h_\delta) + \delta A_{\text{tr}} \geq 0,$$

where  $A_{\text{tr}}$  is the fixed Weil trace of the Cartier pullback data of  $A$ . Choose a Cartier divisor  $H_G \geq A_{\text{tr}}$  on the normal projective variety  $G_{\text{gen}}$ . Such a divisor exists: for a sufficiently large multiple of an ample Cartier divisor  $L_G$ , the coherent divisorial sheaf  $\mathcal{O}_{G_{\text{gen}}}(mL_G - A_{\text{tr}})$  has a nonzero section  $h$ , and  $mL_G + \text{div}(h)$  is a Cartier majorant. Consequently the divisor obtained from (13.12) by replacing  $A_{\text{tr}}$  with  $H_G$  is effective and  $\mathbb{Q}$ -Cartier. Pulling *this* divisor to the valuation  $F$  gives

$$\frac{\text{ord}_F(h_\delta)}{m_\delta} \geq -\delta \text{ord}_F(H_G).$$

Together with (13.11), this implies

$$0 < n_F \leq \delta(\text{ord}_F(H_G) - a_F)$$

for arbitrarily small positive rational  $\delta$ , a contradiction. This proves verticality without assuming  $G_{\text{gen}}$  is  $\mathbb{Q}$ -factorial or pulling back a Weil divisor that is not Cartier.

Next  $N_j$  is very exceptional over  $B$ : it is vertical, and over every prime divisor  $T \subset B$  some prime divisor dominating  $T$  is absent from its support. A component of  $N_j$  dominating a base prime must dominate a boundary prime, because it is the transform of a toroidal component of  $N$ . For a boundary prime  $T$ , minimize  $\phi$  over a positive vector on its base axis. A rational minimizer exists, and its supporting face has  $g = 0$ . Its toroidal divisorial valuation has an lc centre dominating

$T$  on  $P$ , hence on  $P_j$ . That centre is not contained in  $\text{Supp } N_j$ . The pullback of a local Cartier equation of  $T$  vanishes at its generic point. A prime component of this Cartier pullback containing that centre dominates  $T$  and is absent from  $N_j$ . This reasoning also applies when the lc centre has higher codimension; the centre itself need not be a prime divisor. For a non-boundary prime  $T$ , surjectivity and the same Cartier pullback argument supply a dominating prime, and no component of  $N_j$  dominates  $T$ .

Choose rational  $\delta_k \downarrow 0$  and representatives in (13.9) avoiding all components of  $N_j$ . For each such component  $F$ , their restrictions to  $F$  are effective Cartier divisors after clearing denominators, with proper closed supports. Every curve in  $F$  over  $B$  not contained in the countable union of these supports satisfies

$$(N_j + \delta_k A) \cdot C \geq 0 \quad \text{for all } k, \quad N_j \cdot C \geq 0.$$

This is precisely nefness on very general curves of  $F/B$ . The morphism  $P_j \rightarrow B$  is still a projective contraction: its generic function field is regular over that of the normal base, so its finite Stein factor is  $B$ . Apply Shokurov's very-exceptional negativity lemma in the form of [6, Lemma 3.3] to

$$D' = -N_j, \quad (D')^+ = 0, \quad (D')^- = N_j.$$

Its negative part is very exceptional, and  $-D' = N_j$  is nef on the required very general curves. The lemma gives  $D' \geq 0$ . Since  $N_j \geq 0$ , it follows that  $N_j = 0$ .

This vanishing occurs on the fixed *finite* model  $P_j$ . It makes  $K_{P_j} + D_j$  relatively rationally linearly trivial and rules out any subsequent negative step. In particular the proposed infinite sequence cannot occur. The effective dlt structure and the asserted centre correspondence are therefore obtained after finitely many steps, as claimed.  $\square$

### 13.2. Incidence in a prescribed connected component.

**Proposition 13.2** (Lc incidence). *Let  $s$  have nonzero base cone  $\tau_s$ . Let  $\delta_0$  be a source face on which  $g$  vanishes, let  $V = \overline{O_{\delta_0}}$  be its closed stratum, and let  $C_s$  be any connected component of  $V_s$ . The zero face is allowed, with  $V = Y$ . Then there is a source cone  $\beta$  such that*

$$g|_{\beta} = 0, \quad p_*(\beta) = \tau_s, \quad \overline{O_{\beta}} \subset V, \quad O_{\beta} \cap C_s \neq \emptyset.$$

*Proof.* First disregard  $V$  and  $C_s$ . A minimizer for  $\phi$  over a rational vector in  $\text{relint}(\tau_s)$  has a support face on which  $g$  vanishes. The onto-cone refinements imply that this face maps onto  $\tau_s$ . Its stratum occurs over  $s$ : the smooth stratum map has dense open image, properness gives a specialization over  $s$ , and the toroidal face chart at that specialization has open stratum dense in its ordinary fibre branch. This is also the stratum availability used in (T). Since  $Y_s$  is connected, this proves the assertion when  $V = Y$ .

Assume  $\delta_0 \neq 0$ . Closed strata in this strict toroidal model are normal and are unions of strata. At a deeper stratum the boundary divisors containing  $V$  determine a unique face and a unique normal orbit-closure germ. Strictness identifies these divisors globally, so different local face branches of the same closed stratum cannot be confused. In particular their individual ray directions, and coefficients on those directions, agree at incidences.

We separate the chosen component by a pointed étale neighbourhood

$$(B_2, s_2) \longrightarrow (B, s), \quad k(s_2) = k(s) = \kappa.$$

Indeed the Stein factor  $S_V$  of the proper map  $V \rightarrow B$  is finite over  $B$ , also when  $V \rightarrow B$  is nondominant. The points of its fibre over  $s$  correspond to the connected components of  $V_s$ . Over the henselization at  $s$ , the finite algebra splits by the idempotents separating these points. The finitely many idempotents descend to some pointed étale neighbourhood. The associated open-and-closed piece of  $V \times_B B_2$  has precisely  $C_s$  in its pointed fibre. This pullback is normal, and its irreducible components are disjoint. Thus  $C_s$  lies in one such component  $V_2$ , whose fibre at  $s_2$  is exactly  $C_s$  as a topological space. The component  $V_2$  is again a closed stratum.

Shrink to a connected smooth affine neighbourhood of  $s_2$ . The pulled-back  $Y$  remains integral: its generic fibre is geometrically integral and its normal étale pullback has no additional vertical component. It remains strict toroidal with connected fibres. The forms, cones, and gap function pull back, and the pointed ordinary fibres are identified. Use the notation  $Y/B, s$  for this new family during the existence argument. No lift of the trait to  $B_2$  is needed, and no bound on the degree of this neighbourhood will be used.

Apply Lemma 13.1 to obtain the effective dlt crepant contraction  $(P_j, D_j) \rightarrow B$ . On its initial toroidal resolution  $P$ , one lc centre maps onto  $V_2$ : take the centre of a toroidal lc valuation with weights in  $\text{relint}(\delta_0)$ . Another lc centre maps onto the closure of an all-zero cone over  $\tau_s$  furnished by the first paragraph. The latter centre has base image  $\overline{O_{\tau_s}}$ . Transform both to  $P_j$ . Inside the second centre choose an lc centre  $Z$  minimal by inclusion among those whose image contains  $s$ . This is also inclusion-minimal among *all* lc centres with image containing  $s$ : any strictly smaller one would be a subcentre of the second centre as well.

Kollár's linking theorem [27, Theorem 10] now applies. Its hypotheses hold: the morphism is proper with connected fibre over  $s$ , the boundary is effective, the pair is dlt, and  $K_{P_j} + D_j \sim_{\mathbb{Q}, B} 0$ . It yields inside the first centre a subcentre  $Z'$  whose image contains  $s$  and which is  $\mathbb{P}^1$ -linked to  $Z$ . Directly linked centres have the same base image, by the definition of a link [27, Definition 9]; hence so do centres joined by a chain of links. Consequently

$$s \in f(Z') = f(Z) \subset \overline{O_{\tau_s}}.$$

Return these centre inclusions through the finite MMP to  $P$ , using Lemma 13.1, and then map to  $Y$ . On the SNC model the centres are intersections of coefficient-one divisors, so are closures of all-zero strata. A toroidal subdivision maps such a closure to a stratum closure with  $g = 0$ : a relative interior zero vector lies in its old support cone, and nonnegativity and linearity force  $g$  to vanish on that cone. We obtain an all-zero closed stratum contained in  $V_2$ , whose image contains  $s$  and is contained in  $\overline{O_{\tau_s}}$ . Its image is itself the closure of a base stratum, by smoothness of the open stratum map and properness. Since  $s \in O_{\tau_s}$ , these two containments force its image to equal  $\overline{O_{\tau_s}}$ .

Its cone therefore maps onto  $\tau_s$ , and its open stratum meets the ordinary fibre over  $s$ . Explicitly, at a closure point in that fibre the relevant face branch already kills every base boundary equation; the remaining base parameters are smooth stratum parameters and can be eliminated. The open stratum is dense in this fibre branch, so it supplies the required point. The point belongs to  $(V_2)_s = C_s$ . Projecting through the pointed étale neighbourhood returns an all-zero stratum and this component incidence to the original ordinary fibre. This proves the proposition.  $\square$

**13.3. Uniformly short labelled paths.** Return to the original parameter fibre with its bounded Kummer data and the labelled boundary-weight vector  $x$  of  $w$ . Write  $x$  in the primitive ray coordinates of its simplicial support cone  $\delta$ . Retain the coefficients on rays where  $g = 0$  and set the others to zero; denote the resulting vector by  $x_0$ , its support face by  $\delta_0$ , and  $g(x)$  by  $h$ . Finiteness of the ray data gives a constant  $C_0$ , depending only on the bounded families, such that

$$(13.13) \quad \begin{aligned} \|x - x_0\| &\leq C_0 h, & b - p_* x_0 &\geq 0, \\ \|b - p_* x_0\| &\leq C_0 h. \end{aligned}$$

Indeed each discarded coefficient is at most  $h/g(r)$  for its primitive ray  $r$ , the positive numbers  $g(r)$  belong to a fixed finite list, and the ray maps belong to a fixed finite list as well. The middle inequality is coordinatewise in the base cone. We may use the sum of absolute ray coefficients for all norms; the finitely many changes of cone coordinates only change uniform constants.

We construct a path from the labelled  $x$  to a labelled vector defining a valuation  $w_*$  with  $A_\sigma(w_*) = 0$ . If  $h = 0$ , take  $w_* = w_x$  and no path. If the base cone is zero and  $h > 0$ , then  $b = 0$  and the segment from  $x$  to  $x_0$  stays in the same slice and labelled chart. Its length is at most  $C_0 h$ , and its endpoint has zero gap. These cases include the zero face and all zero-dimensional cells.

It remains to consider  $h > 0$  and  $\tau_s \neq 0$ . Put  $V = \overline{O_{\delta_0}}$ , taking  $V = Y$  if  $\delta_0 = 0$ , and let  $C_s$  be the connected component of  $V_s$  containing the label of  $x$ . Proposition 13.2 supplies an all-zero cone  $\beta$  whose labelled stratum meets  $C_s$ .

Partition  $C_s$  by the irreducible components it contains of  $O_\gamma \cap Y_s$ , where the stratum closure lies in  $V$  and  $p_*(\gamma) = \tau_s$ . These smooth stratum fibres cover  $C_s$ . There is a uniform bound  $M$  for the number of their irreducible components. In fact the strata and their maps belong to the fixed finite list of finite-type families. Cover their source and base by finitely many affine charts and embed the source charts in fixed affine spaces over the base charts. The equations have bounded degree, so Bézout bounds the number of geometric components of every chart fibre, and hence of each stratum fibre. This bound concerns the original family before the unbounded étale neighbourhood used only for the existence proof.

Make a graph on these components, joining two when one meets the closure of the other in  $C_s$ . The graph is connected. Otherwise the unions belonging to two groups of connected graph components would both be closed: any point of the closure of one piece belongs to another piece that would then be joined to it. The finite unions would partition the connected space  $C_s$  into disjoint nonempty closed subsets. A simple path therefore joins the starting label to a label of  $\beta$  with at most  $M - 1$  edges. At each edge the strata are comparable by a face inclusion. A point of the deeper stratum in the closure of the other component supplies a chart realizing exactly that face label.

Every cone  $\gamma$  on this path contains the face for  $V$ , with the same ray directions, so we may transport  $x_0$  to it. Surjectivity of  $\gamma \rightarrow \tau_s$  and the fixed ray maps allow us to add weights on rays above each base axis whose sum projects to  $b - p_*x_0$ . Their total size is at most a uniform multiple of  $h$ , by (13.13). This gives a rational lift of  $b$  within uniform distance of  $x_0$ . Because  $b$  is interior to  $\tau_s$ , there is also a rational lift in  $\text{relint}(\gamma)$ . Interpolating with a sufficiently small positive rational coefficient gives a vector  $x_\gamma \in \text{relint}(\gamma)$  with

$$p_*x_\gamma = b, \quad \|x_\gamma - x_0\| \leq C_1 h.$$

The coefficient of interpolation may depend on the chosen lift; the bound does not. Use  $x$  itself for the starting choice.

For each edge join its two chosen vectors by the segment in the larger cone, using the chart at the incidence point. Their common face label is the specified component, and the labelled valuation compatibility in Proposition 12.2 identifies the evaluations when moving to the next chart. Each segment has length at most a fixed multiple of  $h$ , so the total length is at most  $C_2 h$ . The endpoint lies in  $\beta$ , has gap zero, and has excess zero by (TE). Its rational monomial valuation  $w_*$  is divisorial, restricts to  $u$ , and satisfies  $A_\sigma(w_*) = 0$ .

For  $\omega \in \{\psi, \chi\}$  take its comparison ratio  $H_\omega$  with  $\sigma$  at a common tensor index  $p_\omega$ , so that

$$A_\omega(v) = A_\sigma(v) + \frac{v(H_\omega)}{p_\omega}.$$

The two estimates (TV), first for retraction and then along the finitely many labelled segments, give in every case

$$(13.14) \quad \left| \frac{w(H_\omega)}{p_\omega} - \frac{w_*(H_\omega)}{p_\omega} \right| \leq C_3(E(w) + h) = C_3 A_\sigma(w).$$

The constant is uniform because both additional form boundaries are effective on  $G$ , which is exactly the hypothesis needed for the uniform ratio estimates. Neither the number of auxiliary MMP steps nor the degree of the étale neighbourhood enters it.

Write  $q_\omega(v) = v(H_\omega)/p_\omega$ . The assumed inequality at  $w_*$  is  $q_\chi(w_*) - cq_\psi(w_*) \geq 0$ . Thus

$$\begin{aligned} A_\chi(w) - cA_\psi(w) &= (1 - c)A_\sigma(w) + q_\chi(w) - cq_\psi(w) \\ &\geq -(|1 - c| + (1 + c)C_3)A_\sigma(w). \end{aligned}$$

This proves part (3) of Theorem 10.1 for nonzero base restriction with  $\bar{A}_\sigma(u) = 0$ , with precisely the required uniform dependence.

**13.4. Trivial base restriction and the dependency chain.** To treat  $w|_K = 0$ , adjoin one variable  $q$  to both fields and pull  $G$  back to  $K(q)$ . Extend an absolute form  $\omega$  of index  $r_\omega$  by wedging each tensor factor with  $d \log q$ ; denote the resulting index- $r_\omega$  form by  $\omega^{\text{ext}}$ . For every divisorial valuation  $v$  of  $L(q)$ ,

$$(13.15) \quad A_{\omega^{\text{ext}}}(v) = A_\omega(v|_L) + J(v), \quad J(v) \geq 0,$$

where the same  $J$  works for all the forms under consideration and the discrepancy at a trivial valuation is defined to be zero.

Here is the local calculation. If  $v|_L$  is nonzero, extract its prime  $E$  on a smooth model and work over its generic point. On the product with  $\mathbb{P}_q^1$ , the boundary of the extended form is the pullback of the form boundary together with the two reduced sections  $q = 0, \infty$ . Subtracting  $A_\omega(v|_L)$  leaves exactly the discrepancy of  $v$  for the product pair with the reduced pullback of  $E$  and these two sections. That pair is SNC with coefficients one, so its discrepancy  $J(v)$  is nonnegative and does not depend on  $\omega$ . If  $v|_L$  is trivial, work over the generic point of the model instead; the same calculation uses just the two sections. This proves (13.15), including homogeneous scales.

In particular the extended reference form is absolutely lc. If  $A_{\sigma^{\text{ext}}}(v) = 0$ , both  $A_\sigma(v|_L)$  and  $J(v)$  vanish. The hypothesis  $A_\chi \geq cA_\psi$  therefore passes to the extensions on the zero set of the extended reference discrepancy, including trivial restriction to  $L$ . The generic boundaries are still effective, and the geometric Fano and singularity assumptions on  $G$  persist. Moreover

$$\bar{A}_{\sigma^{\text{ext}}}(\text{ord}_{q=0}) = 0 \quad \text{on } K(q) :$$

nonnegativity gives one inequality, and the lift  $\text{ord}_{q=0}$  trivial on  $L$  has zero discrepancy and gives the other.

Given the original nonzero  $w$  trivial on  $K$ , extend it by the Gauss valuation on  $L(q)$  with  $q$  of weight one. This is a rational divisorial valuation: on the product of an extracting prime for  $w$  and the  $q$ -line it is the monomial valuation with those two rational weights. It restricts to  $\text{ord}_{q=0}$  on  $K(q)$  and has  $J = 0$ , since it is monomial for the reduced SNC product pair. The already proved case applied to this valuation yields the desired inequality at  $w$  by (13.15). The total dimension bound has increased by only one, so its constant is still uniform in the original numerical data. For the trivial valuation itself the inequality is  $0 \geq 0$ .

Parts (1) and (2), proved in Section 12, and the preceding argument prove Theorem 10.1 in full. The reductions of Section 10 now prove Theorem 9.1; Section 9 consequently proves Proposition 5.2. The valuative and rounding arguments then give Theorem 3.1, and the global reduction of Section 3 gives Theorem 1.1.

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